

专业英语1,9

第一章 Information Technology, the Internet, and You

信息技术，互联网，还有你

the parts of an information system: **people, procedures, software, hardware, data, and the Internet.**

The way to think about a personal computer is to realize that it is one part of an **information system**. There are several parts of an information system:

- **People** are an essential part of the system. The purpose of information systems is to make people, or **endusers** like you, more productive.

- **Procedures** are rules or guidelines to follow when using software, hardware, and data. They are typically documented in manuals written by computer professionals.

- **Software (programs)** provides step-by-step instructions to control the computer to convert **data** into **information**.

- **Hardware** consists of the physical equipment. It is controlled by software and processes data to create information. 硬件由物理设备组成。它由软件控制，处理数据以创造信息。

- **Data** consists of unprocessed facts, including text, numbers, images, and sound.

Information is data that has been processed by the computer. •数据由未经处理的事实组成，包括文本、数字、图像和声音。信息是计算机处理过的数据。

- The **Internet** allows computers to connect and share information. To efficiently and effectively use the computer, you need to understand **information technology (IT)**, including software, hardware, data, and connectivity•互联网允许计算机连接和共享信息。为了高效、有效地使用计算机，您需要了解信息技术(IT)，包括软件、硬件、数据和连接

1、People are the most important part of an information system. This book contains several features to demonstrate how people just like you use computers. These features include the following: 人是信息系统中最重要的一部分。这本书包含几个功能，以演示人们如何像你一样使用计算机。这些特点包括:

- **Making IT Work for You** presents several interesting and practical applications. Topics include using online office suites and cloud storage. •《让IT为你工作》展示了几个有趣而实际的应用。主题包括使用在线办公套件和云存储。

- **Tips** offer a variety of suggestions on such practical matters as how to improve slow computer performance and how to protect your privacy while on the web. • Tips 提供了各种各样的实用问题的建议，如如何提高缓慢的计算机性能和如何保护您的隐私在网上。

- **Privacy** marginal boxes discuss threats to your personal privacy and suggest ways to protect yourself. • 隐私边缘框讨论对您的个人隐私的威胁，并提出保护您自己的方法。

- **Environment** boxes discuss important and relevant environmental issues. The impact of computers and other technologies is more critical today than ever before. • 环境箱讨论重要的和相关的环境问题。今天，计算机和其他技术的影响比以往任何时候都更加重要。

- **Ethics** boxes pose a variety of different ethical/unethical situations for your consideration. • 伦理框提出了各种不同的伦理/不道德的情况，供您考虑。

- **Careers in IT** presents job descriptions, employment demands, educational requirements, salary ranges, and advancement opportunities.

- IT 行业的职业描述、就业要求、教育要求、工资范围和晋升机会。

2、Software, or programs, consists of system and application software.

软件或程序由系统软件和应用软件组成。

1) System software enables application software to interact with computer hardware. 系统软件使应用软件能够与计算机硬件交互。

- **Operating systems** coordinate resources, provide an interface, and run applications. Three types are

embedded (real-time, RTOS), stand-alone, and network (NOS). • 操作系统协调资源、提供接口、运行应用程序。三种类型是嵌入式(实时, RTOS)、单机和网络(NOS)。

- **Utilities** perform specific tasks to manage computer resources. • 实用程序执行特定的任务来管理计算机资源。

2) Application software includes general-purpose, specialized, and mobile applications. 应用软件包括通用应用、专用应用和移动应用。

- **General purpose**—widely used in nearly all career areas; programs include browsers, word processors, spreadsheets, database management systems, and presentation software. • 通用—广泛应用于几乎所有的职业领域; 程序包括浏览器、文字处理程序、电子表格、数据库管理系统和演示软件。

- **Specialized**—focus more on specific disciplines and occupations; programs include graphics and web authoring. • 专业化——更专注于特定的学科和职业; 程序包括图形和网络创作。

- **Apps (mobile apps, mobile applications)**—designed for mobile devices; most popular are for text messaging, Internet browsing, and connecting to social networks.

•应用程序(移动应用程序, 移动应用程序)–专为移动设备设计; 最受欢迎的是短信、互联网浏览和连接社交网络。 翻译结果来自有道神经网络翻译 (YNMT)

3、Hardware consists of electronic devices that can follow instructions to accept input, process the input, and produce information. 3、硬件由电子设备组成, 它们可以按照指令接受输入、处理输入并产生信息。

Types of Computers: Supercomputer, mainframe, midrange (server), and personal computers (PCs) are four types of computers.

计算机类型:超级计算机、主机、中端(服务器)和个人计算机(pc)是四种类型的计算机。

Personal computers can be **desktop, laptop (notebook computer), tablet, smartphone, and wearable.**

个人电脑可以是台式电脑、笔记本电脑、平板电脑、智能手机和可穿戴电脑。

Personal Computer Hardware: There are four basic categories of hardware devices: 个人电脑硬件:硬件设备有四种基本类别:

- System unit** contains electronic circuitry, including **microprocessors** and **memory**.

Random-access memory (RAM) holds the program and data currently being processed.

- 系统单元包含电子电路, 包括微处理器和内存。随机存取存储器(RAM)存储当前正在处理的程序和数据。

- Input/output devices** are translators between humans and computers. **Input devices** include the **keyboard** and **mouse**. The most common output device is the computer **display (monitor)**.

- 输入/输出设备是人类和计算机之间的翻译。输入设备包括键盘和鼠标。最常见的输出设备是电脑显示器(显示器)。

- **Secondary storage** holds data and programs. Typical media include **hard disks, solid-state storage, and optical discs (CD, DVD, and Blu-ray)**.

- 二级存储保存数据和程序。典型的媒体包括硬盘、固态存储和光盘(CD、DVD和蓝光)。

- Communication devices** allow personal computers to communicate with other computer systems. **Modems** modify audio, video, and other types of data for transmission across the Internet.

- 通信设备允许个人计算机与其他计算机系统通信。调制解调器修改音频、视频和其他类型的数据以进行传输通过互联网的使命。

4、Data is the raw unprocessed facts about something. Common file types include数据是关于某事的未经处理的事实。常见的文件类型包括

- Document files** created by word processors.

- 由文字处理器创建的文档文件。

- Worksheet files** created by spreadsheet programs.

- 通过电子表格程序创建工作表文件。

- Database files** created by database management programs.

- 数据库管理程序创建的数据库文件。
- Presentation files** created by presentation software programs.
- 演示软件程序创建的演示文件。

5、CONNECTIVITY AND MOBILE INTERNET 5、连通性和移动互联网

Connectivity describes the ability of end users to use resources well beyond their desktops. Central to the concept of connectivity is the **network** or communication system connecting two or more computers. : 连通性描述终端用户使用桌面以外的资源的能力。连接概念的核心是网络或通信 连接两台或多台计算机的系统。The **Internet** is the world's largest computer **network**. The **web** provides a multimedia interface to resources available on the Internet. Along with the Internet, three other things are driving the impact of technology 互联网是世界上最大的计算机网络。网络为互联网上的可用资源提供了多媒体界面。除了互联网，还有三件事正在推动技术的影响:

- Cloud computing** uses the Internet and the web to shift many computer activities from a user's computer to computers on the Internet.
- 云计算利用互联网和网络将许多计算机活动从用户的计算机转移到互联网上的计算机上。
- Wireless revolution** has changed the way we communicate and use computer technology. Wireless devices include tablets, smartphones, and watches.
- 无线革命改变了我们沟通和使用计算机技术的方式。无线设备包括平板电脑、智能手机和手表。
- The **Internet of Things (IoT)** is the continuing development of the Internet that allows everyday objects embedded with electronic devices to send and receive data over the Internet
- 物联网(IoT)是互联网的持续发展，它允许嵌入电子设备的日常物品发送和接收互联网数据

习题

第1章 信息技术、互联网和你

雨课堂

Chapter 01 Lesson test

Which part is the most important of an information system?

信息系统的哪个部分是最重要的?

People ✓ 人

Which part of an information system consists of the rules or guidelines for people to follow?

信息系统的哪一部分包含了人们要遵循的规则或指导方针?

Procedures ✓ 程序

Which part is the second most important of an information system?

哪个部分是信息系统的第二重要部分?

Data ✓ 数据

[Software] is another name for a program or programs. The Purpose of software is to convert [data] (unprocessed facts) into [information] (processed facts)

软件是程序或程序的另一个名称项目。软件的目的是将数据(未处理的事实)转换为信息(处理过的事实)

Chapter 1 Test

The most widely used handheld computers are ____

使用最广泛的手持计算机是

smartphones ✓ 智能手机

The purpose of software is to convert data into information

软件的目的是把数据转换成信息。✓

What are the two major software types?

两种主要的软件类型是什么?

System software and application software

系统软件和应用软件

System software includes all of the following except:

系统软件包括以下所有内容除了:

Operating systems 操作系统

Device drivers 设备驱动程序

Utilities 公/实用程序

Browsers 浏览器 ✓

An information system has several parts: [people], [procedures], [software], [hardware], [data], and [the Internet].

一个信息系统有几个部分

There are four types of computers: [supercomputers], [mainframe computers], [midrange computers], and [personal computers].

超级计算机 大型计算机 中型电脑 个人电脑

Chapter 1 Quiz

Which of the following computers is also referred to as a server and is used to support end users for things such as retrieving data from a database or providing application software?

以下哪一种计算机也被称为服务器，用于支持终端用户从数据库检索数据或提供对应用程序软件的访问?

A. Midrange ✓ B. Mainframe C. Supercomputer D. Microcomputer

Antivirus programs protect your computer system from viruses or malicious programs.

This is an example of which type of software?

防病毒程序保护您的计算机系统免受病毒或恶意程序的侵害。这是哪一种软件例子？

A. Browser software B. Operating system C. Application software D. Utilities ✓

Application software is "background" software which helps the computer manage its own internal resources. ✓

应用软件是帮助计算机管理自身内部资源的“后台”软件。

This computer component holds the program and data that is currently being processed.

这个计算机部件保存着当前正在处理的程序和数据。

RAM

Unlike memory this type of storage holds data and programs even after electric power to the computer system has been turned off.

与内存不同的是，这种类型的存储即使在计算机系统断电后也能保存数据和程序。

secondary

[Smartphones] are cell phones with wireless connections to the internet.

智能手机是可以无线上网的手机

[Microblogs], unlike traditional blogs, use only short sentences that take only a few seconds to write.

与传统的博客不同，微博只使用短短几秒钟就能写出来的短句。

[System software] enables the application software to interact with the computer hardware.

系统软件使应用软件能够与计算机硬件交互。

[Program] consists of the step-by-step instructions that tell the computer how to do its work.

程序由告诉计算机如何工作的一步步的指令组成。

What is an operating system? Give two examples.

Operating systems are programs that coordinate computer resources, provide an interface between users and the computer. Such as windows and linux.

操作系统是协调计算机资源，提供用户和计算机之间的接口的程序，如windows和linux。

课后习题

选择

1. The keyboard, mouse display, and system unit are:

键盘、鼠标、显示器和系统单元是: hardware

2. Programs that coordinate computer resources, provide an interface, and run applications are known as:

协调计算机资源、提供接口和运行应用程序的程序被称为:

operating systems

2. A browser is an example of a:

浏览器就是通用应用程序的一个例子

general-purpose application

3. Although not as powerful as a supercomputer, this type of computer is capable of great processing speeds and data storage.

虽然没有超级计算机那么强大,但这种类型的计算机具有很高的处理速度和数据存储能力。

mainframe 大型计算机

4. Apple' s Watch is what type of computer?

苹果手表是什么类型的电脑?

wearable可穿戴

- 6.RAM is a type of:

RAM是存储器的一种

memory

5. Unlike memory, this type of storage holds data and programs even after electrical power to the computer system has been turned off.

与内存不同的是,这种类型的存储即使在计算机系统断电后也能保存数据和程序

secondary

6. The type of file created by word processors to save, for example, memos, term papers, and letters.

由文字处理程序创建的文件类型,如备忘录、学期论文和信件

document

7. Uses the internet and the web to shift many computer activities from a user' scomputer to computers on the internet.

利用互联网和网络将许多计算机活动从用户的计算机转移到互联网上的计算机上。

cloud computing

8. The largest network in the world is [the]: Internet

匹配

9. Consists of the step-by-step instructions that tell the computer how to do its work.

程序由告诉计算机如何工作的一步步的指令组成。

f. program

10. Another name for a program.

软件是程序的另一个名称。

g. software

11. Enables the application software to interact with the computer hardware.
系统软件使应用软件与计算机硬件交互。
i. system software
12. Type of computer that is small enough to fit on top of or alongside a desk yet is too big to carry around.
台式机一种足够小，可以放在桌子上面或旁边，但又太大而不能随身携带的电脑。
a. desktop
13. A container that houses most of the electronic components that make up a computer system.
台式机一种足够小，可以放在桌子上面或旁边，但又太大而不能随身携带的电脑。
j. system unit
14. Devices that translate the processed information from the computer into a form that humans can understand.
输出设备:将计算机处理过的信息转换成人类可以理解的形式。
d. output
15. Unlike hard disks, this type of storage does not have any moving parts, is more reliable, and requires less power.
固态存储不同于硬盘，这种类型的存储没有任何活动部件，更可靠，需要更少的电力。
h. solid-state
16. The most widely used communication device.
调制解调器：最广泛使用的通信设备。
b. modem
17. A type of a file that might contain, for example, audience handouts, speaker notes, and electronic slides.
演示——一种可能包含听众讲义、演讲者笔记和电子幻灯片的文件类型
e. presentation
18. A communications system connecting two or more computers.
网络:连接两台或两台以上计算机的通信系统。
c. network
汉译英
嵌入式操作系统[embedded operating system]
物联网[IoT]
显示器[monitor]
智能手机[smartphone]
专用应用软件[specialized application]

匹配：

1. Consists of the step-by-step instructions that tell the computer how to do its work.

由告诉计算机如何工作的一步一步的指令组成。

program程序

2. Another name for a program.一个程序的另一个名字

software软件

3. Enables the application software to interact with the computer hardware.使应用软件能够与计算机硬件交互

system software系统软件

4. Type of computer that is small enough to fit on top of or alongside a desk yet is too big to carry around.小到可以放在桌子上或旁边，又大到随身携带的一种电脑。

desktop桌面

5. A container that houses most of the electronic components that make up a computer system.容纳组成计算机系统的大部分电子元件的容器 组件。

system unit系统单元

6. Devices that translate the processed information from the computer into a form that humans can understand将计算机处理过的信息转换成人类可以理解的形式的设备

output输出

7. Unlike hard disks, this type of storage does not have any moving parts, is more reliable, and requires less power.与硬盘不同，这种类型的存储没有任何移动部件，更可靠，需要更少的电力。

solid-state固态

8. The most widely used communication device.应用最广泛的通信设备。

modem调制解调器

9. A type of a file that might contain, for example, audience handouts, speaker notes, and electronic slides.

10. 一种文件类型，可能包含，例如，听众的讲义、演讲者的笔记和电子幻灯片。

presentation演讲

11. A communications system connecting two or more computers.连接两台或多台计算机的通信系统。

network网络

第九章 Privacy, Security and Ethics

隐私、安全和道德

The ubiquitous (/ju:'bɪkwɪtəs/, 普遍存在的，无所不在的) use of computers and technology prompts some very important questions about the use of personal data and our right to privacy

计算机和技术的普遍使用引发了一些关于个人数据的使用和我们的隐私权的非常重要的问题

To efficiently and effectively use computers, you need to be aware of

the potential impact of technology on people and how to protect yourself on the web
为了有效地使用计算机，你需要意识到技术对人们的潜在影响以及如何在网络上保护自己
information systems consist of people, procedures, software, hardware, data, and the Internet. T

People

- **Privacy:** What are the threats to personal privacy, and how can we protect ourselves?

隐私:个人隐私面临哪些威胁，我们如何保护自己？

- **Security:** How can access to sensitive information be controlled, and how can we secure hardware and software?

• 安全:如何控制敏感信息的访问，以及如何确保硬件和软件的安全？

- **Ethics:** How do the actions of individual users and companies affect society? Let us begin by examining privacy

• 道德:个人用户和公司的行为如何影响社会？让我们从研究隐私开始

Privacy

Privacy concerns the collection and use of data about individuals. There are three primary privacy issues:

隐私涉及个人数据的收集和使用。有三个主要的隐私问题:

- **Accuracy** relates to the responsibility of those who collect data to ensure that the data is correct.

• 准确性涉及到那些收集数据的人的责任，以确保数据是正确的。

- **Property** relates to who owns data.

• 财产关系到谁拥有数据。

- **Access** relates to the responsibility of those who have data to control who is able to use that data.

• 访问涉及那些拥有数据的人控制谁能够控制谁的责任使用这些数据。

Large Databases

Large organizations are constantly compiling information about us.

大型组织不断收集关于我们的信息。

This ever-growing volume of data is often referred to as **big data**.

这种不断增长的数据量通常被称为大数据。

information resellers or **information brokers** now exists that collects, analyzes, and sells such personal data, create **electronic profiles**,

现在存在收集、分析和出售此类个人数据的信息经销商或信息经纪人,建立电子档案

Personal information is a marketable commodity, which raises many issues:

- **Collecting public, but personally identifying, information**

- 收集公众但个人识别的信息

- **Spreading information without personal consent mistaken identity**

- 未经个人同意传播信息

This is an example of **mistaken identity** in which the electronic profile of one person is switched with another.

这是一个错误身份的例子, 其中一个人的电子档案被切换到另一个人。

- **Spreading inaccurate information**

- 散布不准确的信息

Freedom of Information Act

信息自由法案

Private Networks

employee-monitoring soft 员工监控软件

- **Employers can monitor e-mail legally**

- 雇主可以合法地监控电子邮件

The Internet and the Web

- **illusion of anonymity**

匿名幻想

People are not concerned about privacy when surfing the Internet or when sending e-mails
人们在网上或发送电子邮件时不关心隐私

When you browse the web, critical information is stored on the hard drive in these locations:

当您浏览网页时, 关键信息存储在硬盘驱动器的以下位置:

- **History files**

- 历史文件

- **Temporary Internet files, also known as the browser cache**

- Internet临时文件, 也称为浏览器缓存

- **cookie**

- 饼干 Another way your web activity can be monitored is with

cookies. Cookies are small data files that are deposited on your hard disk from websites you have visited.

- **privacy mode**

- 隐私模式

Incognito Mode ,Private Browsing

隐身模式，私密浏览

In addition, most browsers also offer a **privacy mode**, which ensures that your browsing activity is not recorded on your hard disk. For example, Google Chrome provides **Incognito Mode** accessible from the Chrome menu, and Safari provides **Private Browsing** accessible from the *Safari* option on the main menu.

此外，大多数浏览器还提供隐私模式，这可以确保你的浏览活动不会被记录在硬盘上。例如，谷歌Chrome提供了从Chrome菜单访问的隐身模式，Safari提供了从主菜单的Safari选项访问的私人浏览。

- Privacy Threats

- 隐私威胁

- Web bugs**•网络臭虫 which are invisible images or HTML code hidden within a web page or e-mail message, can be used to transmit information without your knowledge.隐藏在网页或电子邮件中的不可见图像或HTML代码，可以在你不知情的情况下用来传输信息。

- spyware**•间谍软件 The most dangerous type of privacy threat comes in the form of spyware. The term **spyware** is used to describe a wide range of programs that are designed to secretly record and report an individual's activities on the Internet. **Computer monitoring software** is perhaps the most invasive and dangerous type最危险的一种隐私威胁来自间谍软件。间谍软件(spyware)一词用于描述一系列旨在秘密记录和报告个人在互联网上活动的程序。计算机监控软件可能是最具侵入性和危险性的一种

- Computer monitoring software**•计算机监控软件

One type of computer monitoring software, known as a **keylogger**
most invasive and dangerous

- antispyware or spy removal programs**•反间谍软件或间谍清除程序

Online Identity网络身份

the information that people voluntarily post about themselves online
人们自愿在网上发布关于自己的信息

Security

- Hackers

- 黑客

——Gain unauthorized access with malicious intent

——获得未经授权的恶意访问

——Not all hackers are illegal

——并不是所有的黑客都是非法的

- **Cybercrime /computer crime**

- 网络犯罪/计算机犯罪

Cybercrimes can take various forms, including **identity theft, Internet scams, data manipulation, ransomware, and denial of service attacks**

网络犯罪可以采取各种形式，包括身份盗窃、互联网诈骗、数据操纵、勒索软件和拒绝服务攻击

Denial of service A **denial of service (DoS) attack** attempts to slow down or stop a computer system or network by flooding a computer or network with requests for information and data.拒绝服务拒绝服务(DoS)攻击试图减慢或停止

通过向计算机或网络发送大量请求而形成的计算机系统或网络
获取信息和数据。

- **Common Internet scams**•常见的网络诈骗

Advance fee loans

推进费用贷款

Auction fraud

拍卖欺诈

Fake antivirus software

假antivirus软件

Nigerian Scam

尼日利亚诈骗

Social Engineering

Social engineering is the practice of manipulating people to divulge private data.

社会工程是操纵人们泄露私人数据的行为。

One of the most common social engineering techniques is **phishing**Phishing attempts to trick Internet users into thinking a fake but official-looking website or e-mail is legitimate.

网络钓鱼是一种最常见的社交工程技术，它试图欺骗互联网用户，让他们认为一个虚假但看起来像官方的网站或电子邮件是合法的。

Malicious Software

A **cracker** is a computer criminal who creates and distributes malicious programs or **malware**, which is short for **malicious software**

黑客是创建并传播恶意程序或恶意软件的计算机罪犯，恶意软件是恶意软件的简称
三种最常见的病毒

- **Viruses**, migrate through networks and operating systems

- 病毒通过网络和操作系统迁移

- Worms

- 蠕虫

- Trojan horses

- 特洛伊木马

Malicious Hardware

Zombies are computers infected by a virus, worm, or Trojan horse that allows them to be remotely controlled for malicious purposes. A collection of zombie computers is known as a **botnet**, or **robot network**.僵尸是被允许的病毒、蠕虫或特洛伊木马感染的计算机它们可以被恶意远程控制。僵尸的集合计算机被称为僵尸网络或机器人网络。

Measures to Protect Computer Security

restricting access, encrypting data, anticipating disasters, and preventing data loss
限制访问，加密数据，预测灾难，防止数据丢失

1、Restricting Access: 限制访问

biometric scanning 生物识别扫描

fingerprint and iris (eye) scanners 指纹和虹膜(眼睛)扫描仪

- Security suites** provide a collection of utility programs designed to protect your privacy and security while you are on the web. T

- Firewalls** act as a security buffer between a corporation's private network and all

- Password managers** help you create strong passwords.

- 安全套件提供了一组实用程序，旨在保护您的
当你在网络上时，隐私和安全。T

- 防火墙在公司的私有网络 and 所有网络之间起到安全缓冲的作用

- 密码管理器帮助您创建强密码。

2、Encrypting Data: 加密数据

Hypertext transfer protocol secure (https) requires browsers and websites to encrypt all messages.

超文本传输协议安全(https)要求加密所有消息的浏览器和网站

Virtual private networks (VPNs) encrypt connections between company networks and remote users.

实际的私有网络(vpn)加密连接在公司网络和远程用户之间

WPA2 (Wi-Fi Protected Access) is the most widely used wireless network encryption for home wireless networks.

WPA2 (Wi-Fi保护接入)是使用最广泛的无线减少家庭无线网络的网络加密。

3、Anticipating disasters : 预测灾害: Prepare for disasters by ensuring physical security and data security through a disaster recovery plan.通过灾难恢复计划, 确保物理安全和数据安全, 为灾难做好准备。

4、Preventing data loss : 防止数据丢失: Routinely copy data and store it at a remote location.例行地复制数据并将其存储在远程位置。

Ethics

The essential element that controls how computers are used today is *ethics*.

控制计算机运行方式的基本要素今天使用的是道德规范。

Copyright is a legal concept that gives content creators the right to control use and distribution of their work.

版权是一个法律概念, 它赋予内容创造者控制其作品的使用和传播的权利

digital rights management (DRM). DRM encompasses various technologies that control access to electronic media and files.

数字权利管理(DRM)。DRM包含控制对电子媒体和文件访问的各种技术

cyberbullying is the use of the Internet, smartphones, or other devices to send or post content intended to hurt or embarrass another person.

网络欺凌是指利用互联网、智能手机或其他设备发送或发布旨在伤害或使他人尴尬的内容

plagiarism, which means representing some other person's work and ideas as your own without giving credit to the original source.

剽窃, 也就是代表别人的观点 你的作品和想法都是你自己的, 不要注明原始来源

课本习题

1. The three primary privacy issues are accuracy, property, and: A
 - a. access c. ownership
 - b. ethics d. security

三个主要的隐私问题是准确性, 财产和 access访问;ownership所有权 ethics 伦理学;security安全;

2. Highly detailed and personalized descriptions of individuals are electronic: C
 - a. cookies c. profiles
 - b. histories d. web bugs

高度详细和个性化的个人描述是电子的:

曲奇,档案, 历史,网络臭虫

3. Browsers store the locations of sites visited in a: A

- a. history file c. tool bar
- b. menu d. firewall

浏览器将访问过的站点的位置存储在:

A.历史文件c.工具栏

menu菜单;firewall防火墙。

4. The browser mode that ensures your browsing activity is not recorded. C

- a. detect c. privacy
- b. insert d. sleep

确保您的浏览活动不被记录的浏览器模式

5. The information that people voluntarily post in social networking sites, blogs, and photo– and video–sharing sites is used to create their: C

- a. access approval c. online identity
- b. firewall d. phishing

人们自愿在社交网站、博客和照片和视频分享网站是用来创建他们的:

A.访问批准c.在线身份 b,防火墙;d,网络钓鱼

6. Computer criminals who create and distribute malicious programs. B

- a. antispyware c. cyber traders
- b. crackers d. identity thieves

制造和散布恶意程序的计算机罪犯

A.反间谍c.网络商人 d,身份信息窃取者

7. Programs that come into a computer system disguised as something else are called: A

- a. Trojan horses c. web bugs
- b. viruses d. zombies

伪装成其他东西进入计算机系统的程序被称为:

a.特洛伊木马c.网络bug 病毒;僵尸;

8. The use of the Internet, cell phones, or other devices to send or post content intended to hurt or embarrass another person is known as: A

- a. cyberbullying c. social networking discrimination
- b. online harassment d. unethical communication

使用互联网、手机或其他设备发送或发布旨在伤害或使他人尴尬的内容被称为:

A.网络欺凌c.社交网络歧视

c.在线骚扰d.不道德的沟通

9. Special hardware and software used to control access to a corporation's private network is known as a(n): C

- a. antivirus program c. firewall
- b. communication gate d. spyware removal program

用于控制对公司专用网络的访问的特殊硬件和软件称为(n):

A.防病毒程序c.防火墙

B.通信门d.间谍软件删除程序

10. To prevent copyright violations, corporations often use: B

- a. ACT c. VPN
- b. DRM d. WPA2

为了防止侵犯版权，公司经常使用:

1. Privacy concern that relates to the responsibility to ensure correct data collection.
 - a. accuracy 隐私问题涉及到确保正确收集数据的责任。答:准确性
2. Individuals who collect and sell personal data.
 - e. information brokers 收集和出售个人资料的个人。e .信息经纪人
3. Small data files deposited on your hard disk from websites you have visited.
 - c. cookies 从您访问过的网站存储在您硬盘上的小数据文件。c .饼干
4. Wide range of programs that secretly record and report an individual's activities on the Internet.
 - i. spyware各种秘密记录和报告个人在互联网上活动的程序。
我间谍软件。
5. Malicious programs that damage or disrupt a computer system.
 - f. malware破坏或破坏计算机系统的恶意程序。 恶意软件
6. Infected computers that can be remotely controlled.
 - j. zombies 被感染的电脑可以被远程控制。 僵尸
7. Used by scammers to trick Internet users with official-looking websites.
 - g. phishing骗子利用看起来像官方网站的网站来欺骗互联网用户。 钓鱼
8. A type of scanning device such as fingerprint and iris (eye) scanner.
 - b. biometric 一种扫描设备，如指纹和虹膜(眼睛)扫描仪。b生物
9. 9. Process of coding information to make it unreadable except to those who have a key.
 - d. encryption 对信息进行编码，使其只有有密钥的人才能读懂。 加密
10. An ethical issue relating to using another person's work and ideas as your own without giving credit to the original source.
 - h. plagiarism将他人的工作和想法当作自己的而不给予他人的道德问题
信用归原始来源。h .剽窃

专业英语2

第2章 The Internet, the Web, and Electronic Commerce

第2章 The Internet, the Web, and Electronic Commerce

互联网、Web和电子商务

INTERNET AND WEB互联网和网络

Internet 互联网

Launched in 1969 with **ARPANET**, the **Internet** consists of the actual physical network.

1969年，随着阿帕网(ARPANET)的推出，互联网由实际的物理网络组成。

Web 网络

Introduced in 1991, the **web (World Wide Web, WWW)** provides a multimedia interface to Internet resources. Three generations: **Web 1.0** (links existing information, search engines), **Web 2.0** (creates dynamic content and social interaction, social media sites), and **Web 3.0** (identifies relationships between data).

万维网(World Wide web, WWW)于1991年问世，为互联网资源提供了一个多媒体界面。三代:Web 1.0(链接现有信息、搜索引擎)、Web 2.0(创建动态内容和社会互动、社交媒体站点)和Web 3.0(确定数据之间的关系)。

Common Uses 共同使用

The most common uses of the Internet and the web include

互联网和网络最常见的用途包括

- Communication—the most popular Internet activity. •交流——最受欢迎的互联网活动。
- Shopping—one of the fastest-growing Internet activities. •购物——增长最快的互联网活动之一。
- Searching—access libraries and local, national, and international news. •搜索访问图书馆和当地、国内和国际新闻。
- Education—**e-learning** or taking online courses. •在线教育或在线课程。

- Entertainment—music, movies, magazines, and computer games.

•娱乐——音乐、电影、杂志和电脑游戏。

INTERNET ACCESS互联网接入

Once connected to the Internet, your computer seemingly becomes an extension of a giant computer that branches all over the world. 一旦连接到互联网，你的计算机似乎就变成了一个遍布世界各地的巨型计算机的扩展。

Providers 供应商

Internet service providers are connected to the Internet, providing a path for individuals to access the Internet. Connection technologies include **DSL**, **cable**, and **wireless modems**. 互联网服务提供商与互联网连接，为个人访问互联网提供了一条路径。连接技术包括DSL、电缆和无线调制解调器。

Browsers 浏览器

Browsers provide access to web resources. Some related terms are 浏览器提供对web资源的访问。一些相关的术语是

- **URLs—locations or addresses** to web resources; two parts are **protocol** and **domain name**; **top-level domain (TLD)** or **web suffix** identifies type of organization. •url—网站资源的位置或地址; 协议和域名两部分;顶级域名(TLD)或web后缀标识组织类型。

- **HTML**—commands to display **web pages**; **hyperlinks (links)** are connections. Technologies providing interactive, animated websites include **cascading style sheets**, or **CSS** (to control the appearance of web pages); **JavaScript** (executes on the user's computer to trigger interactive features); and **PHP** (like JavaScript except executes on website's computer).

Mobile browsers run on portable devices. •html –命令显示网页;超链接(链接)是连接。提供交互式动画网站的技术包括层叠样式表(CSS)(用于控制网页的外观);JavaScript(在用户的计算机上执行以触发交互特性);和PHP(类似于JavaScript，除了在网站的计算机上执行)。移动浏览器运行在便携式设备上。

WEB UTILITIES网络工具

Web utilities are specialized utility programs that make using the Internet and the web easier and safer. 网络实用程序是专门的实用程序，使使用互联网和网络更容易和更安全。

Filters 过滤器

Filters are used by parents and organizations to block certain sites and to monitor use of the Internet and the web. 过滤器被家长和组织用来屏蔽某些网站，并监控互联网和网络的使用。

File Transfer Utilities 文件传输工具

File transfer utilities copy files to (**downloading**) and from (**uploading**) your computer. There are three types 文件传输实用程序复制文件到(下载)和从(上传)您的计算机。有三种类型

- **File transfer protocol (FTP)** and **secure file transfer protocol (SFTP)** allow you to efficiently copy files across the Internet. •FTP (File transfer protocol)和SFTP (secure File transfer protocol)允许您在Internet上高效地复制文件。
- **BitTorrent** distributes file transfers across many different computers. •BitTorrent在许多不同的计算机上分发文件传输。
- **Web-based file transfer services** make use of a web browser to upload and download files. •基于web的文件传输服务使用web浏览器上传和下载文件。

Internet Security Suite 网络安全套件

An **Internet security suite** is a collection of utility programs designed to protect your privacy and security on the Internet. 一个互联网安全套件是一个实用程序的集合，旨在保护您的隐私和安全在互联网上。

COMMUNICATION沟通

Social Networking 社交网络

Social networking sites connect people and organizations that share a common interest or activity. Common features include **profiles, pages, groups, friends, news feeds, and share settings**. Two well-known sites are **Facebook** and **LinkedIn**. 社交网站将拥有共同兴趣或活动的人和组织联系起来。常见功能包括配置文件、页面、组、朋友、新闻源和共享设置。两个著名的网站是Facebook和LinkedIn。

Blogs, Webcasts, and Wikis 博客、网络广播和维基

Other sites that help individuals communicate across the web are blogs, microblogs, webcasts, podcasts, and wikis. 其他帮助个人通过网络交流的网站有博客、微博、网络广播、播客和维

基。

- **Blogs (web logs)** are typically personal websites to keep in touch with friends and family. Some are like online diaries. Businesses, newspapers, and others also use blogs as a quick publishing method. • 博客(网络日志)是典型的个人网站, 用来与朋友和家人保持联系。有些就像在线日记。企业、报纸和其他机构也使用博客作为一种快速发布的方法。
- **Microblogs**, unlike traditional blogs, use only short sentences that take only a few seconds to write. **Twitter** is the most popular microblogging site. A Twitter message is known as a **tweet**. • 与传统博客不同, 微博只使用几秒钟就能写完的短句。Twitter是最受欢迎的微博网站。Twitter消息被称为tweet。
- **Webcasts** deliver media content over the Internet. Using **streaming** technology (audio and video files continuously downloaded while you are listening to and/or viewing the file content), no files remain on your computer at the conclusion of the webcast. • 网络广播通过互联网传送媒体内容。使用流媒体技术(当你收听和/或观看文件内容时, 音频和视频文件会持续下载), 在网络广播结束时, 文件不会留在你的电脑上。
- **Podcasts**, like webcasts, deliver media content but do not use streaming technology. Before content can be accessed, the entire media file has to be downloaded and saved to your computer. 播客和网络广播一样, 提供媒体内容, 但不使用流媒体技术。在内容可以被访问之前, 整个媒体文件必须被下载并保存到您的计算机。
- A **wiki** is a website designed to allow visitors to use their browsers to add, edit, or delete the site's content. **Wikipedia** is one of the most popular wikis. • wiki是一个允许访问者使用浏览器添加、编辑或删除网站内容的网站。维基百科是最受欢迎的维基之一。

E-mail 电子邮件

E-mail (electronic mail) is the transmission of electronic messages. There are two basic types of e-mail systems: 电子邮件是传递电子信息的一种方式。电子邮件系统有两种基本类型:

- **Client-based e-mail systems** use **e-mail clients** installed on your computer. • 基于客户端的电子邮件系统使用安装在计算机上的电子邮件客户端。
- **Web-based e-mail systems** use **webmail clients** located on the e-mail provider's computer. This is known as **webmail**. A typical e-mail has three basic elements: **header** (including **address**, **subject**, and perhaps **attachment**), **message**, and **signature**. **Spam** is unwanted and unsolicited e-mail that may include a computer **virus** or destructive programs often attached to unsolicited e-mail. **Spam blockers**, also known as **spam filters**, are programs that identify and eliminate spam.

•基于web的电子邮件系统使用位于电子邮件提供商计算机上的网络邮件客户端。这就是所谓的网络邮件。典型的电子邮件有三个基本元素:标题(包括地址、主题,可能还有附件)、消息和签名。垃圾邮件是不需要的和不请自来的电子邮件,可能包括计算机病毒或破坏性程序往往附加到不请自来的电子邮件。垃圾邮件拦截器,也被称为垃圾邮件过滤器,是识别和消除垃圾邮件的程序。

Messaging 消息传递

Although e-mail is the most widely used, there are two other messaging systems: 虽然电子邮件是使用最广泛的,但还有另外两种消息传递系统:

- Text messaging**, also known as **texting** and **SMS (short message service)**, is a process of sending short electronic messages, typically fewer than 160 characters. Texting while driving is very dangerous and illegal in several states. •短信,也被称为短信和SMS(短消息服务),是一个发送短的电子信息的过程,通常小于160个字符。在一些州,开车发短信是非常危险的,而且是违法的。

- **Instant messaging (IM)**—supports live communication between **friends**. Most instant messaging programs also include videoconferencing features, file sharing, and remote assistance.

- 即时通讯(IM) –支持朋友之间的实时通信。大多数即时消息程序还包括视频会议功能、文件共享和远程协助。

SEARCH TOOLS搜索工具

site content. **Spiders** are programs that update these databases. 网站内容。spider是更新这些数据库的程序。

Search Engines 搜索引擎

Search engines are specialized programs to help locate information. To use, enter a keyword or phrase, and a list of search results is displayed. 搜索引擎是帮助定位信息的专门程序。使用时,输入关键字或短语,将显示搜索结果列表。

Specialized Search Engines 专业搜索引擎

Specialized search engines focus on subject-specific websites. 专业搜索引擎专注于特定主题的网站。

Content Evaluation 内容评价

To evaluate the accuracy of information found on the web, consider the following: 要评估在网络上找到的信息的准确性,考虑以下几点:

- Authority**. Is the author an expert? Is the site official or does it present one individual's or organization's opinion? •权威。作者是专家吗?网站是官方的还是个人或组织的意见?

- Accuracy. Has the information been critically reviewed? Does the site provide a method to report

inaccurate information? •精度。信息是否经过严格审查?网站提供报告方法吗不准确的信息?

- Objectivity. Is the information factual or does the author have a bias? Does the author appear to have a personal agenda to convince or form a reader's opinion? •客观性。信息是真实的还是作者有偏见?作者是否有个人意图来说服读者或形成读者的观点?

- Currency. Is the information up to date? Does the site specify when information is updated? Are the site's links operationa•货币。信息是最新的吗?网站是否指定何时更新信息?网站的链接是否可行

ELECTRONIC COMMERCE电子商务

selling of goods over the Internet. Three basic types are 通过互联网销售商品。三种基本类型是

- Business-to-consumer (B2C)** commerce, which involves sales from business to the general public. •企业对消费者(B2C)商务, 涉及从企业到公众的销售。

- Consumer-to-consumer (C2C)** commerce, which involves sales between individuals. **Web auctions** are similar to traditional auctions except buyers and sellers rarely, if ever, meet face to face. •C2C (C2C)商业, 涉及个人之间的销售。网络拍卖与传统拍卖类似, 只是买卖双方很少(如果有的话)面对面。

- Business-to-business (B2B)** commerce, which involves sales from one business to another, typically a manufacturer—supplier relationship. •企业对企业(B2B)商务, 涉及从一个企业到另一个企业的销售, 通常是制造商和供应商的关系。

Security 安全

The two greatest challenges for e-commerce are the development of 电子商务最大的两个挑战是发展

- Safe, secure payment methods. Two types are credit cards and **digital cash**. **Bitcoins** exist only on the Internet. •安全可靠的支付方式。两种类型是信用卡和数字现金。比特币只存在于互联网上。

- Convenient ways to provide required information. •提供所需信息的便捷方式。

CLOUD COMPUTING云计算

Cloud computing shifts many computer activities from the user's computer to other computers on the Internet. The three basic components to cloud computing are clients, the Internet, and service providers.云计算将许多计算机活动从用户的计算机转移到Internet上的其他计算机上。云计算的三个基本组件是客户机、Internet和服务提供商。

INTERNET OF THINGS物联网

The **Internet of Things (IoT)** is a continuing development of the Internet allowing everyday objects embedded with electronic devices to send and receive data over the Internet. Objects include smartphones, wearable devices, and even coffeemakers. For example: 物联网(IoT)是互联网不断发展, 允许嵌入电子设备的日常物品通过互联网发送和接收数据。这些物品包括智能手机、可穿戴设备, 甚至咖啡机。例如:

- The Fitbit is an IoT device (bracelet) that monitors health data and sends that data to a smartphone or personal web page. •Fitbit是一款物联网设备(手环), 可以监测健康数据, 并将数据发送到智能手机或个人网页。
- Apple's Health App is a Web 3.0 application that can access Fitbit data, combine it with other related health data, analyze the data, and report back to you through a smartphone. The Apple Health App reports provide information about users' health, including heart rate, steps taken each day, and an estimate of daily calories burned. •苹果的健康应用程序是一个Web 3.0应用程序, 可以访问Fitbit数据, 将其与其他相关的健康数据结合, 分析数据, 并通过智能手机向你报告。苹果健康应用报告提供了用户的健康信息, 包括心率、每天走的步数和每天消耗的热量。

CAREERS in IT

Webmasters develop and maintain websites and web resources. A bachelor's or associate's degree in computer science or information systems and knowledge of common programming languages and web development software are required. Salary range is \$61,000 to \$83,000. 网站管理员开发和维护网站和网络资源。要求具有计算机科学或信息系统专业的学士学位或大专学历, 熟悉常用编程语言和网络开发软件。薪资范围为61,000美元至83,000美元。

雨课堂

Chapter 2 test

This version of the web evolved to support more dynamic content creation and social Interaction.

这个版本的网页支持更多的动态内容创建和社会互动。

Web 2.0

A web page typically contains_ , which contains the formatting instructions for displaying the web page.

网页通常包含超文本标记语言(HTML),其中包含显示网页的格式说明。

Hypertext Markup Language (HTML)

Which programming language is used within HTML documents to trigger interactive features?

在HTML文档中使用哪种编程语言来触发交互式特性?

JavaScript

In addition to an Internet connection, users need browser software to easily display web pages.

除了网络连接，用户还需要浏览器软件来方便地显示网页。

true

Cascading Style Sheets (CSS) are files inserted into an HTML document that control the appearance of web pages including layout, colors, and fonts.

层叠样式表(CSS)是插入到HTML文档中的文件 用于控制网页的外观，包括布局、颜色和字体。

true

Google Chrome is an example of a web [browser] .

谷歌Chrome是一个网络浏览器的例子。

The [http] protocol is used for web traffic and is the most widely used Internet protocol.

http协议用于web流量，是使用最广泛的 Internet 协议。

In the URL "<http://www.simnetonline.com>", "www.simnetonline.com" is the [domain] name.

在网址“<http://www.simnetonline.com>”中，“www.simnetonline.com”为域名。

Define the parts of a URL.

For browsers to connect to resources on the Internet, the address of the resources must be specified. These addresses are called uniform resource locators or URL . All URLs have at least two basic parts. The protocol is the first part of the address and specifies which protocol (set of rules for exchanging data) to use when connecting to the Internet resources. In the example,<http://www.sohu.com>, the protocol is http.The protocol http is used for web trafficand is the most widely used Internet protocol. The second part of the URL is the domain name for example www.espn.com. This indicates the specific address where the resource is located. The part of the domain name following the dot (.) is the top level domain. It identifies the type of organization. A ".com" top-level domain indicates a commercial site.

当浏览器连接Internet上的资源时，必须指定资源的地址。这些地址被 称为统一资源定位器或URL。所有URL至少有两个基本部分。协议是 地址的第一部分，指定连接到Internet资源时使用的协议(一组交换数据 的规则)。本例中为<http://www.sohu.com>，协议为http。http协议用于 web流量，是使用最广泛的Internet协议。URL的第二部分是域名，例 如www.espn.com。这表示资源所在的特定地址。域名中点(.)后面的部 分是顶级域名。它标识组织的类型。顶级域名“ .com”表示商业网站。

Chapter 2 Learn checking

Launched in 1969 as a U.S. funded project that developed a national computer

network, the Internet was initially called .

1969年，作为一个由美国资助的开发国家计算机网络的项目，因特网最初被称为阿帕网 (ARPANET)

ARPANET

The Internet is a(n) ___ made up of wires, cables, satellites, and rules for exchanging information between computers connected to the network.

因特网是一个物理网络，由电线、电缆、卫星和连接到该网络的计算机之间交换信息的规则组成。

physical network

The most common way to access the Internet is through ___.

最常见的访问Internet的方法是通过Internet服务提供商(ISP)

an Internet service provider (ISP)

A web page typically contains __, which contains the formatting instructions for displaying the web page.

网页通常包含超文本标记语言(HTML)，其中包含显示网页的格式说明。

Hypertext Markup Language (HTML)

___ do not use streaming technology and are used to download media files to your computer.

播客不使用流媒体技术，而是用来下载媒体文件到您的计算机。

Podcasts

When you enter a keyword or phrase into a search engine the results display as a list of these.

当您在搜索引擎中输入关键字或短语时，结果将显示为这些关键字或短语的列表。

Hits 点击数

This type of e-commerce often resembles the electronic version of the classified ads or an auction.

这种类型的电子商务通常类似于分类广告或拍卖的电子版本。

C2C

Programs that automatically start and operate as a part of your browser are called ___.

作为浏览器的一部分自动启动和运行的程序称为插件。

plug-ins

Parents can use this type of program to block access to selected Websites.

家长可以使用这种类型的程序来阻止访问选定的网站。

Filter

This method of file transfer has been used for decades and is frequently used for uploading changes to a Website hosted by an Internet service provider.

这种文件传输方法已经使用了几十年，经常用于上传更改到由互联网服务提供商托管的网站。

FTP

The Internet connects millions of computers and resources across the world via wires, cables, and satellite.

因特网通过电线、电缆和卫星将世界各地数以百万计的计算机和资源连接起来

true

In order to connect to a website, the browser must know only the site's domain name.

为了连接到一个网站，浏览器必须只知道该网站的域名。

false

After a webcast is completed there are no files left on your computer.

网络广播完成后，你的电脑上没有任何文件。

true

Wikis allow many individuals to edit the site's content.

wiki允许许多个人编辑网站的内容。

true

All information found on the web through search tools can be considered as factual.

所有通过搜索工具在网络上找到的信息都可以被认为是真实的。

false

Cloud computing has the potential to free end-users from owning, maintaining, and storing software and data.

云计算有潜力将最终用户从拥有、维护和存储软件和数据中解放出来。

false

Uploading is the process of copying a file from a website to your computer.

上传是将文件从网站上复制到你的电脑上的过程。

false

HTML stands for [Hypertext Markup Language].

A(n) [hyperlink] is a link on a web page that connects to another web page.

超链接是网页上连接到另一个网页的链接。

[Web-based] e-mail accounts do not require an e-mail program to be installed on your computer.

基于web的电子邮件帐户不需要在计算机上安装电子邮件程序。

[text] messaging is the process of sending short electronic messages using a wireless network.

短信是使用无线网络发送短电子信息的过程。

A(n) [search] engine is a specialized program that assists you in locating information on the web and the Internet.

搜索引擎是帮助你在网络和互联网上定位信息的专门程序。

[digital] cash is the Internet's equivalent to traditional cash.

数字现金相当于互联网上的传统现金。

Knowledge of HTML and CSS is considered essential for the job of a(n) [webmaster].

HTML和CSS的知识被认为是必不可少的工作，一个网站管理员。

[Cloud] computing provides access to software, programs, and data from anywhere through an Internet connection.

云计算提供了通过互联网连接从任何地方访问软件、程序和数据的途径。

Explain how a search engine works.

To use a search website, you enter a keyword or phrase reflecting the information you want. The search engine compares your entry against its database and returns a list of hits, or sites that contain the keywords. Each hit includes a link to the referenced web page (or other resource) along with a brief discussion of the information contained at that location. Many searches result in a large number of hits. For example, if you were to enter the keyword music, you would get billions of hits. Search engines order the hits according to those sites that most likely contain the information requested and present the list to you in that order, usually in groups of 10.

要使用搜索网站，你需要输入一个关键字或短语来反映你想要的信息。搜索引擎将您的条目与数据库进行比较，并返回一个点击列表，即包含关键字的站点。每次点击都包含一个链接到引用的网页(或其他资源)，以及对该位置所包含信息的简要讨论。很多搜索都会产生大量的点击。例如，如果你输入关键字music，你将获得数十亿次点击。搜索引擎根据那些最有可能包含信息请求的网站来排序。

Chapter 2 test

This type of e-mail account is more widely used because it frees the user from installing and maintaining an e-mail client on every computer used to access e-mail.

这种类型的电子邮件帐户得到了更广泛的使用，因为它使用户不必在用于访问电子邮件的每一台计算机上安装和维护电子邮件客户机。

Web-based

_ typically broadcast live events and use streaming technology in which audio and video files are continuously downloaded to your computer while you are listening to and/or viewing the content.

网络广播通常是直播事件，并使用流媒体技术，当你听或看内容时，音频和视频文件会不断地被下载到你的电脑上。

Webcasts

Which of the following is a website specially designed to allow visitors to use their browser to add, edit, or delete the site's content?

下列哪项是专门设计让访客使用浏览器添加、编辑或删除网站内容的网站？

Wiki

These special programs continually look for new information and update the databases used by search services.

这些特殊程序不断寻找新的信息，并更新搜索服务使用的数据库。

蜘蛛 Spiders

Online banking is an example of what type of e-commerce?

网上银行是什么类型的电子商务的一个例子？

B2C

Which of the following is not one of the three basic components of cloud computing?

以下哪？

client organizations and end-users 客户组织和最终用户

service providers 服务提供商

the Internet 互联网

wireless connectivity 无线连接 ✓

This method of file transfer distributes file transfers across many different computers.

这种文件传输方法将文件传输分布在许多不同的计算机上

BitTorrent 比特流

Attachments to e-mail messages can be a document or an image.

电子邮件的附件可以是文档或图像

true

Instant messaging allows text only.

即时消息只允许文本。

false

Cloud computing is an example of a social networking site.

云计算是社交网站的一个例子

false

Special web browsers, called [mobile] browsers, are designed to run on portable devices such as smartphones and tablets.

被称为移动浏览器的特殊网络浏览器是专为在智能手机和平板电脑等便携设备上运行而设计的。

Unsolicited e-mail is called [spam] .

未经请求的电子邮件被称为垃圾邮件。

A(n) [microblog] publishes short sentences that only take a few seconds to write, rather than long stories or posts

微博只发布短短几秒钟的句子，而不是长篇故事或帖子

A(n) [wiki] is a website specially designed to allow visitors to use their browser to add, edit, or delete the site's content.

wiki是一种专门设计的网站，允许访问者使用浏览器添加、编辑或删除网站内容。

课后习题

选择

1. The network that connects computers all over the world.
Internet
2. The rules for exchanging data between computers.
protocols
3. Using file transfer utility software, you can copy files to your computer from specially configured servers on the Internet. This is called:
downloading
4. Communities of individuals who share a common interest typically create Facebook:
groups
5. Type of e-mail account that does not require an e-mail program to be installed on a user's computer is:
web-based
6. The most popular microblogging site:
Twitter
7. Using a keyword, a search engine returns a list of related sites known as:
hits
8. This is the Internet's equivalent to traditional cash.
digital cash
9. The continuing Internet development that allows objects to send and receive data over the Internet.
互联网的持续发展，允许对象通过互联网发送和接收数据。
IoT
10. Three basic components to cloud computing are clients, Internet, and _.
service providers
匹配
11. The most popular Internet activity.
c. communicating
12. The most common way to access the Internet is through a(n).

- f. ISP
13. Transmission of electronic messages over the Internet.
d. e-mail
14. Two popular instant messaging services are Google Talk and Facebook _____.
a. chat
15. A business-oriented social networking site.
g. LinkedIn
16. Individual website to keep in touch with friends and family.
j. blog
17. Publishes short sentences that only take a few seconds to write.
h. microblog
18. Maintain huge databases relating to information provided on the Web and the Internet.
b. search services
19. Electronic commerce involving individuals selling to individuals.
b. C2C
20. The basic components of cloud computing are clients, service providers, and the _____.
e. Internet

第3章 Application Software

应用软件

APPLICATION SOFTWARE

The three categories of application software are general purpose, specialized, and mobile.

应用软件分为三类:通用软件、专用软件和移动软件

User Interface 用户界面

You control and interact with a program using a user interface. A graphical user interface (GUI) uses icons selected by a mouse-controlled pointer. A window contains a document, program, or message. Software programs with a traditional GUI have 你可以使用用户界面来控制程序并与之交互。图形用户界面(GUI)使用由鼠标控制的指针选择的图标。窗口包含文档、程序或消息。具有传统 GUI 的软件程序有

- **Menus**—present commands listed on the menu bar.

菜单—显示菜单栏上列出的命令

- **Toolbars**—contain buttons for quick access to commonly used commands.

工具栏—包含快速访问常用命令的按钮

- **Dialog box**—provides additional information or requests user input.

对话框—提供附加信息或请求用户输入

Software programs with a Ribbon GUI have

带有 Ribbon GUI 的软件程序有

- **Ribbons**—replace menus and toolbars.

绶带—替换菜单和工具栏

- **Tabs**—divide ribbons into groups. Contextual tabs automatically appear when needed.

制表符—把绶带分成几组，上下文选项卡会在需要时自动出现。

- **Galleries**—graphically display alternatives before they are selected.

图库—在选择备选方案之前以图形方式显示它们

Common Features 常见功能

Common features include spell checkers, alignment, fonts and font sizes, character effects, and editing options.

常见功能包括拼写检查、对齐、字体和字体大小、字符效果和编辑选项

GENERAL-PURPOSE APPLICATIONS 通用应用程序

General-purpose applications include word processors, spreadsheets, presentation software, and database management systems.

通用应用程序包括字处理器、电子表格、演示软件和数据库管理系统。

Word Processors 文字处理程序

Word processors create text-based documents. Individuals and organizations use word processors to create memos, letters, and reports. Organizations also create newsletters, manuals, and brochures to provide information to their customers. Microsoft Word is the most widely used word processor. Others include Apple Pages, Google Docs, and OpenOffice Writer.

文字处理程序创建基于文本的文档。个人和组织使用文字处理器来创建备忘录、信件和报告。组织还会创建时事通讯、手册和小册子，为客户提供信息。微软的 Word 是使用最广泛的文字处理软件。其他包括 Apple Pages、谷歌 Docs 和 OpenOffice Writer。

Spreadsheets 电子数据表

Spreadsheets organize, analyze, and graph numeric data such as budgets and financial reports. They are widely used by nearly every profession. Microsoft Excel is the most widely used spreadsheet program. Others include Apple Numbers, Google Sheets, and OpenOffice Calc.

电子表格组织、分析和绘制数字数据，如预算和财务报告。它们几乎被每个职业广泛使用。微软的 Excel 是使用最广泛的电子表格程序。其他包括 Apple Numbers、谷歌 Sheets 和 OpenOffice Calc。

GENERAL-PURPOSE APPLICATIONS 通用应用程序

Presentation Software 演示软件

Presentation software are programs that combine a variety of visual objects to create attractive, visually interesting presentations. They are excellent tools to communicate a message and to persuade people. People in a variety of settings and situations use presentation software programs to make their presentations more interesting and professional. Five of the most widely used presentation software

programs are Microsoft PowerPoint, Apple Keynote, Google Slides, OpenOffice Impress, and Prezi.

演示软件是结合各种视觉对象来创建有吸引力的、视觉上有趣的演示文稿的程序。它们是传达信息和说服人们的绝佳工具。人们在不同的环境和情况下使用演示软件程序使他们的演示更有趣和专业。5 个最广泛使用的演示软件程序是 Microsoft PowerPoint、Apple Keynote、谷歌 Slides、OpenOffice Impress 和 Prezi。

Database Management Systems 数据库管理系统

A database is a collection of related data. A database management system (DBMS) or database manager is a program that structures a database. It provides tools to enter, edit, and retrieve data from the database. Organizations use databases for many purposes, including maintaining employee records. Three widely used database management systems designed for personal computers are Microsoft Access, Apple FileMaker, and OpenOffice Base.

数据库是相关数据的集合。数据库管理系统(DBMS)或数据库管理器是一个构建数据库的程序。它提供了输入、编辑和从数据库检索数据的工具。组织使用数据库有很多目的,包括维护员工记录。三个被广泛使用的为个人电脑设计的数据库管理系统是 Microsoft Access、Apple FileMaker 和 OpenOffice Base。

SPECIALIZED APPLICATIONS 专用应用程序

Specialized applications are widely used within specific professions. They include graphics programs, video game design software, and web authoring programs.

专用应用程序广泛应用于特定的职业。它们包括图形程序、电子游戏设计软件和网页创作程序

Graphics Programs 图形程序

Graphics programs are used by graphic arts professionals.

图形程序是由图形艺术专业人员使用的

- **Desktop publishing programs (page layout programs)** mix text and graphics to create professional-quality publications.

桌面出版程序(页面布局程序)混合文本和图形创建专业质量的出版物

- **Image editors (photo editors)** edit digital photographs consisting of thousands

of dots, or pixels, that form bitmap or raster images.

图像编辑器(照片编辑器)编辑由数千个点或像素组成的数字照片,这些点形成位图或栅格图像

- **Illustration programs (drawing programs)** create and edit vector images. **Vector images (vector illustrations)** use geometric shapes.

插图程序(绘图程序)创建和编辑矢量图像。矢量图像(矢量插图)使用几何形状

- **Video editors** edit video to enhance quality and appearance.

视频编辑工器编辑视频以提高质量和外观

Video Game Design Software 电子游戏设计软件

Video game design software helps to organize thoughts and guide users through the game design process, including character development and environmental design.

电子游戏设计软件有助于组织思想,引导用户完成游戏设计过程,包括角色开发和环境设计

Web Authoring Programs 网页制作软件

Web authoring is the process of creating a website. Individuals create online diaries called blogs. **Web authoring programs (web page editors, HTML editors)** create sophisticated commercial websites. Some are WYSIWYG (what you see is what you get) editors.

网络创作是创建一个网站的过程。人们创建在线日记,称为博客。网络编辑程序(网页编辑器,HTML 编辑器)创建复杂的商业网站。有些是 WYSIWYG (所见即所得)编辑器

MOBILE APPS 移动应用程序

Mobile apps (mobile applications, apps) are add-on programs for a variety of mobile devices. Traditional applications include address books, to-do lists, alarms, and message lists. Recently, mobile capabilities have exploded.

移动应用程序(Mobile applications, apps)是针对各种移动设备的附加程序。传统的应用包括地址簿、待办事项列表、警报和消息列表。最近,移动能力出现了爆炸式增长。

Apps 应用程序

Popular apps include those for music, videos, social net-working, shopping, and game playing.

流行的应用程序包括音乐、视频、社交网络、购物和游戏

- Pandora and Spotify provide popular music apps.

Pandora 和 Spotify 提供流行音乐应用程序

- YouTube and Netflix provide streaming video apps.

YouTube 和 Netflix 提供流媒体视频应用程序

- Facebook and Instagram provide social networking apps.

Facebook 和 Instagram 提供社交网络应用程序

- Crossy Road and Final Fantasy are popular game playing apps. Crossy Road includes in-app optional purchases, including the ability to purchase additional characters. Final Fantasy is a **role playing game (RPG)** where gamers can join a team of adventurers and battle across fantastic worlds.

《过马路》和《最终幻想》都是很受欢迎的游戏应用。《过马路》包含应用内部可选购买功能，包括购买额外角色的功能。《最终幻想》是一款角色扮演游戏(RPG)，玩家可以加入冒险团队，在奇幻世界中战斗

App Stores 应用程序商店

An app store is typically a website that provides access to specific mobile apps that can be downloaded either for a nominal fee or free of charge. Two of the best-known stores are Apple's App Store and Google Play. Most of the best-known app stores specialize in applications for a particular line of mobile devices, other less well-known stores provide apps for a wide variety of mobile devices.

应用商店通常是一个网站，提供访问特定的移动应用程序，可以下载象征性的费用或免费。两个最知名的应用商店是苹果的 App Store 和谷歌 Play。大多数知名的应用程序商店专门为特定系列的移动设备提供应用程序，其他不太知名的商店为各种各样的移动设备提供应用程序

App Store	Focus	Site
Apple App Store	Apple devices	itunes.apple.com/us
Google Play	Android devices	play.google.com/store/apps
Amazon	Android devices	http://amazon.com/appstore

SOFTWARE SUITES 软件套件

A **software suite** is a collection of individual application packages sold together.

软件套件是一起出售的单个应用程序包的集合

- Office suites (office software suites or productivity suites) contain professional-grade application programs.

办公套件(办公软件套件或生产力套件)包含专业级应用程序

- Cloud suites (online office suites) are stored on servers and available through the Internet.

云套件(在线办公套件)存储在服务器上, 可以通过互联网使用

- Specialized suites focus on specific applications such as graphics.

专门的套件专注于特定的应用程序, 比如图形

- Utility suites include a variety of programs designed to make computing easier and safer.

实用程序套件包括各种旨在使计算更容易和更安全的程序

CAREERS IN IT 其中的职业

Software engineers analyze users' needs and create application software. A bachelor's or advanced specialized associate's degree in computer science or information systems and extensive knowledge of computers and technology are required. Salary range is \$67,000 to \$100,000.

软件工程师分析用户需求、开发应用软件。要求具备计算机科学或信息系统专业学士学位或高级副学士学位, 并具有广泛的计算机和技术知识。工资范围为 6.7 万至 10 万美元

雨课堂

- Chapter3 Online learning check

- This type of software works with end users, application software, and computer Hardware to handle the majority of technical details.

这类软件与终端用户、应用软件和计算机硬件一起工作, 处理大多数技术细节

System

- A rectangular area that can contain a document, program, or message.

可以包含文档、程序或信息的矩形区域

Window

- Programs that create text based document.

创建基于文本的文档的程序

word processors

- Programs that organize, analyze, and graph numerical data such as budgets and financial reports.

组织、分析和绘制预算和财务报告等数字数据的程序

spreadsheets

- Program that allows you to mix text and graphics to create publications of professional quality.

该程序允许您混合文本和图形，以创建专业质量的出版物

desktop publishing

- The type of image that consists of geometric shapes.

由几何形状组成的图像类型

Vector

- An online diary or commentary.

在线日记或评论

blog

- Programs that combine a variety of visual objects to create attractive, visually interesting presentations.

结合各种视觉对象来创造有吸引力的、视觉上有趣的演示文稿的程序。

presentation software

- Programs typically used to create sophisticated commercial websites.

通常用于创建复杂商业网站的程序

web authoring programs

- Also known as an online office suite.

也被称为在线办公套件

cloud suite

- Most applications use a(n) ____ that displays graphical elements called icons to represent familiar objects.

大多数应用程序使用一个 ____来显示被称为图标的图形元素来表示熟悉的对象

graphical user interface

- This GUI includes tabs, groups and galleries.

这个 GUI 包括选项卡、组和图库

Ribbon

- These are specialized graphics programs for editing or modifying digital photographs.

这些是专门用于编辑或修改数码照片的图形程序

Image editors

- A library of electronic images is called a(n) ____.

电子图像库被称为____

image gallery

- Add-on programs for a variety of mobile devices including smartphones and tablets.

各种移动设备的附加程序，包括智能手机和平板电脑

Mobile apps

- Functions, formulas, charts and what-if analysis are common features of database management systems.

函数、公式、图表和假设分析是数据库管理系统的共同特征

false

- Image editors are used to create and edit vector images.

图像编辑器用于创建和编辑矢量图像

false

- Web authoring programs are used to create sophisticated commercial websites.

Web 创作程序用于创建复杂的商业网站

True

- Cloud suites are stored on your hard drive and are available anywhere you can access the Internet.

云套件存储在您的硬盘驱动器上，可以在任何您可以访问互联网的地方使用

False

- Many apps are written for a particular type of mobile device and will not run

on other types.

许多应用程序是为特定类型的移动设备编写的，不会在其他类型的移动设备上运行

True

- [Application] software is end user software designed to accomplish a variety of different tasks.

[应用程序]软件是设计用来完成各种不同任务的终端用户软件

- In a Ribbon GUI, [tabs] are used to divide the ribbon into major activity areas.

在 Ribbon GUI 中，[标签]用来将 Ribbon 划分为主要的活动区域

- [Spreadsheets] organize, analyze, and graph numeric data such as budgets and financial reports.

[电子表格]组织、分析和绘制数字数据，如预算和财务报告

- Creating a website is called web [authoring] .

创建一个网站叫做 web[创作]

• Chapter 3 test

- __ provide additional information and request user input.

__提供附加信息并请求用户输入

Dialog boxes

- These organize commonly used commands into a set of tabs.

它们将常用命令组织到一组选项卡中

Ribbons

- Programs that allow you to mix text and graphics and focus on flexible page design and layout.

允许混合文本和图形，并专注于灵活的页面设计和布局的程序

Desktop publishing

- Image editors edit images consisting of thousands of pixels that form this type of image.

图像编辑器编辑由数千个像素组成的图像，形成这种类型的图像

Bitmap

- These programs are used to create and edit vector images.

这些程序用于创建和编辑矢量图像

Illustration programs

- Moving text, graphics, or interactive features on a web page are called.

移动文本、图形或网页上的交互功能被称为

Animations

- What-if analysis is a very powerful and simple tool to test the effects of different assumptions in a spreadsheet

假设分析是一个非常强大和简单的工具，可以在电子表格中测试不同假设的影响

True

- A database is a collection of related data.

数据库是相关数据的集合

True

- Presentation graphics software is an excellent tool to communicate a message and to persuade people

演示图形软件是传达信息和说服人们的优秀工具

True

- As opposed to general-purpose applications which are widely used in nearly every discipline, [specialized] applications are more narrowly focused.

与广泛应用于几乎所有学科的通用应用程序相反，[专门化]应用程序的关注范围更窄

- A (n) [database] is a collection of related data the electronic equivalent of a file cabinet.

一个[数据库]是一个相关数据的集合，相当于电子文件柜

- [Tables] make up the basic structure of a relational database with columns containing field data and rows containing record information.

[表格]构成关系数据库的基本结构，列包含字段数据，行包含记录信息

- In an image editor, photographs consist of thousands of dots or [pixels] that form Images.

在图像编辑器中，照片由数千个点或[像素]组成

- [Vector] images are made up of geometric shapes or objects.

[矢量]图像是由几何形状或物体组成的

- Cloud suites are stored at a (n) [server] on the Internet, rather than on your

microcomputer.

云套件存储在互联网上的[服务器]上，而不是存储在微机上

- What is the difference between system software and application software?

系统软件和应用软件的区别是什么？

System software works with end users, application software, and computer hardware to handle the majority of technical details. Application software can be described as end user software designed to accomplish a variety of different tasks.

系统软件与终端用户、应用软件和计算机硬件一起处理大部分技术细节。应用软件可以被描述为设计用来完成各种不同任务的终端用户软件

课后习题

- 选择

- This type of software works with end users, application software, and computer Hardware to handle the majority of technical details.

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web authoring programs

- Also known as an online office suite.

也被称为在线办公套件

cloud suite

- 匹配

1. Toolbars typically appear below the menu bar and include small graphic elements called _.

工具栏通常出现在菜单栏的下方，并包含名为_

a. buttons

2. Simplifies the process of making a selection from a list of alternatives by graphically displaying the effect of alternatives before being selected.

通过在被选中之前以图形化方式显示备选项的效果，简化从备选项列表中进行选择的过程

d. galleries

3. A general purpose program that creates text-based documents.

创建基于文本的文档的通用程序

j. word processor

4. Program that organizes, analyzes, and graphs numeric data.

组织、分析和图形化数字数据的程序

g. spreadsheet

5. A collection of related data.

相关数据的集合

c. database

6. Also known as a photo editor, this specialized graphics program edits or modifies digital photographs.

也被称为照片编辑器，这个专门的图形程序编辑或修改数码照片

e. image editor

7. Image editors create images made up of thousands of dots known as _.

图像编辑器创建的图像由数千个被称为_的点组成

f. pixels

8. A Web site that provides access to specific mobile apps is known as an app _.

提供访问特定移动应用程序的网站被称为 app _

h. store

9. A type of suite that is stored at a server on the Internet and is available anywhere you can access the Internet.

存储在 Internet 上的服务器上的一种套件，可以在任何可以访问 Internet 的地方使用

b. cloud

10. A type of specialized suite that includes a variety of programs designed to make computing easier and safer.

一种专门的套件，包括各种旨在使计算更容易和更安全的程序

i. Utility

第 4 章 SYSTEM SOFTWARE

系统软件

SYSTEM SOFTWARE 系统软件

System software works with end users, application programs, and computer hardware to handle many details relating to computer operations.

Not a single program but a collection or system of programs, these programs handle hundreds of technical details with little or no user intervention.

Four kinds of systems programs are operating systems, utilities, device drivers, and language translators.

系统软件与终端用户、应用程序和计算机硬件一起工作,处理与计算机操作有关的许多细节。不是一个单独的程序,而是程序的集合或系统,这些程序处理数百个技术细节,很少或没有用户干预。

四种系统程序是操作系统、实用程序、设备驱动程序和语言翻译程序。

- Operating systems coordinate resources, provide an interface between users and the computer, and run programs.

操作系统协调资源,提供用户和计算机之间的接口,并运行程序

- Utilities perform specific tasks related to managing computer resources.

实用程序执行与管理计算机资源相关的特定任务

- Device drivers allow particular input or output devices to communicate with the rest of the computer system.

设备驱动程序允许特定的输入或输出设备与计算机系统的其他部分通信

- Language translators convert programming instructions written by programmers into a language that computers can understand and process.

语言翻译人员将程序员编写的程序指令转换成计算机能够理解和处理的语言

OPERATING SYSTEMS 操作系统

Operating systems (software environments, software platforms) handle technical details.

操作系统(软件环境、软件平台)处理技术细节

Functions 功能

Functions include managing resources, providing a user interface (most operating systems use a graphical user interface, or GUI), and running applications. Multitasking allows switching between different applications stored in memory; current programs run in foreground; other programs run in background.

功能包括管理资源、提供用户界面(大多数操作系统使用图形用户界面,或 GUI)和运行应用程序。多任务处理允许在存储在内存中的不同应用程序之间切换;当前程序在前台运行;其他程序在后台运行

Features 特色

Booting starts (cold) or restarts (warm) a computer system. The desktop provides access to computer resources. Common features include icons, pointers, windows, menus, tabs, dialog boxes, help, and gesture control. Data and programs are stored in a system of files and folders.

启动(冷启动)或重新启动(热启动)计算机系统。桌面提供了对计算机资源的访问。常用功能包括图标、指针、窗口、菜单、选项卡、对话框、帮助和手势控制。数据和程序存储在一个由文件和文件夹组成的系统中

Categories 分类

Three categories of operating systems are

有三类操作系统

- Embedded—also known as real-time operating systems (RTOS); used with handheld computers; operating system stored within the device.

嵌入式——也被称为实时操作系统(RTOS);配合手持电脑使用;存储在设备中的操作系统

- Stand-alone (desktop)—controls a single computer; located on the hard disk.

单机(桌面)-控制一台电脑;位于硬盘上

- Network (NOS)—controls and coordinates networked computers; located on the network server.

网络(NOS)-控制和协调联网计算机;位于网络服务器上

Operating systems are often called software environments or software platforms.

操作系统通常被称为软件环境或软件平台

MOBILE OPERATING SYSTEMS 移动操作系统

Mobile operating systems (mobile OS) are embedded in every smartphone and tablet.

These systems are less complicated and more specialized for wireless communication than desktop operating systems.

移动操作系统 (Mobile OS) 嵌入在每一部智能手机和平板电脑中。这些系统比桌面操作系统更简单，更适合于无线通信

Some of the best known are Android, iOS (iPhone OS), and Windows Phone.

其中最著名的是 Android、iOS (iPhone OS) 和 Windows Phone

- **Android** was originally developed by Android, Inc., and later purchased by Google.

It is a widely used mobile OS.

Android 最初由 Android, Inc. 开发，后来被谷歌收购。它是一个广泛使用的移动操作系统

- **iOS (iPhone OS)** was developed by Apple to support iPhone and iPad.

iOS (iPhone OS) 是苹果公司为支持 iPhone 和 iPad 而开发的

- **Windows 10 Mobile** was introduced in 2015 to support a variety of mobile devices, including smartphones and tablets. It can run many powerful programs designed for laptop and desktop computers.

Windows 10 Mobile 于 2015 年推出，支持多种移动设备，包括智能手机和平板电脑。它可以运行许多为笔记本电脑和台式电脑设计的强大程序

Not all mobile applications will run on all smartphones. That is because an app is designed to run on a particular software platform or operating system. Before downloading an app, be sure that it is designed to run with the mobile operating system on your mobile device.

并不是所有的移动应用程序都能在所有的智能手机上运行。这是因为应用程序是为在特定的软件平台或操作系统上运行而设计的。在下载一个应用程序之前，确保它设计为在你的移动设备上的移动操作系统上运行

DESKTOP OPERATING SYSTEMS 桌面操作系统

Windows

Windows is designed to run with many different microprocessors. The most recent

version is Windows 10. Windows 10 was introduced in 2015. It merged Windows' desktop and mobile operating systems. Innovations include Cortana (accepts commands through text or speech), support for Windows apps on desktops and mobile devices, improved gaming with the Xbox gaming environment, a new web browser, and support for Microsoft HoloLens, an augmented reality tool.

Windows 被设计成可以在许多不同的微处理器上运行。最新的版本是 Windows 10。Windows 10 于 2015 年推出。它合并了 Windows 的桌面和移动操作系统。创新包括 Cortana (通过文本或语音接受命令), 支持桌面和移动设备上的 Windows 应用程序, 通过 Xbox 游戏环境改善游戏, 新的网页浏览器, 以及支持增强现实工具微软 HoloLens。

macOS

macOS, an innovative, powerful, easy-to-use operating system, runs on Macintosh computers. The two most recent versions are macOS Sierra and High Sierra. Sierra, which was announced in 2016, introduced Siri, improved integration of iOS devices, and cloud computing. High Sierra, introduced in 2017, includes a new file system and better video compression.

macOS 是一种创新的、功能强大的、易于使用的操作系统, 运行在 Macintosh 电脑上。最新的两个版本是 macOS Sierra 和 High Sierra。Sierra 于 2016 年宣布推出 Siri, 改进了 iOS 设备和云计算的集成。2017 年推出的 High Sierra 包括一个新的文件系统和更好的视频压缩

DESKTOP OPERATING SYSTEMS 桌面操作系统

UNIX and Linux

UNIX was originally designed to run on minicomputers in network environments. Now, it is widely used by servers on the web, mainframe computers, and very powerful personal computers. There are many different versions of UNIX. One version, Linux, a popular and powerful alternative to the Windows operating system, is open-source software. Google's Chrome OS is based on Linux. It integrates with web servers to run applications and to perform other traditional operating system functions. Chrome OS is a popular choice for inexpensive notebook computers using cloud computing and cloud storage. One limitation of these computers is that their efficiency is

dependent upon the speed of their Internet connection.

UNIX 最初设计用于在网络环境中的小型计算机上运行。现在，它被广泛应用于网络服务器、大型计算机和功能强大的个人计算机。UNIX 有许多不同的版本。Linux 是开源软件的一个版本，它是 Windows 操作系统的一个流行而强大的替代品。谷歌的 Chrome OS 是基于 Linux 的。它与 web 服务器集成以运行应用程序和执行其他传统操作系统功能。Chrome OS 是使用云计算和云存储的廉价笔记本电脑的热门选择。这些计算机的一个限制是它们的效率取决于它们的互联网连接速度

Virtualization

Virtualization allows a single physical computer to support multiple operating systems. Using a special program (virtualization software) allows the single physical computer to operate as two or more separate and independent computers known as virtual machines. Host operating systems run on the physical machine. Guest operating systems operate on virtual machines. Parallels creates and runs virtual machines on Apple' s macOS X.

虚拟化允许一台物理计算机支持多个操作系统。使用一个特殊的程序(虚拟化软件)可以让一台物理计算机作为两台或两台以上独立的计算机运行，这两台计算机被称为虚拟机。主机操作系统运行在物理机器上。来宾操作系统在虚拟机上操作。Parallels 在苹果的 macOS X 上创建和运行虚拟机

UTILITIES 实用工具

Utilities make computing easier. The most essential are troubleshooting (diagnostic), antivirus, backup, and file compression.

实用程序使计算变得更容易。最重要的是故障排除(诊断)、反病毒、备份和文件压缩

Windows Utilities

Windows operating systems are accompanied by several utility programs, File History, Disk Cleanup, and Storage:

Windows 操作系统伴随着几个实用程序，文件历史，磁盘清理和存储：

- **File History** makes a copy of all files that are in the libraries, contacts, and favorites and on the desktop; protects from the effects of disk failure.

文件历史使所有文件的副本，在库，联系人，收藏夹和桌面；防止磁盘故障的影响

- **Disk Cleanup**, a troubleshooting utility, identifies and eliminates nonessential files/frees up disk space and improves system performance.

磁盘清理，一个故障排除实用程序，识别和消除不必要的文件/释放磁盘空间和提高系统性能

- **Storage** eliminates unused files and applications.

存储消除不使用的文件和应用程序

Utility Suites

Utility suites combine several programs into one package. Computer viruses are dangerous programs.

实用程序套件将多个程序组合成一个包。计算机病毒是危险的程序

CAREERS in IT 其中的职业

Computer support specialists provide technical support to customers and other users.

Degrees in computer science or information systems are preferred plus good analytical and communication skills. Salary range is \$35,000 to \$51,000.

计算机支持专家为客户和其他用户提供技术支持。计算机科学或信息系统专业优先，具备良好的分析和沟通能力。工资范围在 35,000 美元到 51,000 美元之间

雨课堂

- Chapter 04 Online Test

- Which of the following is not a system software program type?

下列哪一项不是系统软件程序类型？

Utilities 实用工具

Device drivers 设备驱动

Language translators 语言翻译程序

Sectors 分区 ✓

- Which of the following is not a function of the operating system?

下列哪项不是操作系统的功能？

Managing resources 资源管理

Providing Internet access 提供互联网接入 ✓

Providing a user interface 提供用户界面

Running applications 运行应用

- This type of boot occurs when the computer is already on and you restart it without turning off the power.

这种类型的启动发生在计算机已经打开并且在没有关闭电源的情况下重新启动它的时候

Warm boot 热启动

- This operating system is widely used by servers on the web, mainframe computers, and very powerful personal computers.

这种操作系统广泛应用于网络服务器、大型计算机和功能强大的个人计算机

UNIX

- When a single physical computer runs this type of software, it operates as though it were two or more separate and independent computers.

当一台物理计算机运行这种类型的软件时，它就像两台或两台以上独立的计算机一样运行

Virtualization 虚拟化

- When a file is broken up into small parts that are stored wherever space is available, the hard disk is said to become ____.

当一个文件被分解成小的部分，存储在任何有空间的地方，硬盘被称为____

Fragmented

- Before sending a large file over the Internet, you might want to use one of these programs.

在通过 Internet 发送大文件之前，您可能需要使用这些程序中的一个

File compression program

- A utility program that makes a copy of all files or selected files that are in the libraries, contacts, and favorites on the desktop.

一种实用程序，可以复制桌面上的库、联系人和收藏夹中的所有文件或选定文件

Backup and Restore

- Starting or restarting a computer is called tasking the system.

启动或重新启动计算机被称为系统任务

false

- The operating system allows the user to interact with the application programs and computer hardware through a user interface.

操作系统允许用户通过用户界面与应用程序和计算机硬件进行交互

true

- The operating system is often referred to as the software platform.

操作系统通常被称为软件平台

true

- Utility suites come with one program only.

实用程序套件只提供一个程序

false

- Windows Update makes it easy to update the device drivers on your computer.

Windows Update 使更新您的计算机上的设备驱动程序变得容易

true

- [Icons] are graphic representations for a program, type of file, or function.

[图标]是程序、文件类型或函数的图形表示

- [Network] operating systems are used to control and coordinate computers that are linked together.

[网络]操作系统是用来控制和协调连接在一起的计算机的

- [Virtualization] refers to the process whereby a single physical computer can support multiple operating systems.

[虚拟化]指的是一个物理计算机可以支持多个操作系统的过程

- Chapter4 test

- This system software is responsible for managing your computer's resources including memory, processing, and storage.

这个系统软件负责管理计算机的资源，包括内存、处理和存储

Operating system

- In many application programs, this feature is used to provide information or request input.

在许多应用程序中，此特性用于提供信息或请求输入

Dialog boxes

- Most operating systems support the ability to switch between different applications. This is called .

大多数操作系统都支持在不同应用程序之间切换的能力。这就是所谓的 multitasking

- The operating system is often referred to as the software environment or software.

操作系统通常被称为软件环境或软件 platform

- This version of UNIX is an open source operating system that is an alternative to Windows.

这个版本的 UNIX 是一种开源操作系统，可以替代 Windows Linux

- In a virtualized environment, this operating system runs on the physical machine.

在虚拟化环境中，这个操作系统运行在物理机器上

Host

- The windows troubleshooting utility that identifies and eliminates nonessential files is called .

调用识别和消除非必要文件的 windows 故障排除实用程序

Disk Cleanup

- A computer can become infected with viruses in all the following ways, except by .

计算机可以通过下列所有方式感染病毒，但除外

opening attachments to e-mail messages 打开电子邮件的附件

downloading software from the Internet 从互联网下载软件

downloading files from the Web 从 Web 下载文件

creating a new folder on your computer 在计算机上创建一个新文件夹 ✓

- This type of program allows communication between a peripheral device such as a mouse or printer, and the rest of the computer system .

这种类型的程序允许外围设备(如鼠标或打印机)与计算机系统的其他部分之间通信

Device driver

- System software cannot handle technical details without user intervention.

没有用户干预, 系统软件无法处理技术细节

false

- In a network environment the network server computer coordinates all communications between the other computers.

在网络环境中, 网络服务器计算机协调其他计算机之间的所有通信

true

- Linux is a popular and powerful alternative to the Windows operating system.

Linux 是 Windows 操作系统的流行和强大的替代品

true

- When a single physical computer runs a special program known as virtualization software, it operates as though it were two or more separate and independent computers known as virtual machines.

当一台物理计算机运行一个被称为虚拟化软件的特殊程序时, 它就像两台或两台以上被称为虚拟机的独立计算机一样运行

true

- For all devices, the appropriate drivers are automatically selected and installed when the device is first connected to the computer.

对于所有设备, 当设备第一次连接到计算机时, 会自动选择并安装适当的驱动程序

false

- A (n) [language translator] converts the programming instructions into a language that the computer understands and processes.

[语言翻译器]把程序指令转换成计算机能理解和处理的语言

- Smartphone operating systems are a type of [embedded] operating system.

智能手机操作系统是一种[嵌入式]操作系统

- Disk Cleanup, File History, and DiskDefragmenter are all examples of [utility] programs.

磁盘清理、文件历史和磁盘碎片整理程序都是[实用程序]的例子

课后习题

- 选择

1. Programs that perform specific tasks related to managing computer resources.

执行与管理计算机资源相关的特定任务的程序

h. utilities

2. Restarting a running computer without turning off the power.

在不关闭电源的情况下重新启动正在运行的计算机

j. warm boot

3. Type of operating system that controls and coordinates networked computers.

控制和协调联网计算机的一种操作系统

f. NOS

4. An operating system is often referred to as the software environment or____.

一个操作系统通常被称为软件环境或____

g. platform

5. Switching between different applications.

在不同应用程序之间切换

e. multitasking

6. A type of software that allows a single physical computer to operate as though it were two or more separate and independent computers.

一种允许一台物理计算机像两台或两台以上独立的计算机一样运行的软件

i. virtualization

7. Mobile operating system that is owned by Google and is one of the fastest-growing mobile operating systems.

它是谷歌拥有的移动操作系统，是增长最快的移动操作系统之一进入翻译页面

a. Android

8. Type of program that guards computer systems from viruses and other damaging programs.

保护计算机系统免受病毒和其他破坏性程序侵害的一种程序

b. antivirus

9. The operating system operating on each virtual machine ____.

在每台虚拟机上运行的操作系统____

d. guest

10. Program that works with the operating system to allow communication between a device and the rest of a computer system is called a device.

与操作系统一起工作，使设备与计算机系统的其他部分进行通信的程序称为设备

c. driver

第 5 章 THE SYSTEM UNIT

系统处理单位

SYSTEM UNIT 系统处理单位

System unit (system chassis) contains electronic components. The most common personal computers are desktop, laptop, tablet, smartphones, and wearable computers.

系统单元(系统机箱)包含电子元器件。最常见的个人电脑有台式电脑、笔记本电脑、平板电脑、智能手机和可穿戴电脑。

Desktop 桌面

Desktop system units are located in a separate case; tower unit (tower computer) has vertical system unit; all-in-one computer combines system unit and monitor.

桌面系统单元位于单独的情况下;塔单元(塔计算机)有垂直系统单元;一体机结合了系统单元和监视器

Laptop 笔记本电脑

Laptop system units contain secondary storage devices and input devices. Specialty laptops include two-in-one laptops (touch screen and fold flat), gaming laptops (high-end graphics and fast processors), ultrabooks, also known as ultraportables,

and mini notebooks (lighter, thinner, longer battery life).

笔记本电脑系统单元包含二级存储设备和输入设备。专业笔记本电脑包括二合一笔记本电脑(触摸屏和折叠平板), 游戏笔记本电脑(高端图形和快速处理器), 超极本, 也被称为超便携电脑, 和迷你笔记本电脑(更轻, 更薄, 电池寿命更长)

Tablet 平板电脑

Tablet system units are located behind the monitor. They are smaller, lighter, and generally less powerful than laptops and use a virtual keyboard. Mini tablets are a smaller version of a traditional tablet.

平板系统单元位于显示器后面。它们比笔记本电脑更小、更轻, 通常性能也不如笔记本电脑, 而且使用虚拟键盘。迷你平板电脑是传统平板电脑的缩小版

Smartphones 智能手机

Smartphones are the most popular mobile device; extend cell phones by providing computing power to capture video, send audio and video, run apps, connect to the Internet, and more.

智能手机是最受欢迎的移动设备;通过提供计算能力来扩展手机捕捉视频、发送音频和视频、运行应用程序、连接互联网等等

Wearable Computers 可穿戴计算机

Wearable computers (wearable devices) contain an embedded computer on a chip. The most common wearable computers are smartwatches and activity trackers.

可穿戴计算机(可穿戴设备)包含一个嵌入在芯片上的计算机。最常见的可穿戴电脑是智能手表和活动追踪器

Components 组件

Each type of system unit has the same basic components, including system board, microprocessor, and memory.

每一种系统单元都有相同的基本组件, 包括系统板、微处理器和内存

SYSTEM BOARD 主板

The system board (mainboard or motherboard) controls all communication for the computer system. All external and internal devices and components connect to it.

系统板(主板或主板)控制计算机系统的所有通信。所有外部和内部的设备和组件都连接到它

- **Sockets** provide connection points for chips (silicon chips, semiconductors, integrated circuits). Chips are mounted on chip carriers.

插座为芯片(硅芯片, 半导体, 集成电路)提供连接点。芯片安装在芯片载体上

- **Slots** provide connection points for specialized cards or circuit boards.

槽为专用卡或电路板提供连接点

- **Bus lines** provide pathways to support communication.

总线线路提供了支持通信的通道

MICROPROCESSOR 微处理器

In most personal computers, the central processing unit (CPU) or processor is contained on a single chip called the microprocessor. It has two basic components: a control unit and ALU.

在大多数个人计算机中, 中央处理单元(CPU)或处理器包含在一块叫做微处理器的芯片上。

它有两个基本组成部分:控制单元和 ALU

Microprocessor Chips 微处理器芯片

A word is the number of bits that can be accessed by the microprocessor at one time.

Clock speed represents the number of times the CPU can fetch and process data or instructions in a second.

一个字是微处理器一次可以访问的比特数。时钟速度表示 CPU 在一秒钟内获取和处理数据或指令的次数

Multicore processors can provide multiple independent CPUs. Parallel processing requires programs that allow multiple processors to work together to run large complex programs.

多核处理器可以提供多个独立的 cpu。并行处理需要允许多个处理器一起工作的程序来运行大型复杂的程序

Specialty Processors 专业处理器

Specialty processors include graphics coprocessors, also known as GPU or graphics processing unit (processes graphic images), and processors in automobiles (monitor fuel efficiency, satellite entertainment, and tracking systems).

专业处理器包括图形协处理器, 也被称为 GPU 或图形处理单元(处理图形图像), 以及汽车上

的处理器(监测燃料效率、卫星娱乐和跟踪系统)

MEMORY 存储器

Memory holds data, instructions, and information. There are three types of memory chips.

内存保存数据、指令和信息。有三种类型的存储芯片

RAM

RAM (random-access memory) chips are called temporary or volatile storage because their contents are lost if power is disrupted.

RAM(随机存取存储器)芯片被称为临时存储器或易失存储器, 因为如果电源中断, 它们的内容就会丢失

- **Cache memory** is a high-speed holding area for frequently used data and information.

高速缓存是高速保存常用数据和信息的区域

- **DIMM (dual in-line memory module)** is used to expand memory.

双列直插式内存模块用于内存扩展

- **Virtual memory** divides large programs into parts that are read into RAM as needed.

虚拟内存将大型程序划分为需要时读入RAM的部分

ROM

ROM (read-only memory) chips are nonvolatile storage and control essential system operations.

ROM(只读存储器)芯片是非易失性存储和控制基本系统操作

Flash Memory

Flash memory does not lose its contents when power is removed.

当电源被拔出时, 闪存不会丢失它的内容

EXPANSION SLOTS AND CARDS 扩展插槽和卡

Most computers allow users to expand their systems by providing expansion slots on their system boards to accept expansion cards.

大多数计算机允许用户通过提供系统板上的扩展插槽来扩展系统, 以接受扩展卡

Examples of expansion cards include graphics cards, network interface cards (NIC; network adapter cards), and wireless network cards.

扩展卡的例子包括图形卡、网络接口卡 (NIC; 网络适配器卡) 和无线网卡

SD cards are fingernail-size expansion cards used primarily for laptop computers, tablets, and smartphones. They connect to expansion slots within these devices.

SD 卡是指甲大小的扩展卡，主要用于笔记本电脑、平板电脑和智能手机。它们连接到这些设备中的扩展槽

BUS LINES 总线线路

Bus lines, also known as buses, provide data pathways that connect various system components. Bus width is the number of bits that can travel simultaneously.

总线线路，也称为总线，提供连接各种系统组件的数据通道。总线宽度是可以同时传输的比特数

System buses connect CPU and memory. Expansion buses connect CPU and slots.

系统总线连接 CPU 和内存。扩展总线用于连接 CPU 和槽位

Expansion Buses 扩展总线

Three principal expansion bus types:

三种主要的扩展总线类型：

- **USB (universal serial bus)** can connect from one USB device to another or to a common point (hub) and then onto the system board.

USB(通用串行总线)可以从一个USB设备连接到另一个USB设备或连接到一个公共点(集线器)，然后连接到系统板上

- **FireWire bus** is similar to USB bus but more specialized.

火线总线类似于 USB 总线，但更专业

- **PCI Express (PCIe) bus** is widely used; provides a single dedicated path for each connected device.

PCI Express (PCIe)总线应用广泛;为每个连接的设备提供单个专用路径

PORTS 端口

Ports are connecting sockets on the outside of the system unit.

端口是连接系统单元外部的套接字

Standard Ports 标准端口

Four standard ports are

四个标准端口是

- **USB (universal serial bus)**—one USB port can connect several devices to system unit USB-A on PCs; USB-B on peripherals; USB-C gaining popularity.

USB(通用串行总线)——一个USB端口可以将多个设备连接到pc上的系统单元USB-A;USB-B 外设;USB-C 越来越受欢迎

- **HDMI (high definition multimedia interface)**—provides high-definition video and audio.

HDMI(高清多媒体接口)–提供高清视频和音频

- **Thunderbolt**—provides high-speed connections to up to seven Thunderbolt devices at once, such as external hard drives and monitors.

Thunderbolt —提供高速连接，最多可同时连接 7 个 Thunderbolt 设备，如外部硬盘驱动器和监视器

- **Ethernet**—high-speed networking port that has become a standard for many of today' s computers.

以太网高速网络端口，已成为当今许多计算机的标准

Specialized Ports 专用端口

Five specialty ports are eSATA (external serial advanced technology attachment) for high-speed connections to large secondary storage devices, MIDI for digital music, Mini DisplayPort (MiniDP, mDP) for large monitors, VGA (Video Graphics Adapter) and DVI (Digital Video Interface) for connecting to monitors, and FireWire for high-speed connection to devices such as camcorders and secondary storage.

5 个专用端口为 eSATA(外部串行先进技术附件)，用于高速连接到大型二级存储设备，MIDI 用于数字音乐，Mini DisplayPort (MiniDP, mDP)用于大型监视器，VGA(视频图形适配器)和 DVI(数字视频接口)用于连接到监视器，以及 FireWire 用于高速连接到摄像机和二级存储等设备

Cables 线缆

Cables are used to connect external devices to the system unit via ports.

线缆用于连接外部设备和系统单元的接口

POWER SUPPLY 电源供应

Power supply units convert AC to DC and power desktops. AC adapters power laptops and tablets and recharge batteries. Some smartphones use wireless charging platforms.

电源模块支持交流转直流和台式电源。交流适配器为笔记本电脑和平板电脑供电，并为电池充电。一些智能手机使用无线充电平台

ELECTRONIC REPRESENTATION 电子显示

Human voices create analog (continuous) signals; computers only recognize digital electronic signals.

人类声音产生模拟(连续)信号;计算机只识别数字电子信号

Numeric Representation 数字表示法

Data and instructions can be represented electronically with a two-state or binary system of numbers (0 and 1). Each 0 or 1 is called a bit. A byte consists of 8 bits. Hexadecimal system (hex) uses 16 digits to represent binary numbers.

数据和指令可以用两种状态或二进制数字系统(0和1)来表示。每一个0或1被称为一个比特。一个字节由8位组成。十六进制系统(hex)使用16位数字来表示二进制数字

Character Encoding 字符编码

Character encoding standards assign unique sequences of bits to each character. Three standards are

字符编码标准为每个字符分配唯一的位序列。三个标准是

- **ASCII—American Standard Code for Information Interchange.** Historically used for personal computers.

美国信息交换标准代码。过去用于个人电脑

- **EBCDIC—Extended Binary Coded Decimal Interchange Code.** Historically used for mainframe computers.

扩展的二进制编码十进制交换码。历史上用于大型计算机

- **Unicode**—16-bit code, most widely used standard.

unicode - 16位代码，使用最广泛的标准

CAREERS IN IT 其中的职业

Computer technicians repair and install computer components and systems.

Certification in computer repair or an associate's degree from professional schools is required. Salary range is \$33,000 to \$49,000.

计算机技术员修理和安装计算机部件和系统。需要计算机维修证书或专业学校的大专学历。

工资范围是 33000 美元到 49000 美元

雨课堂

• Chapter 05 Test (System Unit)

- With these systems, input and output devices are located outside the system unit.

在这些系统中，输入和输出设备位于系统单元之外

Desktop systems

- Which of the following is not a character encoding standard?

下列哪项不是字符编码标准？

CISC

- On the motherboard the connection points for chips are referred to as__.

在主板上，芯片的连接点被称为__

sockets

- Integrated circuits must be mounted on, which are then plugged into the motherboard.

集成电路必须安装在上面，然后插入主板

chip carriers

- Which of the following is a function of the control unit?

下列哪项是控制单元的功能？

It directs the movement of electronic signals between memory and the arithmetic-logic unit.

它指导电子信号在存储器和算术逻辑单元之间的运动

- In most personal computer systems, the CPU is contained on a single chip called the _.

在大多数个人计算机系统中，CPU 被包含在一个叫做_的单芯片上

microprocessor

- For multicore processors to be used effectively, computers must understand how to divide tasks into parts that can be distributed across each core—an operation called .

为了有效地使用多核处理器,计算机必须了解如何将任务划分为可以分布在每个核上的部分——这一操作被称为

parallel processing

- These specialty processors are designed to improve specific computing operations such as displaying 3-D images and encrypting data.

这些特殊处理器被设计用于改进特定的计算操作,如显示 3-D 图像和加密数据

Coprocessors

- Frequently-accessed information is held in ____.

经常访问的信息保存在____中

cache memory

- Which of the following can be used to connect several devices to the system unit and are widely used to connect keyboards, mice, printers, storage devices, and a variety of specialty devices?

下列哪一项可用于将多个设备连接到系统单元,并广泛用于连接键盘、鼠标、打印机、存储设备和各种专用设备?

USB port

- Which of the following ports provide high-speed connections for external hard disk drives optical disks, and other large secondary storage devices?

以下哪个端口为外部硬盘驱动器、光盘和其他大型二级存储设备提供高速连接?

eSATA

- The system unit, also known as the system chassis, is a container that houses most of the electronic components that make up a computer system.

系统单元,也被称为系统机箱,是一个容器,容纳了组成计算机系统的大部分电子元件

true

- The alu tells the rest of the computer system how to carry out a program's

instructions.

alu 告诉计算机系统的其他部分如何执行程序指令

false

- Parallel processing requires a computer to perform one complete task before beginning the next task.

并行处理要求计算机在开始下一个任务之前完成一个完整的任务

false

- Plug and Play has become a generic term that is associated with the ability to plug any device into a computer and have it play or work immediately
即插即用已经成为一个通用术语,它指的是将任何设备插入计算机并立即播放或工作的能力

true

- The number of bits that can travel simultaneously down a bus is known as the word size.

在总线上可以同时传输的比特数被称为字的大小

false

- eSTAT ports provide high-speed connections for external hard disk drives, optical disks and other large secondary storage devices.

eSTAT 端口为外部硬盘驱动器、光盘和其他大型二级存储设备提供高速连接。

true

- HDMI ports make it possible to use a computer as a video jukebox or an hd video recorder.

HDMI 接口使电脑成为视频点唱机或高清录像机成为可能

true

- The [system unit], also known as the system chassis, a container that houses most of the electronic components that make up a computer system.

[系统单元], 也被称为系统机箱, 是容纳构成计算机系统的大部分电子元件的容器

- The [binary] system consists of only two digits—0 and 1

二进制系统只有两个数字—0 和 1

- A (n) [word] is the number of bits that can be accessed at one time by the CPU.

一个[字节]是 CPU 一次可以访问的比特数

- [Multicore] processors allow a single computer to run two or more operations at the same time.

[多核]处理器允许一台计算机同时运行两种或多种操作

- [Cache] memory improves processing by acting as a temporary high-speed holding area between the memory and the CPU.

[高速缓冲存储器]内存通过充当内存和 CPU 之间的临时高速保持区来提高处理能力

- [Virtual] memory uses the hard drive to substitute for RAM.

[虚拟]内存使用硬盘驱动器来代替 RAM

- [System] buses connect the CPU to memory on the system board.

[系统]总线用于连接 CPU 和系统板上的内存

- [Wireless] network cards allow computers to be connected without cables

[无线]网卡使计算机不需要电缆就可以连接

• Chapter 05 Online Test (System Unit)

- This type of computer is a thin slab that is almost all monitor with the system unit located behind the monitor.

这种类型的计算机是一个薄板，几乎所有的显示器和系统单元位于显示器后面

Tablet

- This character encoding scheme allows non-English characters and special characters to be represented.

这种字符编码方案允许表示非英语字符和特殊字符

Unicode

- Tiny circuit-boards etched onto squares of sand-like material, called silicon, are called ____.

微小的电路板蚀刻在方形的沙状材料上，称为硅，被称为____

chips

- Pathways that support communication among the various electronic components on the system board are called ____.

支持系统板上各种电子元件之间通信的通道称为____

bus lines

- Which of the following allows a computer to run two or more operations simultaneously?

下列哪个选项允许计算机同时进行两项或多项操作？

Multicore chip

- A type of specialty processor devoted exclusively to protecting your privacy.
一种专门保护您隐私的特殊处理器

Cryptoprocessor

- With this type of memory, large programs are divided into parts and the parts are stored on a secondary device, usually a hard disk.

使用这种类型的内存，大的程序被分成几个部分，这些部分存储在辅助设备上，通常是硬盘

Virtual

- Which of the following offers a combination of the features of RAM and ROM?
下列哪一项结合了 RAM 和 ROM 的特性？

Flash Memory

- Most personal computers allow users to expand their systems by providing these on the system board.

大多数个人计算机允许用户通过在系统板上提供这些来扩展他们的系统。

Expansion slots

- Which of the following ports is used for connecting musical instruments to a sound card?

下面哪个端口用于连接乐器和声卡？

MIDI

- Laptops and tablets use ___ that are typically located outside the system unit.
笔记本电脑和平板电脑使用___，通常位于系统单元之外

AC adapters

- ASCII supports Chinese and Japanese.

ASCII 支持中文和日文

false

- The control unit directs the movement of electronic signals between the memory

and the ALU.

控制单元指导电子信号在存储器和 ALU 之间的移动。

ture

- Virtual memory is stored on a RAM chip.

虚拟内存存储在 RAM 芯片上。

false

- Graphics cards connect the system board to secondary storage.

显卡连接系统板和二级存储。

false

- USB ports are used to connect keyboards, mice, printers, storage devices, and a variety of specialty devices.

USB 接口用于连接键盘、鼠标、打印机、存储设备和各种专业设备。

true

- Firewire ports are used to connect keyboards, mice, printers, and storage devices.

火线接口用于连接键盘、鼠标、打印机和存储设备

false

- Computers require direct current (DC) to power their electronic components and to represent data and instructions.

计算机需要直流电 (DC) 来驱动其电子元件，并表示数据和指令。

true

- A(n) [64] -bit-word computer can access eight bytes of information at a time.

一个[64]位字计算机一次可以访问 8 个字节的信息

- [Logical] operations consist of making comparisons.

[逻辑]操作包括做比较

- The system board is also known as the [motherboard] .

系统板也被称为[主板]

- The most widely used character encoding standard today is [Unicode] .

今天使用最广泛的字符编码标准是[万国码]

- The processing speed of a microprocessor is typically represented by its [clock]

speed.

微处理器的处理速度通常用它的时钟速度来表示

- [Sound] cards accept audio input from a microphone and convert it into a form that can be processed by the computer.

[声音]卡接受来自麦克风的音频输入，并将其转换成可由计算机处理的形式

- Network interface cards (NIC) are connected through [expansion] slots on the motherboard.

网卡通过主板上的扩展插槽连接。

- The number of bits that can travel simultaneously down a bus is known as the bus [width].

在总线上可以同时传输的比特数被称为总线[宽度]。

课后习题

- 选择

1. This container houses most of the electrical components for a computer system.

这个容器容纳了计算机系统的大部分电子元件

system unit

2. The most popular mobile device.

最受欢迎的移动设备

smartphone

3. The mainboard or motherboard is also known as the:

主板或主板也被称为:

system board

4. How many bytes can a 32-bit-word computer access at one time?

一台 32 位字的计算机一次可以访问多少个字节?

4

5. In a microcomputer system, the central processing unit is contained on a single:

在微型计算机系统中，中央处理单元包含在一个单处理器上:

chip

6. This type of memory divides large programs into parts and stores the parts on a secondary storage device.

这种类型的内存将大程序分成若干部分，并将这些部分存储在一个二级存储设备上。

virtual

7. Also known as NIC, this adapter card is used to connect a computer to a: 也被称为 NIC，这种适配器卡用于连接计算机到：

network

8. This provides a pathway to connect parts of the CPU to each other.

这提供了一个连接 CPU 各部分的通道。

bus

9. The specialized port that provides very high-speed connections for large secondary storage devices.

为大型二级存储设备提供高速连接的专用端口。

eSATA

10. Computers can only recognize this type of electronic signal.

计算机只能识别这种电子信号。

digital

• 匹配

1. A type of multiprocessor chip that provides two or more separate and independent CPUs.

一种多处理器芯片，提供两个或多个独立的 cpu。

c. multicore

2. A type of memory that is volatile or loses its contents when power is turned off.

一种当电源关闭时易失性或内容丢失的存储器。

f. random access

3. System board component that provides a connection point for specialized cards or circuit boards.

为专用卡或电路板提供连接点的系统板组件。

g. slots

4. Provide connection points for chips.

为芯片提供连接点。

h. sockets

5. A type of memory that improves processing by acting as a temporary high-speed holding area between the memory and the CPU.

一种内存类型，通过在内存和 CPU 之间充当临时的高速保持区来提高处理能力。

a. cache

6. A type of memory that provides a combination of features of RAM and ROM.

一种结合了 RAM 和 ROM 特性的存储器

b. flash

7. Another name for a network adapter card.

网络适配器卡的另一种名称。

d. NIC

8. This bus connects the CPU to memory on the system board.

该总线将 CPU 与系统板上的内存连接起来。

i. system

9. This port can be used to connect many USB devices to the system.

该端口可用于连接多个 USB 设备到系统。

j. USB

10. A socket for external devices to connect to the system unit

外部设备连接到系统单元的插座

e. port

第 6 章 INPUT AND OUTPUT

输入和输出

KEYBOARDS 键盘

Input is any data or instructions that are used by a computer. Input devices translate words, numbers, sounds, images, and gestures that people understand into a form that the system unit can process. These include keyboards and pointing, scanning, image

capturing, and audio-input devices.

输入是指计算机使用的任何数据或指令。输入设备将人们理解的单词、数字、声音、图像和手势转换成系统单元可以处理的形式。这些设备包括键盘和指向、扫描、图像捕捉和音频输入设备

Keyboards convert numbers, letters, and special characters that people understand into electrical signals. These signals are sent to, and processed by, the system unit.

键盘将数字、字母和人们能理解的特殊字符转换成电信号。这些信号被发送到系统单元，并由系统单元进行处理

Keyboards 键盘

There are four basic categories of keyboards: traditional, laptop, virtual, and thumb.

键盘有四种基本类型:传统键盘、笔记本电脑键盘、虚拟键盘和拇指键盘

- **Traditional keyboards**, used on desktop and larger computers. Standard keyboard has 101 keys. Toggle keys turn features on and off. Combination keys perform actions when combinations of keys are held down.

传统键盘，用于台式电脑和更大的电脑。标准键盘有 101 个键。切换键可以打开和关闭功能。当按下组合键时，组合键执行操作

- **Laptop keyboards**, used on laptop computers. Smaller than traditional keyboard with fewer keys. Typically does not have numeric keypad or standard location for function and navigation keys.

笔记本键盘，用于笔记本电脑。比传统键盘更小，按键更少。通常没有数字键盘或功能键和导航键的标准位置

- **Virtual keyboard**, primarily used on mobile devices and tablets. Does not have a physical keyboard. Keys displayed on screen and selected by pressing a key's image.

虚拟键盘，主要用于移动设备和平板电脑。没有物理键盘。在屏幕上显示的键，通过按下一个键的图像来选择

- **Thumb keyboard**, used on smartphones and other small mobile devices. Designed primarily for communicating via texting and connecting to the web.

拇指键盘，用于智能手机和其他小型移动设备。主要用于通过短信和连接到网络进行交流

POINTING DEVICES 指向设备

Pointing devices provide an intuitive interface with the system unit by accepting physical movements or gestures and converting them into machine-readable input. 通过接受物理动作或手势并将其转换为机器可读的输入，指向设备为系统单元提供了一个直观的界面

Mouse 鼠标

A **mouse** controls a pointer that is displayed on the monitor. The mouse pointer usually appears in the shape of an arrow. Some mice have a wheel button that rotates to scroll through information on the monitor. Optical mouse is the most widely used. A cordless or wireless mouse uses radio waves or infrared light waves. A touch pad operates by touching or tapping a surface. It is widely used instead of a mouse with laptops and some types of mobile devices.

鼠标控制显示在显示器上的指针。鼠标指针通常以箭头的形状出现。有些鼠标有一个滚轮按钮，可以旋转来滚动显示器上的信息。光学鼠标是应用最广泛的。无线鼠标使用无线电波或红外线光波。触摸板通过触摸或轻击表面来操作。在笔记本电脑和一些移动设备上，它被广泛用于代替鼠标

Touch Screens 触摸屏

Touch screens allow users to select actions by touching the screen with a finger or penlike device. Multitouch screens accept multiple-finger commands.

触摸屏允许用户用手指或笔状设备触摸屏幕来选择动作。多点触控屏幕接受多手指指令

Game Controllers 游戏控制器

Game controllers provide input to computer games. Widely used controllers include gaming mice, joysticks, gamepads, and motion-sensing devices.

游戏控制器为电脑游戏提供输入。广泛使用的控制器包括游戏鼠标、操纵杆、手柄和动作感应设备。

Stylus 触控笔

A **stylus** is typically used with tablets and mobile devices. Often, a stylus interacts with handwriting recognition software to translate handwritten notes into a form

that the system unit can process.

触控笔通常用于平板电脑和移动设备。通常，手写笔与手写识别软件交互，将手写笔记转换为系统单元可以处理的形式。

SCANNING DEVICES 扫描设备

Scanning devices move across text and images to convert them into a form that the system unit can process.

扫描设备在文本和图像之间移动，将它们转换成系统单元可以处理的形式。

Optical Scanners 光学扫描仪

An **optical scanner (scanner)** converts documents into machine-readable form. The four basic types are flatbed, document, portable, and 3D.

光学扫描仪将文档转换成机器可读的形式。四种基本类型是平板、文档、便携式和 3D。

Card Readers 读卡器

Card readers interpret encoded information located on a variety of cards. The most common is the magnetic card reader that reads information from a thin magnetic strip on the back of a card. Chip cards contain microchips to encrypt data and improve security.

读卡器解读各种卡片上的编码信息。最常见的是磁卡读卡器，它从卡背面的一个薄磁条读取信息。芯片卡包含微芯片，可以加密数据，提高安全性

Bar Code Readers 条形码读取器

Bar code readers or scanners (either handheld wand readers or platform scanners) read bar codes on products. There are a variety of different codes, including the UPC and MaxiCode codes.

条形码阅读器或扫描仪(手持棒式阅读器或平台扫描仪)可以读取产品上的条形码。有各种不同的代码，包括 UPC 和 MaxiCode 代码。

RFID Readers 动物标识读码器

RFID readers read RFID (radio-frequency identification) tags. These tags are widely used for tracking lost pets, production, and inventory and for recording prices and product descriptions.

RFID 阅读器用于读取 RFID(射频识别)标签。这些标签广泛用于跟踪丢失的宠物，生产和库

存，并记录价格和产品描述。

Character and Mark Recognition Devices 字符和标记识别设备

Character and mark recognition devices are scanners that are able to recognize special characters and marks. Three types are magnetic-ink character recognition (MICR), optical-character recognition (OCR), and optical-mark recognition (OMR). 字符和标记识别设备是能够识别特殊字符和标记的扫描器。三种类型是磁墨字符识别 (MICR)、光学字符识别 (OCR) 和光学标记识别 (OMR)。

IMAGE CAPTURING DEVICES 图像捕捉设备

Image capturing devices create or capture original images. These devices include digital cameras and webcams.

图像捕捉设备创建或捕获原始图像。这些设备包括数码相机和网络摄像头

Digital Cameras 数码相机

Digital cameras record images digitally and store them on a memory card or in the camera's memory. Most digital cameras record video too.

数码相机以数字方式记录图像，并将它们存储在存储卡或相机的内存中。大多数数码相机也记录视频。

Webcams 网路摄影机

Webcams are specialized digital video cameras that capture images and send them to a computer for broadcast over the Internet. Webcams are built into many smartphones and tablets, while others are attached to the computer monitor.

网络摄像头是一种专门的数字摄像机，它捕捉图像并将其发送到计算机上，通过互联网播放。许多智能手机和平板电脑都内置了网络摄像头，还有一些则连接到电脑显示器上。

AUDIO-INPUT DEVICES 音频输入设备

Audio-input devices convert sounds into a form that can be processed by the system unit. By far the most widely used audio-input device is the microphone.

音频输入设备将声音转换成可由系统单元处理的形式。到目前为止，使用最广泛的音频输入设备是麦克风。

Voice Recognition Systems 语音识别系统

Voice recognition systems use a microphone, a sound card, and special software.

Siri, Cortana, and Google Now are digital assistants that use voice recognition. Specialized portable voice recorders are widely used by doctors, lawyers, and others to record dictation. Some systems are able to translate dictation from one language to another, such as from English to Japanese.

语音识别系统使用麦克风、声卡和特殊软件。Siri、Cortana 和谷歌 Now 都是使用语音识别的数字助手。专业的便携式录音机被医生、律师和其他人广泛用于记录口述。有些系统能够将听写从一种语言翻译成另一种语言，比如从英语翻译成日语

MONITORS 显示器

Output is processed data or information. Output devices are hardware used to provide or create output. Monitors (display screens) are the most-used output device.

输出的是处理过的数据或信息。输出设备是用于提供或创建输出的硬件。显示器是最常用的输出设备。

Features 特点

Monitor clarity is a function of resolution (matrix of pixels, or picture elements), dot pitch, contrast ratio, size (active display area), and aspect ratio.

显示器清晰度是分辨率(像素矩阵或图像元素)、点间距、对比度、大小(活动显示区域)和纵横比的函数。

Flat-Panel Monitors 平面监视器

Flat-panel monitors are the most widely used monitor. Three basic types are LCD (liquid crystal display), LED (light-emitting diode), and OLED (organic light-emitting diode).

平板显示器是应用最广泛的显示器。三种基本类型是 LCD(液晶显示器)、LED(发光二极管)和 OLED(有机发光二极管)

E-book Readers 电子阅读

E-books (electronic books) are traditional printed books in electronic format. E-book readers (e-readers) store and display e-books and other electronic media. They use e-ink technology.

电子书(electronic books)是传统的纸质电子书。电子书阅读器(e-reader)存储和显示电子书和其他电子媒体。他们使用电子墨水技术。

Other Monitors 其它显示器

Other types of monitors include digital (interactive) whiteboards, ultra-high-definition television (UHDTV), and digital projectors.

其他类型的显示器包括数字(交互式)白板、超高清电视(UHDTV)和数字投影仪。

PRINTERS 打印机

Printers translate information processed by the system unit and present the information on paper.

打印机翻译由系统单元处理的信息，并将信息显示在纸上

Features 特点

Basic features include resolution measured in dpi (dots per inch), color capability (most common black ink selection is grayscale), speed, memory, and duplex (both sides of paper) printing.

基本特性包括分辨率 dpi(每英寸点数)、颜色能力(最常见的黑色墨水选择是灰度)、速度、存储和双面(纸张两面)打印。

Inkjet 喷墨打印机

Inkjet printers spray ink at high speed onto the surface of paper. They are the most widely used type of printer, reliable, quiet, and inexpensive. The most costly aspect of inkjet printers is replacing the ink cartridges.

喷墨打印机以高速将油墨喷到纸张表面。它们是使用最广泛的打印机类型，可靠，安静，廉价。喷墨打印机最昂贵的部分是更换墨盒。

Laser 激光

Laser printers use technology similar to photocopying machine. There are two categories: personal and shared.

激光打印机使用类似于复印机的技术。有两类:私人的和共享的。

3D Printer 3D 打印机

3D printers (additive manufacturing) create objects by adding layers of material onto one another. They have been available for decades; however, recent price reductions have increased their popularity.

3D 打印机(增材制造)通过将材料层层叠加来创造物体。它们已经存在了几十年;然而，最近

的降价增加了它们的受欢迎程度。

Other Printers 其他打印机

There are several other types of printers, including cloud printers (Google Cloud Print is a cloud printing service), thermal printers (use heat elements to produce images on heat-sensitive paper), and plotters (use data from graphics tablets and other graphical devices).

还有其他几种类型的打印机,包括云打印机(谷歌云打印是一种云打印服务)、热敏打印机(使用热元件在热敏纸上生成图像)和绘图仪(使用来自图形平板电脑和其他图形设备的数据)。

AUDIO-OUTPUT DEVICES 音频输出设备

Audio-output devices translate audio information from the computer into sounds that people can understand. The most widely used are speakers and headphones. These devices connect either by cable to an audio jack on the system unit or by a wireless connection. Bluetooth technology is widely used to connect wireless devices.

音频输出设备将计算机的音频信息转换成人们可以理解的声音。使用最广泛的是扬声器和耳机。这些设备要么通过电缆连接到系统单元的音频插孔,要么通过无线连接。蓝牙技术被广泛用于连接无线设备。

COMBINATION INPUT AND OUTPUT DEVICES 组合输入和输出设备

Combination devices combine input and output capabilities. Devices within this category include:

组合设备结合了输入和输出能力。这类设备包括:

- **Headsets** combine the functionality of microphones and headphones; integral part of serious video game systems.

耳机结合了麦克风和耳机的功能;电子游戏系统的重要组成部分。

- **Multifunctional devices (MFD)** typically combine capabilities of scanner, printer, fax, and copy machine.

多功能设备(MFD)通常结合了扫描仪、打印机、传真和复印机的功能。

- **Virtual reality (VR)** creates 3D simulated immersive experiences. VR hardware includes head-mounted displays and controllers.

虚拟现实(VR)创造三维模拟沉浸式体验。VR 硬件包括头戴显示器和控制器。

- **Drones (unmanned aerial vehicles, UAV)** take input from a controller and send output video and sound to the user.

无人机(unmanned aerial vehicles, UAV)从控制器获取输入,并将输出的视频和声音发送给用户。

- **Robots** use cameras, microphones, and other sensors as inputs; outputs can be as complex as exploring damaged nuclear reactors to as simple as taking a photo.

机器人使用摄像头、麦克风和其他传感器作为输入;输出可能复杂到探索受损的核反应堆,也可能简单到拍张照片。

ERGONOMICS 人体工程学

Ergonomics is the study of human factors related to things people use.

人体工程学是研究与人们使用的事物相关的人为因素。

Recommendations 建议书

Some recommendations to avoid physical discomfort associated with heavy computer use:

避免因大量使用电脑而引致身体不适的建议:

- To avoid eyestrain and headache, take a 15-minute break every hour; keep everything you're focusing on at the same distance; and clean the screen periodically.

为了避免眼睛疲劳和头痛,每小时休息 15 分钟;让你关注的所有东西都在同一个距离上;定期清洁屏幕。

- To avoid back and neck pain, use adjustable equipment; adjust chairs for height, angle, and back support; position monitors at eye level or slightly below. Use footrest, if necessary, to reduce leg fatigue.

为避免背部和颈部疼痛,应使用可调节的器械;调整椅子的高度、角度和背部支撑;将监视器置于眼睛水平或稍低的位置。必要时可使用脚凳,以减少腿部疲劳。

- To avoid repetitive strain injury (RSI) and carpal tunnel syndrome, use ergonomically correct keyboards; take frequent, short rest breaks; and gently massage hands.

为了避免重复性劳损(RSI)和腕管综合症,使用符合人体工程学的正确键盘;经常进行短暂的休息;轻轻按摩双手。

Portable Computers 便携式电脑

The design of portable computers presents ergonomic challenges.

便携式电脑的设计提出了人体工程学的挑战。

- Laptops do not allow correct positioning of keyboard and screen; raise level of laptop and use external keyboard.

笔记本电脑不允许正确定位键盘和屏幕;提高笔记本电脑的档次, 使用外接键盘。

- Tablets with virtual keyboards cause improper alignment of user's head; take frequent breaks, move while working, use cover or stand, use external keyboard.

带有虚拟键盘的平板电脑会导致使用者头部不正确对齐;经常休息, 工作时移动, 使用盖子或支架, 使用外接键盘。

- Smartphones require extensive use of thumbs; keep wrists straight, head up, and shoulders straight and use other fingers.

智能手机需要大量使用拇指;保持手腕伸直, 抬头, 肩膀伸直, 使用其他手指。

CAREERS IN IT 其中的职业

Technical writers prepare instruction manuals, technical reports, and other documents. An associate's or a bachelor's degree in communication, journalism, or English and a specialization in, or familiarity with, a technical field are required. Salary range is \$54,000 to \$71,000.

技术写手准备说明书、技术报告和其他文档。传媒、新闻或英语专业的大专或学士学位, 以及技术领域的专业知识或熟悉程度。薪资范围为 54,000 至 71,000 美元。

雨课堂

- Chapter 6 Online Test (Input and Output)

- ___ is any data or instructions that are used by a computer.

___是指计算机使用的任何数据或指令。

Input

- A keyboard key, like Caps Lock, that turns a feature on or off is called a ___ key.

一个键盘键, 如 Caps Lock, 打开或关闭功能被称为___键。

toggle

- This type of mouse emits and senses light to detect mouse movement.

这种类型的鼠标通过发光和感知光线来探测鼠标的移动。

Optical

- Which of the following is not a type of scanning device?

下列哪一项不是一种扫描设备？

Stylus

- Bar code readers use ____ embedded in them to read bar codes.

条形码阅读器使用嵌入其中的____来读取条形码。

photoelectric cells

- ____-input devices convert sounds into a form that can be processed by the system unit.

____输入设备将声音转换成系统单元可以处理的形式

Audio

- The distance between each pixel.

每个像素之间的距离。

Dot pitch

- The size, or ____, is measured by the diagonal length of a monitor's viewing area.

尺寸，或____，是测量的对角线长度的显示器的可视区域。

active display area

- Specialized devices with a large display connected to a computer projector and are widely used in classrooms and corporate boardrooms.

一种与电脑投影仪相连的大型显示器的专门设备，广泛用于教室和公司会议室。

Digital or interactive whiteboards

- Which of the following printer features enables you to print on both sides of a sheet of paper?

下列哪一项打印机功能可使你在一张纸的两面都打印？

Duplex

- An optical mouse uses light in order to detect movement.

光学鼠标利用光来探测移动。

true

- Optical scanners recognize individual letters or images.

光学扫描仪可以识别单个字母或图像。

false

- A magnetic card reader does not require the card to actually make contact with the reader.

磁卡读卡器并不要求磁卡与读卡器实际接触。

false

- The monitor size is measured by the diagonal length of the viewing area.

显示器的尺寸是通过观察区域的对角线长度来测量的。

true

- Digital or interactive whiteboards display a computer's desktop and can be controlled using a special pen or a finger.

数字或交互式白板显示在电脑桌面上，可以用特殊的笔或手指控制。

true

- Shared laser printers typically support color, are more expensive, and are used by a group of people.

共享激光打印机通常支持颜色，更贵，并由一群人使用。

true

- Wireless audio-output connections typically use Bluetooth technology.

无线音频输出连接通常使用蓝牙技术。

true

- Flatbed, document, and portable are types of [scanners] that accept documents and convert them into machine-readable form.

平板、文档和便携式是接受文档并将其转换为机器可读形式的[扫描仪]类型。

- Supermarkets use [bar] code readers to scan codes printed on product containers.

超市使用条形码阅读器扫描产品容器上的条形码。

- [RFID] tags are tiny chips that can be embedded into almost anything and contain

information used to track and locate lost pets; to monitor production and update inventory; to record prices, product descriptions, and locations of retail items.

RFID 标签是一种几乎可以嵌入任何东西的微型芯片，其中包含用于追踪和定位丢失宠物的信息；监控生产和更新库存；记录价格、产品描述和零售商品的位置。

- [Laser] printers use a technology similar to that used in a photocopy machine.
激光打印机使用一种类似于复印机的技术。
- [Cloud] printers are printers connected to the Internet that provide printing services to others on the Internet.

[云]打印机是连接到互联网的打印机，在互联网上向其他人提供打印服务。

- [Internet] telephones are specialized input and output devices for receiving and sending voice communication.

[互联网]电话是接收和发送语音通信的专用输入和输出设备。

- [Aspect] ratio indicates the proportional relationship between a display's width and height.

[纵横比]表示显示器的宽度和高度之间的比例关系。

- Chapter 6 Test (Input and Output)

- The arrangement of keys on a keyboard, QWERTY reflects the keyboard layout by taking the letters of the first six alphabetic characters found on the top row of keys

键盘上键的排列，QWERTY 通过。取最上面一排键的前六个字母中的字母

- Type of keyboard that uses a touch screen as the input device.
使用触摸屏作为输入设备的键盘类型。

Virtual

- The study of human factors related to things people use.

与人们使用的事物有关的人为因素的研究。

Ergonomics

- These screens can be touched with more than one finger, which allows for interactions such as rotating graphical objects on the screen with your hand or zooming in and out by pinching and stretching your fingers.

这些屏幕可以用多个手指触摸，这就允许了一些交互，比如用手旋转屏幕上的图形对象，或者通过捏和拉伸手指来放大和缩小屏幕。

Multitouch

- This reading device is used in banks to read the numbers on the bottom of checks and deposit slips.

这种读卡器用于银行读取支票和存款单底部的数字。

MICR

- OMR, OCR, and MICR are all types of__ .

OMR、OCR 和 MICR 都是__的类型

character and mark recognition devices

字符和标记识别装置

- This indicates the monitor's ability to display colors by comparing the light intensity of the brightest white to the darkest black.

这表明显示器通过比较最亮的白色和最暗的黑色的光强度来显示颜色的能力。

Contrast ratio

- The proportional relationship between a display's width and height.

显示器宽度和高度之间的比例关系。

Aspect ratio

- Which of the following statements is incorrect?

下列哪个陈述是不正确的？

Printer speed is measured in the number of words printed per minute.

打印机速度是以每分钟打印的字数来衡量的。

- Printers connected to the Internet that provide printing services to others on the internet are called__ .

连接到互联网上向互联网上其他人提供打印服务的打印机被称为__。

cloud printers

- Function keys provide shortcuts for specific tasks.

功能键为特定任务提供快捷方式。

true

- A stylus uses pressure to draw images on a screen.

触控笔使用压力在屏幕上绘制图像。

true

- Multitouch screens allow for interaction such as rotating graphical objects on the screen with your hand.

多点触控屏幕允许交互，比如用手旋转屏幕上的图形对象。

true

- An MICR device senses the presence or absence of a mark, such as a pencil mark.

MICR 设备能够感知标记的存在或缺失，例如铅笔标记。

false

- 3-D HDTV requires special viewing glasses in order to achieve a three-dimensional viewing experience.

3d 高清电视需要特殊的观看眼镜来实现三维观看体验。

true

- Resolution for a printer is a measure of the clarity of images produced, measured in dpi (dots per inch).

打印机的分辨率是对所产生图像清晰度的度量，以 dpi (每英寸点数) 为单位。

true

- Thermal printers use heat elements to produce images on heat-sensitive paper and are used at ATMs and gasoline pumps.

热敏打印机使用热元件在热敏纸上打印图像，用于自动取款机和汽油泵。

true

- The mouse, joysticks, touch screens, and styluses are all types of [pointing] devices.

鼠标、操纵杆、触摸屏和触控笔都是各种类型的[指向]设备。

- The most common card reader is the [magnetic] card reader which reads encoded information stored on a thin magnetic strip located on the back of the card.

最常见的读卡器是磁卡读卡器，它读取存储在卡背面的一个薄磁条上的编码信息。

- Character and [mark] recognition devices are scanners that are able to recognize special characters and marks.

字符和[标记]识别设备是能够识别特殊字符和标记的扫描仪。

- Digital [whiteboards] connect to computers and project computer output and can control the computer with the use of a special pen.

数字[白板]连接到计算机和项目计算机输出，并可以使用一种特殊的笔控制计算机。

- [Thermal] printers are widely used with ATMs and Expresses.

[热敏]打印机广泛应用于自动取款机和快速取款机。

- [Plotters] are special-purpose printers typically found in architectural and engineering environments.

[绘图仪]是一种特殊用途的打印机，通常用于建筑和工程环境。

- [Contrast] ratio indicates a monitor's ability to display colors by comparing the light intensity of the brightest white to the darkest black.

[对比度]比率表示显示器通过比较最亮的白色和最暗的黑色的光强度来显示颜色的能力。

课后习题

- 选择

1. Most keyboards use an arrangement of keys known as:

大多数键盘使用一种被称为:

QWERTY

2. The device that controls a pointer displayed on the monitor.

控制显示器上显示的指针的设备。

mouse

3. The type of screen that can be touched with more than one finger and supports zooming in and out by pinching and stretching your fingers.

一种可以用多个手指触摸的屏幕，支持通过捏和拉伸手指来放大和缩小屏幕。

multitouch

4. Flatbed and document are types of:

平板和文件是:

scanners

5. Device used by banks to automatically read those unusual numbers on the bottom of checks and deposit slips.

银行使用的一种自动读取支票和存款单底部不寻常数字的设备。

MICR

6. The most widely used audio-input device.

使用最广泛的音频输入设备。

microphone

7. The monitor feature that indicates the ability to display colors.

显示器功能，指示显示颜色的能力。

contrast ratio

8. Mobile devices able to store and display electronic media.

能够存储和显示电子媒体的移动设备。

e-book readers

9. Virtual reality strives to create this type of experience.

虚拟现实努力创造这种体验。

immersive

10. The study of human factors related to things people use is:

与人们使用的事物相关的人为因素的研究是：

ergonomics

• 匹配

1. Pressing this key turns a feature on or off.

按下这个键可以开启或关闭某个功能。

i. toggle key

2. Input device that controls a pointer that is displayed on the monitor.

控制显示在监视器上的指针的输入设备。

e. mouse

3. A penlike device commonly used with tablet PCs and PDAs.

一种笔状设备，通常用于平板电脑和 pda。

h. stylus

4. Bar code readers use either handheld wand readers or platform.

条形码阅读器使用手持读卡器或平台。

g. scanners

5. Bar code system used by many electronic cash registers.

许多电子收银机使用的条形码系统。

j. UPC

6. Records images digitally on a on a disk or its memory

以数字方式将图像记录在磁盘或其存储器上

b. digital camera

7. The distance between each pixel.

每个像素之间的距离。

c. dot pitch

8. A monitor feature that is measured by the diagonal length of the viewing area.

一种显示器特性，其由观察区域的对角线长度测量。

a. active-display area

9. Special-purpose printers for creating maps, images, and architectural and engineering drawings.

用于创建地图、图像、建筑和工程图纸的专用打印机。

f. plotters

10. Another name for an unmanned aerial vehicle (UAV).

无人驾驶飞机(UAV)的另一个名字。

d. drone

第 7 章 SECONDARY STORAGE

二级存储器

STORAGE 存储

RAM is primary storage. Most RAM is volatile, meaning that it loses its contents whenever power is disrupted. Secondary storage provides nonvolatile storage. Secondary storage retains data and information after the computer system is turned off.

RAM 是主存储器。大多数 RAM 是不稳定的，这意味着只要电力中断，它就会丢失内容。二级存储器提供非易失性存储器。二级存储保留数据和信息后，计算机系统关闭

Writing is the process of saving information to secondary storage devices.

Reading is the process of accessing information.

写入是将信息保存到二级存储设备的过程。阅读是获取信息的过程。

Important characteristics include

重要的特征包括

- **Media**—actual physical material that retains data and programs.
媒体—保留数据和程序的实际物理材料。
- **Capacity**—how much a particular storage medium can hold.
容量——一种特定的存储介质所能容纳的容量。
- **Storage devices**—hardware that reads and writes to storage media.
存储设备——对存储媒体进行读写的硬件。
- **Access speed**—time required to retrieve data from a secondary storage device.
从二级存储设备检索数据所需的访问速度-时间。

HARD DISKS 硬盘

Hard disks use rigid metallic platters that provide a large amount of capacity. They store data and programs by altering the electromagnetic charges on the platter's surface. Files are organized according to

硬盘使用的是提供大容量的刚性金属盘片。它们通过改变盘面上的电磁电荷来存储数据和程序。文件是根据

- **Tracks**—concentric rings on the platter.
轨迹-盘子上的同心环。
- **Sectors**—wedge-shaped sections.
行业楔形部分
- **Cylinders**—run through each track of a stack of platters.
圆筒-通过盘片堆栈的每个轨道。

Density refers to how tightly electromagnetic charges can be packed next to one another on the disk.

密度是指电磁电荷在磁盘上相邻排列的紧密程度。

A **head crash** occurs when the hard disk makes contact with the drive's read-write heads.

当硬盘与驱动器的读写磁头接触时，就会发生磁头碰撞

Two types are internal and external hard disks.

硬盘分为内部硬盘和外部硬盘。

Internal Hard Disk 内置硬盘

Internal hard disks are located within the system unit. Used to store programs and data files.

内部硬盘位于系统单元内部。用于存储程序和数据文件。

HARD DISKS 硬盘

External Hard Drives 外置硬盘

Unlike internal hard disks, external hard drives are removable. External drives use the same basic technology as internal disks.

与内部硬盘不同，外部硬盘驱动器是可移动的。外部驱动器使用与内部磁盘相同的基本技术。

Performance Enhancements 性能优化

Three ways to improve hard-disk performance are disk caching, RAID, and file compression and decompression.

提高硬盘性能的三种方法是磁盘缓存、RAID 和文件压缩和解压缩。

- **Disk caching**—provides a temporary high-speed holding area between a secondary storage device and the CPU; improves performance by anticipating data needs and reducing time to access data from secondary storage.

磁盘缓存——在二级存储设备和 CPU 之间提供一个临时的高速保存区域；通过预测数据需求和减少从二级存储访问数据的时间来提高性能。

- **RAID (redundant array of inexpensive disks)**—several inexpensive hard-disk drives are connected together; improves performance by providing expanded storage, fast access, and high reliability.

RAID(廉价磁盘冗余阵列)——几个廉价的硬盘驱动器连接在一起；通过扩展存储空间、快速访问和高可靠性提高性能。

- **File compression and decompression**—files are compressed before storing and then decompressed before being used again; improves performance through efficient storage.

文件压缩和解压缩文件在存储之前被压缩，在再次使用之前被解压缩;通过高效存储提高性能。

SOLID-STATE STORAGE 固态存储

Solid-state storage devices have no moving parts; provide access to flash memory (solid-state storage).

固态存储设备没有移动部件;提供对闪存(固态存储)的访问。

Solid-State Drives 固态硬盘

Solid-state drives (SSDs) are similar to internal hard-disk drives except they use solid-state memory; are faster, more durable, and more expensive; and generally provide less capacity. Hybrid systems contain SSD and magnetic disks.

固态硬盘(SSDs)类似于内部硬盘驱动器，除了它们使用固态存储器;更快，更耐用，更昂贵;通常提供更少的容量。混合系统包括 SSD 硬盘和硬盘。

Flash Memory Cards 闪存卡

Flash memory cards are small solid-state storage devices that are widely used with portable devices. They are used with a variety of specialized input devices, including digital cameras to store and transfer images and digital media players like the iPod to store and transfer music and video files.

闪存卡是一种小型固态存储设备，广泛应用于便携式设备。它们与各种专门的输入设备一起使用，包括用于存储和传输图像的数码相机，以及用于存储和传输音乐和视频文件的数字媒体播放器，如 iPod。

USB Drives USB 设备

USB drives (flash drives) are so small that they fit onto a key ring. These drives connect to a computer's USB port and are widely used to transfer data and information between computers, specialty devices, and the Internet.

USB 驱动器(闪存驱动器)是如此之小，他们可以装在一个钥匙环。这些驱动器连接到计算机的 USB 端口，被广泛用于在计算机、专用设备和互联网之间传输数据和信息。

OPTICAL DISCS 光碟片

Optical discs use laser technology. 1s and 0s are represented by pits and lands.

Optical disc drives project light and measure the reflected light.

光盘使用激光技术。1s 和 0s 用坑和地表示。光盘驱动器投射光和测量反射光。

The most widely used optical discs are

使用最广泛的光盘是

- **Compact discs (CDs)** were the first; typical storage 700 MB; often used to store music.

cd 是最早的;典型存储 700mb;通常用来储存音乐。

- **Digital versatile discs (DVDs)** are standard optical discs; typical storage 4.7 GB; often used to store movies and software.

数码多功能光盘(dvd)是标准的光盘;典型存储 4.7 GB;通常用于存储电影和软件。

- **Blu-ray discs (BDs)** are the newest; designed to store hi-def (high-definition) video; typical storage 25 GB; often used to store hi-def video and video games.

蓝光光碟(BDs)是最新的;用于存储高清(高清)视频;典型存储 25gb;通常用来储存高清视频和视频游戏。

Each of these optical discs has three basic formats:

每一张光盘都有三种基本格式:

- **Read-only (ROM for read-only memory)** discs cannot be written on or erased by the user.

只读磁盘不能被用户写入或擦除

- **Write-once (R for recordable)** discs can be written on once. After that, they can be read many times but cannot be written on or erased.

一次写入(R 表示可记录)磁盘可以写入一次。之后,它们可以被多次读取,但不能被写入或擦除。

- **Rewritable (RW for rewritable or RAM for random-access memory)** discs are similar to write-once discs except that the disc surface is not permanently altered when data is recorded.

可重写磁盘(RW 指可重写磁盘, RAM 指随机存取存储器)类似于一次写入磁盘,只是在记录数据时磁盘表面不会永久改变。

CLOUD STORAGE 云端储存

With cloud computing, the Internet acts as a “cloud” of servers. Cloud storage

(online storage) is supplied by these servers.

有了云计算，互联网就像服务器的“云”。云存储(在线存储)由这些服务器提供。

- Cloud servers provide storage, processing, and memory.

云服务器提供存储、处理和内存。

- Advantages for users include less maintenance, fewer hardware upgrades, and easy file sharing and collaboration.

对用户的好处包括更少的维护，更少的硬件升级，以及容易的文件共享和协作。

- Disadvantages for users include slower access speed and less control over file security

用户的缺点包括访问速度较慢，对文件安全的控制较少

MASS STORAGE DEVICES 电脑中的大量储存装置

Mass storage refers to the tremendous amount of secondary storage required by large organizations. Mass storage devices are specialized high-capacity secondary storage devices.

大容量存储是指大型组织所需要的大量二级存储。大容量存储设备是专门的高容量二级存储设备。

Most large organizations have established a strategy called an enterprise storage system to promote efficient and safe use of data.

大多数大型组织已经建立了一种称为企业存储系统的策略，以促进数据的高效和安全使用。

Mass storage devices that support this strategy are file servers, network attached storage (NAS), RAID systems, and organizational cloud storage. A storage area network (SAN) is a method of using enterprise-level remote storage systems as if they were local to your computer.

支持该策略的海量存储设备包括文件服务器、NAS (network attached storage)、RAID 系统和组织云存储。存储区域网络(SAN)是一种使用企业级远程存储系统的方法，就好像它们是您的计算机的本地存储一样。

CAREERS in IT 其中的职业

Disaster recovery specialists are responsible for recovering systems and data after

a disaster strikes an organization. Bachelor's or associate's degree in information systems or computer science, experience, and additional skills in the areas of networking, security, and database administration are desirable. Salary range is \$73,000 to \$92,000.

灾难恢复专家负责在灾难发生后恢复系统和数据。信息系统或计算机科学本科或大专学历，有网络、安全、数据库管理方面的经验和额外技能者优先。薪资范围为 73,000 美元至 92,000 美元。

雨课堂

- Chapter 7 Test (Secondary Storage)

- Hard disks store and organize files using all the following, except

硬盘存储和组织文件使用所有以下，除了

tracks

sectors

cylinders

paths ✓

- Which of the following is true about an internal hard disk?

关于内部硬盘，下列哪项是正确的？

Fixed amount of storage

- Which of the following is not an example of secondary storage?

下列哪一个不是二级存储的例子？

RAM

- Storage that retains its data after the power is turned off is referred to as _.

在电源关闭后保留其数据的存储称为_。

nonvolatile storage

- The of a secondary storage device indicates how much data the storage medium can hold.

二级存储设备的表示该存储介质可以容纳多少数据。

capacity

- Which of the following characteristics about RAID is incorrect?

关于 RAID，下面哪个特征是不正确的？

one high-capacity hard disk drive

- Three ways to improve the performance of a hard disk include__.

提高硬盘性能的三种方法包括__。

disk caching RAID, and file compression

磁盘缓存 RAID 和文件压缩

- Smartphones use which of these storage technologies?

智能手机使用哪一种存储技术？

Flash memory card

- Solid-state storage devices are different than hard disks in that they .

固态存储设备与硬盘的不同之处在于它们。

have no moving parts

- This allows you to upload and store documents on the Internet.

这允许您在 Internet 上上传和存储文档。

Cloud storage

- Media are the actual physical material that holds the data and programs.

媒体是保存数据和程序的实际物理材料。

true

- Secondary storage provides temporary or volatile storage.

二级存储提供临时或易失性存储。

false

- A smoke particle, fingerprint, dust, or human hair could cause a hard disk

head crash.

烟雾颗粒、指纹、灰尘或头发都可能导致硬盘磁头崩溃。

true

- External hard drives are used primarily to complement an internal hard disk.

外接硬盘主要用于补充内部硬盘。

true

- Disk caching uses a combination of hardware and software.

磁盘缓存结合使用硬件和软件。

true

- RAID connects several inexpensive hard-disk drives to one another.

RAID 将几个便宜的硬盘驱动器连接到一起。

true

- In order to access your documents and files that are stored in the cloud you must have an Internet connection.

为了访问存储在云中的文档和文件，你必须有互联网连接。

true

- File servers are appropriate devices to use in an enterprise storage system.

文件服务器是适用于企业存储系统的设备。

true

- [Access] speed is the measure of the amount of time required by a storage device to retrieve data and programs.

[访问]速度是存储设备检索数据和程序所需时间的度量。

- Disk [caching] improves hard-disk performance by anticipating data needs and reducing time to access data from secondary storage.

磁盘[缓存]通过预测数据需求和减少从二级存储访问数据的时间来提高硬盘性能。

- File [decompression] expands compressed files.

文件 [解压]展开压缩文件

- USB drives are also known as [flash] drives.

USB 驱动器也被称为[闪存]驱动器。

- Due to its expanded storage capability, fast access speed, and high reliability, [RAID] is often used by Internet servers and large organizations.

[RAID]由于其扩展的存储能力、快速的访问速度和高可靠性，经常被互联网服务器和大型组织使用。

- Typically a personal computer uses a (n) [internal] hard disk to store the operating system and software applications.

一般来说，个人计算机使用一个内部硬盘来存储操作系统和软件应用程序。

- With [cloud], or online storage, you can access your files from any computer with an Internet connection.

有了[云]或在线存储，你可以从任何连接互联网的计算机上访问你的文件。

- Chapter 7 Online Test(Secondary Storage)

- On a hard disk each track is divided into invisible wedge shaped sections called ____.

在硬盘上，每个磁道被分成看不见的楔形部分，称为____。

sectors

- The hard disk's disk surface is scratched and some or all of the data is destroyed when there is a ____.

当硬盘出现故障时，硬盘表面会被划伤，部分或全部数据被破坏。

head crash

- The process of accessing information from secondary storage devices is referred to as ____.

从二级存储设备访问信息的过程称为____

reading

- Which of the following measures the amount of time required by the storage device to retrieve data and programs?

以下哪项可以测量存储设备检索数据和程序所需的时间？

Access speed

- One way to improve the storage capacity of a hard disk is to ____.

提高硬盘存储容量的一种方法是____。

use file compression

- This process improves system performance by acting as a temporary high-speed holding area between a secondary storage device and the CPU.

该进程在二级存储设备和 CPU 之间充当一个临时的高速保持区，从而提高系统性能。

Disk caching

- Which of these is not an example of solid state storage?

下面哪个不是固态存储的例子？

Digital versatile disc

- Optical discs use these to represent data.

盘用这些来表示数据。

Pits and lands

- What is it called when servers on the Internet supply applications as a service, rather than a product?

当互联网上的服务器将应用程序作为服务而不是产品提供时，它被称为什么？

Cloud computing

- Which of these is not a mass storage device?

哪个不是大容量存储设备？

RAID system

File server

Optical disc ✓

Organizational cloud storage

- The process of saving information to a secondary storage device is referred to as embedding.

将信息保存到二级存储设备的过程称为嵌入。

false

- Hard disks use tracks, sectors, and cylinders to store and organize files.

硬盘使用磁道、扇区和柱面来存储和组织文件。

true

- A hard crash typically occurs on an optical disk when it makes contact with particles on its surface.

当光盘与表面的粒子接触时，通常会发生硬碰撞。

false

- External hard disks use different technology than internal hard disks.

外置硬盘与内置硬盘采用不同的技术。

false

- Disk caching improves hard-disk performance by compressing files.

磁盘缓存通过压缩文件来提高硬盘性能。

false

- File compression is only available on hard disk drives.

文件压缩只能在硬盘驱动器上使用。

false

- Mass storage devices are specialized high-capacity secondary storage devices designed to store large amounts of data for an organization.

大容量存储设备是专门为组织存储大量数据而设计的高容量二级存储设备。

true

- If you have used Google Docs to create a word processing document, you have used cloud computing.

如果您曾经使用谷歌 Docs 创建过字处理文档，那么您已经使用过云计算。

true

- File [compression] programs scan files for ways to reduce the amount of required storage.

文件[压缩]程序扫描文件以减少所需的存储空间。

- A(n) [head crash] occurs when a read/write head makes contact with the hard disk's surface or with particles on its surface.

当[读写头]接触到硬盘表面或硬盘表面上的颗粒时，就会发生撞头现象。

- Tracks are divided into invisible wedge-shaped sections called [sectors] .

轨道被划分为无形的楔形部分，称为[扇区]。

- A good way to share files with others is to use a(n) [cloud] , or online, storage service.

与他人共享文件的一个好方法是使用(n)[云]，或在线存储服务。

课后习题

- 选择

1. RAM is sometimes referred to as:

RAM 有时被称为:

a. primary storage

2. The actual physical material that holds the data and programs.

保存数据和程序的实际物理材料。

capacity

3. Measures how tightly the magnetic charges can be packed next to one another on the disk.

测量磁盘上相邻磁荷的紧密程度。

density

4. When a read-write head makes contact with the hard disk's surface, it causes a head:

当读写头接触到硬盘表面时，会产生如下现象：

crash

5. This hard-disk performance enhancement anticipates data needs.

这种硬盘性能的提高预期了数据需求。

disk caching

6. This type of storage uses pits and lands to represent 1s and 0s.

这种存储方式使用坑和地来表示 1 和 0

optical

7. DVD stands for:

DVD 代表：

digital versatile disc

数字通用光盘

8. USB drives are also known as:

USB 驱动器也被称为

flash drives

9. An organizational strategy to promote efficient and safe use of data across the networks.

一个组织策略，以促进有效和安全的使用数据的网络。

enterprise storage system

10. Specialized high-capacity secondary storage devices designed to meet organizational demands.

专为满足组织需求而设计的高容量二级存储设备。

mass storage devices

• 匹配

1. Provides permanent or nonvolatile storage.

提供永久或非易失性存储。

e. secondary storage

2. Hardware that reads data and programs from storage media.

从存储介质读取数据和程序的硬件。

i. storage devices

3. Concentric rings on a hard-disk platter.

硬盘盘片上的同心环。

j. tracks

4. Each track is divided into invisible wedge-shaped sections called _ .

每条轨道都被分成看不见的楔形段，称为_。

f. sectors

5. Increases storage capacity by reducing the amount of space required to store data and programs.

通过减少存储数据和程序所需的空间量来增加存储容量。

b. file compression

6. Optical disc most common on today' s personal computers.

光盘在今天的个人电脑上最常见。

a. DVD

7. The next generation of optical discs.

下一代的光盘。

c. hi-def

8. Similar to internal hard-disk drives except they use solid-state memory.

类似于内部硬盘驱动器，只不过它们使用的是固态存储器。

g. solid-state drives

9. Mass storage device widely used for home and small business storage.

大容量存储设备广泛应用于家庭和小型企业存储。

d. network attached storage

10. Architecture to link remote storage devices to computers such that the

devices are as available as locally attached drives.

将远程存储设备连接到计算机的架构，使这些设备与本地连接的驱动器一样可用。

h. storage area network

第 8 章 COMMUNICATIONS AND NETWORKS

通讯和网络

COMMUNICATIONS 通讯

Communications is the process of sharing data, programs, and information between two or more computers. Applications include e-mail, texting, videoconferencing, and electronic commerce.

通信是在两台或两台以上的计算机之间共享数据、程序和信息的过程。应用包括电子邮件、短信、视频会议和电子商务。

Connectivity 连接

Connectivity is a concept related to using computer networks to link people and resources. You can link or connect to large computers and the Internet, providing access to extensive information resources.

连接是一个与使用计算机网络连接人和资源有关的概念。您可以连接或连接到大型计算机和互联网，从而获得广泛的信息资源。

The Wireless Revolution 无线革命

Mobile devices like smartphones and tablets have brought dramatic changes in connectivity and communications. These wireless devices are becoming widely used for computer communication.

像智能手机和平板电脑这样的移动设备给连接和通信带来了巨大的变化。这些无线设备正被广泛用于计算机通信

Communication Systems 通信系统

Communication systems transmit data from one location to another. There are four basic elements:

通信系统将数据从一个地点传送到另一个地点。有四个基本要素：

- Sending and receiving devices originate or accept messages.

发送和接收设备产生或接收消息。

- Connection devices act as an interface between sending and receiving devices and the communication channel.

连接设备充当发送、接收设备和通信通道之间的接口。

- Data transmission specifications are rules and procedures for sending and receiving data.

数据传输规范是发送和接收数据的规则和程序。

- Communication channel is the actual connecting or transmission medium for messages.

通信信道是消息的实际连接或传输介质。

COMMUNICATION CHANNELS 通讯频道

Communication channels carry data from one computer to another.

通信通道将数据从一台计算机传送到另一台计算机。

Physical Connections 物理连接

Physical connections use a solid medium to connect sending and receiving devices. Connections include twisted pair cable (telephone lines and Ethernet cables), coaxial cable, and fiber-optic cable.

物理连接使用固体介质来连接发送和接收设备。连接包括双绞线(电话线和以太网电缆)，同轴电缆和光纤电缆。

Wireless Connections 无线连接

Wireless connections do not use a solid substance to connect devices. Most use radio waves.

无线连接不使用固体物质来连接设备。大多数使用无线电波。

- Bluetooth—transmits data over short distances; widely used for a variety of wireless devices.

蓝牙可以短距离传输数据;广泛用于各种无线设备。

- Wi-Fi (wireless fidelity)—uses high-frequency radio signals; most home and business wireless networks use Wi-Fi.

Wi-Fi(无线保真)-使用高频无线电信号;大多数家庭和商业无线网络使用 Wi-Fi。

- Microwave—line-of-sight communication; used to send data between buildings;

longer distances require microwave stations.

微波视线沟通;用于在建筑物之间发送数据;较长的距离需要微波站。

- WiMax (Worldwide Interoperability for Microwave Access)—extends the range of Wi-Fi networks using microwave connections.

WiMax(微波访问全球互操作性)-使用微波连接扩展 Wi-Fi 网络的范围

- Cellular—uses cell towers to send and receive data within relatively small geographic regions or cells.

蜂窝网络——使用蜂窝基站在相对较小的地理区域或蜂窝内发送和接收数据。

- Satellite—uses microwave relay stations; GPS (global positioning system) tracks geographic locations.

卫星微波中继站;GPS(全球定位系统)跟踪地理位置。

- Infrared—uses light waves over a short distance; line-of-sight communication.

红外线——利用短距离的光波;视距通信

CONNECTION DEVICES 连接设备

Many communication systems use standard telephone lines and analog signals.

Computers use digital signals.

许多通信系统使用标准电话线和模拟信号。计算机使用数字信号。

Modems 调制解调器

Modems modulate and demodulate. Transfer rate is measured in megabits per second.

Three types are DSL, cable, and wireless (wireless wide area network, WWAN).

调制解调器调制和解调。传输速率以每秒兆比特为单位。三种类型是 DSL、电缆和无线(无线广域网, WWAN)。

Connection Service 连接服务

T1, T3 (DS3), and OC (optical carrier) lines provide support for very-high-speed, all-digital transmission for large corporations. More affordable technologies include dial-up, DSL(digital subscriber line), ADSL (widely used), cable, fiberoptic service (FiOS), satellite, and cellular services. 4G(fourth-generation mobile telecommunications) use LTE(Long Term Evolution) connections. 5G (fifth-generation mobile telecommunications) are being developed, with speeds that

rival home Internet connections; expected available by 2020.

T1、T3 (DS3) 和 OC(光载波) 线路为大公司提供非常高速的全数字传输支持。更实惠的技术包括拨号、DSL(数字用户线路)、ADSL(广泛使用)、电缆、光纤服务(FiOS)、卫星和蜂窝服务。4G(第四代移动通信) 使用 LTE(长期演进) 连接。5G(第五代移动通信) 正在开发中, 其速度可与家庭互联网连接媲美; 预计到 2020 年。

DATA TRANSMISSION 数据传输

Bandwidth measures a communication channel's width or capacity. Four bandwidths are voiceband (low bandwidth), medium band, broadband (high-capacity transmissions), and baseband. Protocols are rules for exchanging data. Widely used Internet protocols include https and TCP/IP. IP addresses (Internet protocol addresses) are unique numeric Internet addresses. DNS (domain name server) converts text-based addresses to and from numeric IP addresses. Packets are small parts of messages. 带宽测量通信信道的宽度或容量。四种带宽分别是: 声带(低带宽)、中带、宽带(大容量传输) 和基带。协议是交换数据的规则。目前广泛使用的互联网协议包括 https、TCP/IP 等。IP 地址(互联网协议地址) 是唯一的数字互联网地址。DNS(域名服务器) 将基于文本的地址转换为数字的 IP 地址。包是消息的一小部分。

NETWORKS 网络

Computer networks connect two or more computers. Some specialized network terms include

计算机网络连接两台或多台计算机。一些专门的网络术语包括

- Node—any device connected to a network.
节点-连接到网络的任何设备。
- Client—node requesting resources.
客户机节点请求的资源。
- Server—node providing resources.
服务器节点提供资源。
- Directory server—specialized node that manages resources.
管理资源的目录服务器专用节点。
- Host—any computer system that provides access to its resources over a network.

主机——通过网络提供对其资源的访问的任何计算机系统。

- Router—a node that forwards data packets from one network to another network.

路由器—将数据包从一个网络转发到另一个网络的节点。

- Switch—node that coordinates direct flow of data between other nodes. Hub is an older device that directed flow to all nodes.

交换节点，协调其他节点之间的直接数据流。Hub 是一种较老的设备，它将流定向到所有节点。

- NIC (network interface card)—LAN adapter card for connecting to a network.

NIC(网络接口卡)—用于连接网络的 LAN 适配器。

- NOS (network operating system)—controls and coordinates network operations.

NOS(网络操作系统)——控制和协调网络操作。

- Network administrator—network specialist responsible for network operations.

网络管理员-网络专家，负责网络运营

NETWORK TYPES 网络类型

Networks can be citywide or even international, using both wired and wireless connections.

网络可以是全市的，甚至是国际的，使用有线和无线连接。

- Local area networks (LANs) connect nearby devices. Network gateways connect networks to one another. Ethernet is a LAN standard. These LANs are called Ethernet LANs.

局域网(Local area network)连接附近的设备。网络网关之间的网络连接。以太网是局域网的标准。这些局域网被称为以太网局域网。

- Home networks are LANs used in homes.

家庭网络是在家庭中使用的局域网

- Wireless LANs (WLANs) use a wireless access point (base station) as a hub. Hotspots provide Internet access in public places.

无线局域网(wlan)使用无线接入点(基站)作为集线器。热点在公共场所提供互联网接入。

- Personal area networks (PANs) are wireless networks for PDAs, cell phones, and other wireless devices.

个人区域网络 (pan) 是用于 pda、手机和其他无线设备的无线网络。

- Metropolitan area networks (MANs) link office buildings within a city, spanning up to 100 miles.

都市区域网络 (MANs) 将城市内的办公大楼连接起来，跨度可达 100 英里。

- Wide area networks (WANs) are the largest type. They span states and countries or form worldwide networks. The Internet is the largest wide area network in the world.

广域网 (wan) 是其中最大的一种。它们跨越各州和国家，或形成全球网络。互联网是世界上最大的广域网。

NETWORK ARCHITECTURE 网络体系结构

Network architecture describes how networks are arranged and resources are shared.

网络架构描述了如何安排网络和共享资源。

Topologies 拓扑结构

A network's topology describes the physical arrangement of a network.

网络拓扑描述了网络的物理布局

- Bus network—each device is connected to a common cable called a bus or backbone.
总线网络——每个设备都连接到一个被称为总线或主干的公共电缆上。
- Ring network—each device is connected to two other devices, forming a ring.
环网——每个设备连接到另外两个设备，形成一个环。
- Star network—each device is connected directly to a central network switch; most common type today.

星型网络——每个设备都直接连接到一个中央网络交换机;今天最常见的一种。

- Tree (hierarchical) network—a central node is connected to subordinate nodes forming a treelike structure.

树形(层次)网络——中心节点与下级节点连接，形成树形结构。

- Mesh network—newest; each node has two or more connecting nodes.
网最新的;每个节点有两个或多个连接节点。

Strategies 策略

Every network has a strategy, or way of sharing information and resources. Common

network strategies include client/server and peer-to-peer.

每个网络都有共享信息和资源的策略或方式。常见的网络策略包括客户机/服务器和点对点。

- Client/server (hierarchical) network—central computers coordinate and supply services to other nodes; based on specialization of nodes; widely used on the Internet; able to handle very large networks efficiently; powerful network management software available.

客户/服务器(分层)网络中心计算机协调并向其他节点提供服务;基于节点的专业化;在互联网上广泛使用;能够有效地处理非常大的网络;强大的网络管理软件。

- Peer-to-peer network—nodes have equal authority and act as both clients and servers; widely used to share games, movies, and music over the Internet; easy to set up and use; lacks security controls.

点对点网络节点具有同等的权限,同时充当客户端和服务端;广泛用于在互联网上分享游戏、电影和音乐;易于设置和使用;缺乏安全控制。

ORGANIZATIONAL NETWORKS 组织网络

Internet Technologies 互联网技术

Internet technologies support effective communication using intranets and extranets.

因特网技术支持使用内部网和外部网进行有效的通信。

- Intranet—private network within an organization; uses browsers, websites, and web pages. Typical applications include electronic telephone directories, e-mail addresses, employee benefit information, internal job openings, and much more.

内联网—组织内部的专用网络;使用浏览器、网站和网页。典型的应用包括电子电话簿、电子邮件地址、员工福利信息、内部职位空缺等等。

- Extranet—like intranet except connects more than one organization; typically allows suppliers and others limited access to their networks.

类似于外联网的内部网除了连接多个组织外;通常允许供应商和其他人有限地访问其网络

Network Security 网络安全

Three technologies commonly used to ensure network security are firewalls, intrusion detection systems, and virtual private networks.

防火墙、入侵检测和虚拟专用网是保证网络安全的常用技术。

- Firewall—controls access; all communications pass through proxy server.

防火墙—控制访问;所有通信都通过代理服务器进行。

- Intrusion detection systems (IDS)—work with firewalls; use sophisticated statistical techniques to recognize and disable network attacks.

入侵检测系统(IDS) -与防火墙一起工作;使用复杂的统计技术来识别和禁用网络攻击。

- Virtual private network (VPN)—creates secure private connection between remote user and organization' s internal network.

虚拟私有网络(VPN)—在远程用户和组织内部网络之间创建安全的私有连接。

CAREERS in IT 其中的职业

Network administrators manage a company' s LAN and WAN networks. Bachelor' s or associate' s degree in computer science, computer technology, or information systems and practical networking experience required. Salary range is \$46,000 to \$66,000.

网络管理员管理公司的局域网和广域网。本科或大专学历, 计算机科学, 计算机技术, 或信息系统和实际网络经验。工资范围是 46000 美元到 66000 美元。

雨课堂

- Chapter 8 Online Test(Communications and Networks)

• In a communication system, which among the following originate and accept messages in the form of data, information, and/or instructions?

在通信系统中, 下列哪一种系统以数据、信息和/或指令的形式产生和接收消息?

Sending and receiving devices

- Which physical connection is the fastest?

哪个物理连接是最快的?

Fiber-optic cable

- Which among the following is considered a line-of-sight communication medium?

以下哪一种被认为是视距通信媒介?

Microwave

- Using this type of communications channel, several users can simultaneously use a single connection for high-speed data transfer.

使用这种通信通道，多个用户可以同时使用一个连接进行高速数据传输。

Broadband

- ___ is the process of breaking down information sent or transmitted across the Internet into small parts.

___是将通过互联网发送或传输的信息分解为小部分的过程。

Packetization

- The ___ is the central node that coordinates the flow of data by sending messages directly between the sender and receiver nodes.

___是通过在发送方和接收方节点之间直接发送消息来协调数据流的中心节点。

switch

- This device enables connectivity between two LANS or a LAN and a larger network.

该设备可以实现两个局域网或一个局域网和一个更大的网络之间的连接。

Network gateway

- A network topology in which the central node is connected to two or more subordinate nodes that, in turn, are connected to other subordinate nodes.

一种网络拓扑结构，其中中心节点连接到两个或多个从属节点，从属节点依次连接到其他从属节点。

Tree

- Organizations use the following to allow suppliers and others limited access to their networks.

组织使用以下方法来允许供应商和其他人有限地访问其网络。

Extranets

- All communications between a company's internal networks and the outside world pass through this server.

公司内部网络与外部世界之间的所有通信都通过该服务器。

Proxy

- TCP/IP protocol involves identifying, sending, and receiving devices and breaking information into small parts for transmission across the Internet.

TCP/IP 协议包括识别、发送和接收设备，并将信息分解成小的部分在互联网上传输。

true

- Network operating systems (NOS) control and coordinate the activities of all computers and other devices on a network.

网络操作系统(NOS)控制和协调网络上所有计算机和其他设备的活动。

true

- The most common standard in LANs is called Ethernet.

局域网中最常见的标准是以太网。

true

- The client/server network strategy can handle very large networks efficiently.

客户机/服务器网络策略可以有效地处理非常大的网络。

true

- In a peer-to-peer network, only the central host computer supplies the resources, the other computers request resources from this host.

在点对点网络中，只有中央主机提供资源，其他计算机向该主机请求资源。

false

- Intrusion detection systems can recognize signs of a network attack and disable access before an intruder can do damage.

入侵检测系统可以识别网络攻击的迹象，并在入侵者造成破坏之前禁用访问。

true

- A network [gateway] is a device that allows one LAN to be linked to other LANs or to larger networks.

网络[网关]是允许一个局域网连接到其他局域网或更大的网络的设备。

- A LAN set up for a household's personal use is called a(n) [home] network.

为家庭个人使用而建立的局域网称为家庭网络。

- [Wide] area networks are countrywide and worldwide networks.

广域网络是指全国范围和全球范围的网络。

- Chapter 8 Test(Communications and Networks)

- One of the most dramatic changes in connectivity and communications in the past few years has been__ .

在过去的几年里，连接和通信方面最显著的变化之一是……

widespread use of mobile devices with wireless Internet connectivity

广泛使用具有无线互联网连接的移动设备

- In a communication system, which among the following convert messages into packets that can travel across the communication channel?

在通信系统中，下列哪一种方法将消息转换成可以通过通信通道传输的包？

Connection devices

- Microwave communication uses high-frequency _ .

微波通信使用高频_。

radio waves

- The capacity of a communication channel is measured in__ .

通信通道的容量以__为度量单位。

bandwidth

- Every computer on the Internet has a unique numeric address called a(n)__ .

互联网上的每台计算机都有一个唯一的数字地址，称为__。

IP address

- The essential features of this include identifying, sending, and receiving devices and breaking information into small parts for transmission across the Internet.

它的基本特征包括识别、发送和接收设备，并将信息分成小部分在互联网上传输。

TCP/IP

- In a computer network, this node that requests and uses resources available from other nodes.

在计算机网络中，向其他节点请求和使用可用资源的节点。

Client

- A wireless access point that provides Internet access in a public place such as a coffeeshop, library, bookstore, or university.

在咖啡馆、图书馆、书店或大学等公共场所提供互联网接入的无线接入点。

Hotspot

- This type of network strategy uses central servers to coordinate and supply

services to other nodes on the network.

这种类型的网络策略使用中心服务器来协调网络上的其他节点并向其提供服务。

Client/server

- Remote users can connect to an organization's network through this type of secure private connection.

远程用户可以通过这种安全的私有连接连接到组织的网络。

VPN

- Fiber-optic cable transmits data as pulses of light through tiny tubes of glass.

光纤电缆通过微小的玻璃管以光脉冲的形式传输数据。

true

- Packetization refers to breaking information into small parts.

分组化是指将信息分解成小的部分。

true

- WANs are widely used by organizations to link personal computers and to share printers and other resources.

广域网被许多组织广泛用于连接个人电脑、共享打印机和其他资源。

false

- Network architecture describes how a computer network is configured and what topologies and strategies are employed.

网络架构描述了计算机网络是如何配置的，以及采用什么拓扑和策略。

true

- With respect to peer-to-peer network, there is an abundant amount of powerful management software that monitors the network activities.

对于点对点网络，有大量功能强大的管理软件来监控网络活动。

false

- A firewall is a specialized technology designed to protect an organization's network against external threats.

防火墙是一种专门用于保护组织网络免受外部威胁的技术。

true

- The actual connecting or transmission medium that carries the message in a

communication system is called the communication [channel].

在通信系统中承载信息的实际连接或传输介质称为通信[信道]。

- [Fiber optic] cable transmits data as pulses of light through tiny tubes of glass.

[光缆]通过微小的玻璃管以光脉冲的形式传输数据。

- A (n) [local] area network is a network where the nodes are in close physical proximity to each other.

一个[本地]区域网络是一个节点在物理上彼此接近的网络。

课后习题

- 选择

1. The concept related to using computer networks to link people and resources.

这个概念是指使用计算机网络来连接人和资源。

connectivity

2. A high-frequency transmission cable that delivers television signals as well as connects computers in a network.

一种高频传输电缆，既能传输电视信号，又能连接网络中的计算机。

coaxial

3. A short-range radio communication standard that transmits data over short distances of up to approximately 33 feet.

一种短程无线电通信标准，可在近 33 英尺的短距离内传输数据。

Bluetooth

4. The speed with which a modem transmits data is called its:

调制解调器传输数据的速度称为：

transfer rate

5. The bandwidth typically used for DSL, cable, and satellite connections to the Internet.

通常用于 DSL、电缆和卫星连接到 Internet 的带宽。

broadband

6. Every computer on the Internet has a unique numeric address called a(n):

互联网上的每台计算机都有一个唯一的数字地址，称为：

IP address

7. Sometimes referred to as a LAN adapter, these expansion cards connect a computer to a network.

有时被称为局域网适配器，这些扩展卡将计算机连接到网络。

NIC

8. A device that allows one LAN to be linked to other LANs or to larger networks.

一种允许一个局域网连接到其他局域网或更大的网络的设备。

network gateway

9. Typically using Wi-Fi technology, these wireless access points are available from public places such as coffee shops, libraries, bookstores, colleges, and universities.

这些无线接入点通常使用 Wi-Fi 技术，可在咖啡馆、图书馆、书店、学院和大学等公共场所使用。

hotspots

10. Bus, ring, star, tree, and mesh are five types of network:

总线、环形、星形、树形和网状五种类型的网络：

topologies

- 匹配

1. Type of network topology in which each device is connected to a common cable called a backbone.

网络拓扑的一种类型，其中每个设备都连接到一根称为骨干网的普通电缆上

b. bus

2. A widely used Internet protocol.

种广泛使用的互联网协议。

i. TCP/IP

3. Uses high-frequency radio waves.

使用高频无线电波。

d. microwave

4. Signals that are continuous electronic waves.

连续的电子信号。

a. analog

5. Rules for exchanging data between computers.

计算机之间交换数据的规则。

h. protocols

6. Any device that is connected to a network.

任何连接到网络的设备。

f. node

7. A computer specialist responsible for efficient network operations and implementation of new networks.

负责有效的网络操作和新网络的实施的计算机专家。

e. network administrator

8. This network, also known as a hierarchical network, is often used to share corporatewide data.

这个网络，也被称为层次网络，通常用于共享全公司的数据。

j. tree

9. In this network, nodes have equal authority and can act as both clients and servers.

在这个网络中，节点具有同等的权限，可以同时充当客户端和服务端。

g. peer-to-peer

10. Work with firewalls to protect an organization's network.

使用防火墙来保护组织的网络。

c. intrusion detection systems

第9章 PRIVACY, SECURITY, AND ETHICS

隐私、安全和道德

PRIVACY 隐私

Privacy concerns the collection and use of data about individuals. There are three primary privacy issues: accuracy(who is responsible to ensure data is correct), property(who owns data), access (who controls access to data).

隐私涉及个人数据的收集和使用。有三个主要的隐私问题:准确性(谁负责确保数据的正确性)、属性(谁拥有数据)、访问(谁控制对数据的访问)。

Large Databases 大数据库

Large organizations are constantly compiling information about us. This ever-growing volume of data is often referred to as big data. Information resellers (information brokers) collect and sell personal data. Electronic profiles are compiled from databases to provide highly detailed and personalized descriptions of individuals.

大型组织不断收集我们的信息。这种不断增长的数据量通常被称为大数据。信息转售商(信息经纪人)收集和出售个人数据。电子档案从数据库中编译而来,提供对个人非常详细和个性化的描述。

Mistaken identity occurs when an electronic profile of one person is switched with another. The Freedom of Information Act entitles individuals access to governmental records relating to them.

当一个人的电子档案与另一个人交换时,就会发生错误的身份。《信息自由法》使个人有权查阅与他们有关的政府记录。

Private Networks 专用网

Many organizations monitor employee e-mail and computer files using special software called employee monitoring software.

许多组织使用一种叫做员工监控软件的特殊软件来监控员工的电子邮件和电脑文件。

The Internet and the Web 互联网和网络

Many people believe that, while using the web, little can be done to invade their privacy. This is called the illusion of anonymity.

许多人认为,在使用网络时,几乎没有什么可以侵犯他们的隐私。这被称为匿名错觉。

PRIVACY

Information stored by browsers includes history files (record sites visited) and temporary Internet files or browser cache (contain website content and display instructions). Cookies store and track information. Privacy mode (Private Browsing) ensures that your browsing activity is not recorded.

浏览器存储的信息包括历史文件(记录访问过的网站)和 Internet 临时文件或浏览器缓存(包含网站内容和显示说明)。cookie 存储和跟踪信息。隐私模式(隐私浏览)确保您的浏览活动不被记录。

Spyware secretly records and reports Internet activities. Keyloggers (one type of computer monitoring software) records every activity and keystroke. Antispyware (spy removal programs) detects and removes various privacy threats.

间谍软件秘密记录和报告互联网活动。键盘记录器(一种计算机监控软件)记录每一个活动和击键。反间谍软件(间谍清除程序)检测和删除各种隐私威胁。

Online Identity 网络身份

Many people post personal information and sometimes intimate details of their lives without considering the consequences. This creates an online identity. With the archiving and search features of the web, this identity is indefinitely available to anyone who cares to look for it.

许多人在不考虑后果的情况下发布个人信息,有时甚至是他们生活的私密细节。这就产生了一个在线身份。有了网络的存档和搜索功能,这个身份对任何想要寻找它的人来说都是无限可用的。

Major Laws on Privacy 有关私隐的主要法例

The Gramm-Leach-Bliley Act protects personal financial information; the Health Insurance Portability and Accountability Act (HIPAA) protects medical records; and the Family Educational Rights and Privacy Act (FERPA) restricts disclosure of educational records.

《金融金融法案》保护个人财务信息;《健康保险携带与责任法案》(HIPAA)保护医疗记录;《家庭教育权利和隐私法》(FERPA)限制了教育记录的公开。

SECURITY 安全性

Computer security focuses on protecting information, hardware, and software from unauthorized use as well as preventing damage from intrusions, sabotage, and natural disasters.

计算机安全的重点是保护信息、硬件和软件不被未经授权的使用,以及防止入侵、破坏和自然灾害造成的损害。

Cybercrime 网络犯罪

Cybercrime (computer crime) is a criminal offense involving a computer or network.

网络犯罪(计算机犯罪)是涉及计算机或网络的刑事犯罪。

- Identity theft—illegal assumption of someone's identity for the purposes of economic gain.

身份盗窃——以经济利益为目的非法获取某人的身份。

- Internet scams—use Internet to create financial and legal problems for people.

网络诈骗——利用互联网为人们制造财务和法律问题。

- Data manipulation—unauthorized access of a network to copy files.

数据操纵，非法访问网络复制文件。

- Ransomware—malicious software that encrypts computer data and ransoms owner for access.

勒索软件-恶意软件，加密计算机数据和赎金的所有者访问。

- Denial of service (DoS) attack—stops or slows computer network by repeatedly requesting service. Distributed denial of service (DDoS) coordinates several computers.

拒绝服务(DoS)攻击通过反复请求服务来停止或减慢计算机网络的速度。分布式拒绝服务(DDoS)协调多台计算机。

Social Engineering 社会工程

Social engineering practice of manipulating people to divulge private data. Phishing attempts to trick people into believing a fake website is real.

操纵人们泄露私人数据的社会工程实践。网络钓鱼试图欺骗人们相信一个虚假网站是真实的。

Malicious software 恶意软件

Crackers create and distribute malware (malicious software). Three most common types are

破坏者创建和分发恶意软件(恶意软件)。三种最常见的类型是

- Viruses migrate through networks and operating systems; can alter and/or delete files.

病毒通过网络和操作系统迁移;可以修改和/或删除文件。

- Worms repeatedly replicate themselves, clogging computers and networks.

蠕虫反复复制自己，堵塞了计算机和计算机。

- Trojan horses appear harmless but contain malicious programs.

特洛伊木马看似无害，但包含恶意程序。

Malicious hardware 恶意的硬件

Cyber criminals can use computer hardware to steal information. Three types of malicious hardware:

网络罪犯可以利用电脑硬件窃取信息。三种恶意硬件：

- Zombies—remotely controlled computers used for malicious purposes; a collection is known as a botnet (robot network).

僵尸——用于恶意目的的远程控制计算机；集合被称为僵尸网络（机器人网络）。

- Rogue Wi-Fi hotspots—network that appears to be a legitimate free Wi-Fi hot. Once connected, any input by users is captured.

流氓 Wi-Fi 热点-网络似乎是一个合法的免费 Wi-Fi 热点。连接之后，将捕获用户的任何输入。

- Infected USB flash drives—USB drives that are often free that contain a virus.

受感染的 USB 闪存驱动器-通常是免费的，但包含病毒的 USB 驱动器。

Measures to Protect Computer Security 保护计算机安全的措施

Computer Fraud and Abuse Act makes it a crime for unauthorized persons to view data across state lines using computers; prohibits unauthorized use of computers owned by government or federally insured financial institution. Ways to protect computer security include limiting access, encrypting, and anticipating disasters.

《计算机欺诈与滥用法案》规定，未经授权的人使用计算机跨州查看数据是犯罪；禁止未经授权使用政府或联邦保险金融机构拥有的计算机。保护计算机安全的方法包括限制访问、加密和预测灾难

- Access can be restricted through biometric scanning devices and passwords (secret words or phrases; dictionary attacks use thousands of words to attempt to gain access; Picture Password uses series of gestures; facial recognition to allow access); security suites, firewalls, and password managers help with security tasks.

可以通过生物识别扫描设备和密码（秘密单词或短语；字典攻击使用数千个单词来试图获得访问权限；图片密码使用一系列手势；面部识别允许访问）；安全套件、防火墙和密码管理器有

助于执行安全任务。

- Encrypting is coding information to make it unreadable except to those who have the encryption key. Hypertext transfer protocol secure (https) requires browsers and websites to encrypt all messages. Virtual private networks (VPNs) encrypt connections between company networks and remote users. WPA2 (Wi-Fi Protected Access) is the most widely used wireless network encryption for home wireless networks. 加密就是对信息进行编码，使其只有拥有加密密钥的人才能读懂。超文本传输安全协议 (https) 要求浏览器和网站对所有信息进行加密。虚拟私有网络 (vpn) 对公司网络和远程用户之间的连接进行加密。WPA2 (Wi-Fi 保护接入) 是目前应用最广泛的家庭无线网络加密技术。
- Anticipating disasters involves physical security, data security, and disaster recovery plans.

灾难预测包括物理安全、数据安全和灾难恢复计划。

- Preventing data loss involves protecting data by screening job applicants, guarding passwords, and auditing and backing up data.

防止数据丢失包括求职者筛选、密码保护、数据审计和备份等。

ETHICS 行为准则

What do you suppose controls how computers can be used? You probably think first of laws. Of course, that is right, but technology is moving so fast that it is very difficult for our legal system to keep up. The essential element that controls how computers are used today is ethics.

你认为是什么控制计算机的使用？你可能首先想到的是法律。当然，这是对的，但技术发展如此之快，我们的法律系统很难跟上。当今控制计算机使用方式的基本因素是道德。

Ethics are standards of moral conduct. Computer ethics are guidelines for the morally acceptable use of computers in our society. We are all entitled to ethical treatment. This includes the right to keep personal information, such as credit ratings and medical histories, from getting into unauthorized hands.

伦理是道德行为的标准。计算机伦理是我们社会道德上可以接受的使用计算机的准则。我们都有权得到合乎道德的待遇。这包括保护个人信息 (如信用评级和医疗记录) 不落入未经授权的人之手的权利。

Copyright and Digital Rights Management 版权和数字版权管理

Copyright is a legal concept that gives content creators the right to control use and distribution of their work. Materials that can be copyrighted include paintings, books, music, films, and even video games.

版权是一个法律概念，它赋予内容创作者控制其作品的使用和传播的权利。可以被版权保护的材料包括绘画、书籍、音乐、电影，甚至电子游戏。

Software piracy is the unauthorized copying and distribution of software. The software industry loses over \$60 billion annually to software piracy. Two related topics are the Digital Millennium Copyright Act and digital rights management.

软件盗版是指未经授权复制和分发软件。软件行业每年因软件盗版损失超过 600 亿美元。两个相关的主题是数字千年版权法和数字权利管理。

- Digital Millennium Copyright Act makes it illegal to deactivate or disable antipiracy technologies, to copy, resale, or give away commercial programs or to sell or use programs or devices that are illegally copying software.

《数字千年版权法》规定禁用或禁用反盗版技术，复制、转售或赠送商业程序，或出售或使用非法复制软件的程序或设备都是非法的。

- Digital rights management (DRM) is a collection of technologies designed to prevent copyright violations. Typically, DRM is used to (1) control the number of devices that can access a given file and (2) limit the kinds of devices that can access a file.

数字版权管理 (DRM) 是旨在防止侵犯版权的一系列技术。通常，DRM 用于 (1) 控制可以访问给定文件的设备数量，(2) 限制可以访问文件的设备类型。

Today, many legal sources for digital media exist, including

今天，数字媒体存在许多合法来源，包括

- Television programs that can be watched online, often for free, on television-network-sponsored sites.

可在线观看的电视节目，通常在电视网络赞助的网站上免费观看。

- Sites like Pandora that allow listeners to enjoy music at no cost.

像 Pandora 这样的网站可以让听众免费享受音乐。

- Online stores that legally sell music and video content. A pioneer in this area

is Apple' s iTunes Music Store.

合法出售音乐和视频内容的在线商店。这一领域的先驱是苹果的 iTunes 音乐商店。

Cyberbullying 网上欺凌

Cyberbullying is the use of the Internet, smartphones, or other devices to send or post content intended to hurt or embarrass another person. It includes sending repeated unwanted e-mails to an individual, ganging up on victims in electronic forums, posting false statements, maliciously disclosing personal data, and sending any type of threatening or harassing communication. It can lead to criminal prosecution.

网络欺凌是指利用互联网、智能手机或其他设备发送或发布旨在伤害或羞辱他人的内容。它包括反复向个人发送不必要的电子邮件，在电子论坛上联合攻击受害者，发布虚假声明，恶意披露个人数据，以及发送任何类型的威胁或骚扰通信。这可能导致刑事起诉。

Plagiarism 剽窃

Plagiarism is the illegal and unethical representation of some other person' s work and ideas as your own without giving credit to the original source. Examples include cutting and pasting web content into a report or paper.

剽窃是非法和不道德地将他人的工作和想法作为自己的，而不注明原始来源。例如，将 web 内容剪切和粘贴到报告或纸张中。

Recognizing and catching plagiarists is relatively easy. For example, services such as Turnitin are dedicated to preventing Internet plagiarism. This service examines a paper' s content and compares it to a wide range of known public electronic documents including web page content. Exact duplication or paraphrasing is readily identified.

识别和抓住剽窃者相对容易。例如，像 Turnitin 这样的服务致力于防止互联网剽窃。这项服务检查一篇论文的内容，并将其与包括网页内容在内的广泛的已知公共电子文件进行比较。准确的重复或释义是很容易识别的。

CAREERS IN IT 其中的职业

IT security analysts are responsible for maintaining the security of a company' s network, systems, and data. Employers look for candidates with a bachelor' s or

associate's degree in information systems or computer science and network experience. Salary range is \$58,000 to \$86,000.

IT 安全分析师负责维护公司网络、系统和数据的安全。雇主希望应聘者具有信息系统或计算机科学学士或大专学历，并具有网络经验。薪资范围为 58,000 至 86,000 美元。

雨课堂

课后习题

• 选择

1. The three primary privacy issues are accuracy, property, and:

三个主要的隐私问题是准确性、财产和:

access

2. Highly detailed and personalized descriptions of individuals are electronic:

高度详细和个性化的个人描述是电子的:

a. cookies c. profiles

b. histories d. web bugs

3. Browsers store the locations of sites visited in a:

浏览器将访问过的站点的位置存储在:

a. history file c. tool bar

b. menu d. firewall

4. The browser mode that ensures your browsing activity is not recorded.

确保您的浏览活动不被记录的浏览器模式。

a. detect c. privacy

b. insert d. sleep

5. The information that people voluntarily post in social networking sites, blogs, and photo- and video-sharing sites is used to create their:

人们自愿在社交网站、博客、照片和视频分享网站上发布的信息被用来创建他们的:

a. access approval c. online identity

b. firewall d. phish

6. Computer criminals who create and distribute malicious programs.

创建和传播恶意程序的计算机罪犯。

- a. antispies c. cyber traders
- b. crackers d. identity thieves

7. Programs that come into a computer system disguised as something else are called:

伪装成其他程序进入计算机系统的程序被称为:

- a. Trojan horses c. web bugs
- b. viruses d. zombies

8. The use of the Internet, cell phones, or other devices to send or post content intended to hurt or embarrass another person is known as:

使用互联网、手机或其他设备发送或发布旨在伤害或使他人尴尬的内容被称为:

- a. cyberbullying c. social networking discrimination
- b. online harassment d. unethical communication

9. Special hardware and software used to control access to a corporation's private network is known as a(n):

用于控制对公司专用网络的访问的特殊硬件和软件被称为:

- a. antivirus program c. firewall
- b. communication gate d. spyware removal program

10. To prevent copyright violations, corporations often use:

为了防止侵犯版权, 公司经常使用:

- a. ACT c. VPN
- b. DRM d. WPA2

• 匹配

1. Privacy concern that relates to the responsibility to ensure correct data collection.

对隐私的关注涉及确保正确收集资料的责任。

a. accuracy

2. Individuals who collect and sell personal data.

收集和出售个人资料的个人。

e. information brokers

3. Small data files deposited on your hard disk from websites you have visited.

从您访问过的网站存储在您硬盘上的小数据文件。

c. cookies

4. Wide range of programs that secretly record and report an individual's activities on the Internet.

各种秘密记录和报告个人在互联网上活动的程序。

i. spyware

5. Malicious programs that damage or disrupt a computer system.

破坏或破坏计算机系统的恶意程序。

f. malware

6. Infected computers that can be remotely controlled.

被感染的电脑可以被远程控制。

j. zombies

7. Used by scammers to trick Internet users with official-looking websites.

骗子利用看起来像官方网站的网站来欺骗网民。

g. phishing

8. A type of scanning device such as fingerprint and iris (eye) scanner.

一种扫描设备，如指纹和虹膜(眼睛)扫描仪。

b. biometric

9. Process of coding information to make it unreadable except to those who have a key.

对信息进行编码，使其只有有钥匙的人才能读懂的过程。

d. encryption

10. An ethical issue relating to using another person's work and ideas as your own without giving credit to the original source.

将他人的作品和想法当作自己的而不注明出处的道德问题。

h. plagiarism

第 10 章 INFORMATION SYSTEMS

信息系统

ORGANIZATIONAL INFORMATION FLOW 组织的信息流动

Information flows in an organization through functional areas and between management levels.

信息在组织中通过职能区域和管理级别流动。

Functions 功能

Most organizations have separate departments to perform five functions:

大多数组织都有独立的部门来执行五项职能：

- Accounting—tracks all financial activities and generates periodic financial statements.

会计——跟踪所有财务活动并生成定期财务报表。

- Marketing—advertises, promotes, and sells the product (or service).

营销——做广告、推广和销售产品(或服务)。

- Production—makes the product (or service) using raw materials and people to turn out finished goods.

生产——利用原材料和人力生产成品，制造产品(或服务)。

- Human resources—finds and hires people; handles such matters as sick leave, retirement benefits, evaluation, compensation, and professional development.

人力资源——寻找和雇佣员工;处理病假、退休福利、评估、薪酬、职业发展等事宜。

- Research—conducts product research and development; monitors and troubleshoots new products.

研究-进行产品研究和开发;监控和解决新产品的问题。

Management Levels 管理层次

The three basic management levels are

三个基本的管理层次是

- Top level—concerned with long-range planning and forecasting.

高层负责长期计划和预测。

- Middle level—deals with control, planning, decision making, and implementing long-term goals.

中层处理控制、计划、决策和执行长期目标。

- Supervisors—control operational matters, monitor day-to-day events, and supervise workers.

主管负责控制操作事项，监控日常事件，并监督工人。

Information Flow 信息流

Information flows within an organization in different directions.

信息在组织内以不同的方向流动。

- For top-level managers, the information flow is primarily upward from within the organization and into the organization from the outside.

对于高层管理者来说，信息流主要是从组织内部向上，从外部进入组织。

- For middle-level managers, the information flow is horizontal and vertical within departments.

对于中层管理者来说，部门内部的信息流是横向和纵向的。

- For supervisors, the information flow is primarily vertical.

对于主管来说，信息流主要是垂直的。

INFORMATION SYSTEMS 信息系统

Transaction Processing Systems 交易处理系统

Transaction processing systems (TPSs), sometimes called data processing systems (DPSs), record day-to-day transactions. Accounting activities include sales order processing, accounts receivable, inventory and purchasing, accounts payable, payroll, and general ledger. General ledger is used to produce income statements and balance sheets.

事务处理系统(TPSs)，有时也被称为数据处理系统(DPSs)，记录日常事务。会计活动包括销售订单处理、应收账款、库存和采购、应付账款、工资和总账。总账用于编制损益表和资产负债表。

Management Information Systems 管理信息系统

Management information systems (MISs) produce predetermined reports (periodic, exception, demand).

管理信息系统(MISs)产生预定的报告(定期、例外、需求)。

Decision Support Systems 决策支持系统

Decision support systems (DSSs) focus on unanticipated questions; consist of the user, system software, data (internal and external) and decision models. Three types

are strategic, tactical, and operational. Group -decision -support systems (GDSS) support a team of decision makers.

决策支持系统(DSSs)关注意外问题;由用户、系统软件、数据(内部和外部)和决策模型组成。三种类型是战略、战术和作战。群体决策支持系统(GDSS)为决策者团队提供支持。

Executive Support Systems 主管支持系统

Executive support systems (ESSs) are similar to MIS or DSS but easier to use. ESSs are designed specifically for top-level decision makers.

执行支持系统(ess)类似于 MIS 或 DSS, 但更容易使用。ESSs 是专门为高层决策者设计的。

Other Information Systems 其它信息系统

Many other systems are designed to support information workers who create, distribute, and communicate information. Three such systems are

许多其他系统被设计用来支持创建、分发和交流信息的信息工作者。三个这样的系统是

- Office automation systems (OASs), which support data workers who are involved with distribution and communication of information. Project managers and videoconferencing systems are OASs.

办公自动化系统(OASs), 支持参与信息分发和通信的数据工作者。项目经理和视频会议系统都是 OASs。

- Knowledge work systems (KWSs), which support knowledge workers, who create information. Many engineers use computer-aided design/computer-aided manufacturing (CAD/CAM) systems.

知识工作系统(KWSs), 支持创建信息的知识工作者。许多工程师使用计算机辅助设计/计算机辅助制造(CAD/CAM)系统。

- Expert (knowledge-based) systems, which are a type of knowledge work system. They use knowledge bases to apply expert knowledge to specific user problems.

专家(知识型)系统, 是知识工作系统的一种。他们利用知识库将专家知识应用于特定的用户问题。

CAREERS in IT 其中的职业

Information systems managers oversee a variety of other computer professionals. Strong leadership and communication skills are required; experience as a consultant

and/or manager is desired. Salary range is \$62,000 to \$100,000.

信息系统经理监督其他各种各样的计算机专业人员。具有较强的领导和沟通能力;有顾问和/或经理经验者优先。工资范围为 6.2 万美元至 10 万美元。

雨课堂

课后习题

• 选择

1. Which of the basic organizational functions records all financial activity from billing customers to paying employees?

哪个基本的组织职能记录所有的财务活动，从帐单客户到支付员工？

a. accounting

2. What managerial level has information flow that is vertical, horizontal, and external?

什么样的管理级别拥有垂直、水平和外部的信息流？

a. top

3. Which computer-based information system uses data from TPS and analytical tools to support middle managers?

哪个基于计算机的信息系统使用来自 TPS 和分析工具的数据来支持中层管理？

a. ESS

4. Accounts payable refers to money the company owes its suppliers for materials and services it has:

应付账款是指公司因其拥有的材料和服务而欠供应商的钱：

d. received

5. What accounting activity keeps track of all summaries of all transactions?

什么会计活动记录所有交易的摘要？

b. general ledger

6. What accounting statement lists the overall financial condition of an organization?

什么会计报表列出了一个组织的整体财务状况？

a. balance sheet

7. What type of report is produced at regular intervals?

什么类型的报告会定期提出?

d. periodic

8. A DSS consists of four parts: user, system software, decision models, and:

决策支持系统由四部分组成:用户、系统软件、决策模型和:

b. data

9. What type of worker is involved with the distribution, communication, and creation of information?

什么类型的工人参与信息的分配、交流和创造?

c. information

10. What type of program is designed to schedule, plan, and control project resources?

设计什么类型的计划来安排、计划和控制项目资源?

c. project managers

• 匹配

1. Function that plans, prices, promotes, sells, and distributes the organization' s goods and services.

计划、定价、促销、销售和分配组织的商品和服务的职能。

c. marketing

2. Managerial level where information flow is vertical and horizontal.

信息是纵向和横向流动的管理层次。

d. middle

3. Computer-based information system that uses data from TPS to support middle-level managers.

基于计算机的信息系统, 使用来自 TPS 的数据来支持中层管理人员。

e. MIS

4. The accounting activity that records the customer requests for the company' s products or services is sales order _.

记录客户对公司产品或服务的要求的会计活动是销售订单_。

g. processing

5. The accounting activity concerned with calculating employee paychecks.

与计算员工工资有关的会计活动。

f. payroll

6. MIS produces this type of report.

管理信息系统产生这种类型的报告。

h. standardized

7. A type of report that calls attention to unusual events.

一种对不寻常事件引起注意的报告类型。

b. exception

8. Type of software that works behind the scenes to handle detailed operating procedures.

这种软件在幕后工作，处理详细的操作程序。

i. system

9. Type of worker who is involved with the distribution and communication of information.

参与信息分发和交流的工人类型。

a. data

10. Computer system that allows people located at various geographic locations to communicate and have in-person meetings.

允许位于不同地理位置的人们进行通信和面对面会议的计算机系统。

j. videoconferencing

第 11 章 DATABASES

数据库

DATA ORGANIZATION 数据结构

Data is organized by the following groups:

数据按以下几组组织：

- Character—the most basic logical element, consisting of individual numbers, letters, and special characters.

字符—最基本的逻辑元素，由单个数字、字母和特殊字符组成。

- Field—next level, consisting of a set of related characters, for example, a person's last name. A data field represents an attribute (description or characteristic) of some entity (person, place, thing, or object).

字段的下一级，由一组相关的字符组成，例如，一个人的姓氏。数据字段表示某些实体(人、地点、事物或对象)的属性(描述或特征)。

- Record—a collection of related fields; for example, a payroll record consisting of fields of data relating to one employee.

记录——相关字段的集合;例如，由与一个员工相关的数据字段组成的工资单记录。

- Table—a collection of related records; for example, a payroll table consisting of all the employee records.

表格-收集相关记录;例如，包含所有员工记录的工资单表。

- Database—an integrated collection of related tables; for example, a personnel database contains all related employee tables.

数据库——相关表的集成集合;例如，人事数据库包含所有相关的员工表。

Key Field 关键字域

A key field (primary key) is the field in a record that uniquely identifies each record.

键字段(主键)是记录中唯一标识每条记录的字段。

- Tables can be related (connected) to other tables by key fields.

表可以通过关键字段关联(连接)到其他表。

- Key fields in different files can be used to integrate the data in a database.

不同文件中的关键字段可以用于将数据集成到一个数据库中。

- Common key fields are employee ID numbers and driver's license numbers.

常见的关键字段是员工身份证号码和驾照号码。

Batch versus Real-Time Processing 批处理与实时处理

Traditionally, data is processed in one of two ways: batch or real-time processing.

传统上，数据处理有两种方式:批处理或实时处理。

- Batch processing—data is collected over time and then processed later all at one time (batched). For example, monthly credit card bills are typically created by

processing credit card purchases throughout the past month.

批量处理——随着时间的推移收集数据，然后在稍后一次处理所有数据(批量)。例如，每月的信用卡账单通常是通过处理过去一个月的信用卡购买而生成的。

- Real-time processing (online processing)—data is processed at the same time the transaction occurs; direct access storage devices make real-time processing possible. For example, a request for cash using an ATM machine initiates a verification of funds, approval or disapproval, disbursement of cash, and an update of the account balance.

实时处理(在线处理)——在事务发生的同时处理数据;直接访问存储设备使实时处理成为可能。例如，使用ATM机提取现金的请求将启动资金验证、批准或不批准、支付现金和更新帐户余额。

DATABASES 数据库

A database is a collection of integrated data—logically related files and records.

数据库是与数据逻辑相关的集成文件和记录的集合。

Need for Databases 需要数据库

Advantages of databases are sharing data, improved security, reduced data redundancy, and higher data integrity.

数据库的优点是共享数据、提高安全性、减少数据冗余和提高数据完整性。

Database Management 数据库管理

A database management system (DBMS) is the software for creating, modifying, and gaining access to the database. A DBMS consists of five subsystems:

数据库管理系统(DBMS)是用于创建、修改和获得对数据库的访问权的软件。DBMS由五个子系统组成:

- DBMS engine provides a bridge between logical and physical data views.

DBMS引擎提供了逻辑数据视图和物理数据视图之间的桥梁。

- Data definition subsystem defines the logical structure of a database using a data dictionary or schema.

数据定义子系统使用数据字典或模式定义数据库的逻辑结构。

- Data manipulation subsystem provides tools for data maintenance and data analysis;

tools include query-by-example and structured query language (SQL).

数据操作子系统为数据维护 and 数据分析提供工具;工具包括实例查询和结构化查询语言。

- Application generation subsystem provides tools for creating data entry forms with specialized programming languages.

应用程序生成子系统提供了用专门的编程语言创建数据输入表单的工具。

- Data administration subsystem manages the database; database administrators (DBAs) are computer professionals who help define processing rights.

数据管理子系统管理数据库;数据库管理员 (dba) 是帮助定义处理权限的计算机专业人员。

DBMS STRUCTURE 数据库管理系统结构

DBMS programs are designed to work with specific data structures or database models. These models define rules and standards for all the data in the database. Five principal database models are hierarchical, network, relational, multidimensional, and object-oriented.

DBMS 程序设计用于处理特定的数据结构或数据库模型。这些模型为数据库中的所有数据定义规则和标准。五种主要的数据库模型分别是层次、网络、关系、多维和面向对象。

Hierarchical Database 层次数据库

Hierarchical database uses nodes to link and structure fields and records; entries may have one parent node with several child nodes in a one-to-many relationship. 层次数据库使用节点对字段和记录进行链接和结构化;条目可能具有一对多关系中的一个父节点和几个子节点。

Network Database 网络数据库

Network database is like hierarchical except a child node may have more than one parent in a many-to-many relationship; additional connections are called pointers. 网络数据库类似于层次结构, 只是一个子节点可能在多对多关系中有一个以上的父节点;其他连接称为指针。

Relational Database 关系数据库

Relational database data is stored in tables (relations); related tables must have a common data item (key field). A table and its data are called a relation.

关系数据库数据存储存在表(关系)中;相关的表必须有一个公共数据项(关键字段)。表及其数

据称为关系。

DBMS STRUCTURE 数据库管理系统结构

Multidimensional Database 多维数据库

Multidimensional databases extend two-dimensional relational tables to three or more dimensions, sometimes called a data cube.

多维数据库将二维关系表扩展为三维或更多维，有时称为数据多维。

Multidimensional databases offer more flexible structures than relational databases, providing a more intuitive way of modeling data.

多维数据库提供了比关系数据库更灵活的结构，提供了一种更直观的数据建模方法。

Object-Oriented Database 面向对象的数据库

Object-oriented databases store data, instructions, and unstructured data. Data is organized using classes, objects, attributes, and methods.

面向对象的数据库存储数据、指令和非结构化数据。数据是用类、对象、属性和方法组织起来的。

- Classes are general definitions.

类是通用定义。

- Objects are specific instances of a class that can contain both data and instructions to manipulate the data.

对象是类的特定实例，可以包含数据和操作数据的指令。

- Attributes are the data fields an object possesses.

属性是对象所拥有的数据字段。

- Methods are instructions for retrieving or manipulating attribute values.

方法是检索或操作属性值的指令。

TYPES OF DATABASES 类型的数据库

There are four types of databases:

数据库有四种类型：

- Individual (personal computer) database: Used by one person.

个人(个人电脑)数据库:供一人使用。

- Company database: Stored on central server; accessed by multiple people.

公司数据库:存储在中央服务器;多人访问

- Distributed database: Spread out geographically; accessed by communications links.

分布式数据库:地理分布;通过通信链接访问。

- Commercial databases (information utilities and data banks): Enormous; for particular subjects.

商业数据库(信息设施和数据库):庞大;为特定主题。

DATABASE USES AND ISSUES 数据库使用和问题

Databases offer a great opportunity for increased productivity; however, security is always a concern.

数据库为提高生产力提供了很大的机会;然而,安全性始终是一个问题。

Strategic Uses 策略性应用

Data warehouses support data mining. Data mining is a technique for searching and exploring databases for related information and patterns.

数据仓库支持数据挖掘。数据挖掘是一种搜索和探索数据库相关信息和模式的技术。

Security 安全性

Two important security concerns are illegal use of data and unauthorized access. Most organizations use firewalls to protect their internal networks.

两个重要的安全问题是非法使用数据和未经授权的访问。大多数组织使用防火墙来保护他们的内部网络。

CAREERS in IT 其中的职业

Database administrators use database management software to determine the most efficient ways to organize and access a company's data. They are also responsible for database security and system backup. A bachelor's degree in computer science or information systems and technical experience are required. Salary range is \$56,000 to \$90,000.

数据库管理员使用数据库管理软件来确定组织和访问公司数据的最有效方法。它们还负责数据库安全和系统备份。计算机科学或信息系统学士学位和技术经验。工资范围为5.6万美元至9万美元。

雨课堂

课后习题

- 选择

1. Facts or observations about people, places, things, and events are:

关于人、地方、事物和事件的事实或观察是:

a. data

2. The most basic logical data element such as a single letter, number, or special character is known as a(n):

最基本的逻辑数据元素，如单个字母、数字或特殊字符被称为:

a. character

3. Each record in a database has at least one distinctive field, called the:

数据库中的每条记录至少有一个独特的字段，称为:

a. key field

4. One element of database security is to provide only authorized users with:

数据库安全的一个要素是只向授权用户提供:

c. passwords

5. The bridge between the logical and physical views of the data is provided by:

数据的逻辑视图和物理视图之间的桥梁由:

a. DBMS

6. Highly trained computer specialists who interact with the data administration subsystem are known as:

与数据管理子系统交互的训练有素的计算机专家被称为:

c. database administrators

7. In a network database, each child node may have more than one parent node; this is known as a:

在网络数据库中，每个子节点可以有一个以上的父节点;这被称为:

b. many-to-many relationship

8. Connections between parent nodes and child nodes are provided by:

父节点和子节点之间的连接方式如下：

d. pointers

9. Two of the most significant advantages of multidimensional databases over relational databases are processing speed and:

多维数据库相对于关系数据库的两个最显著的优势是处理速度和：

a. conceptualization

10. Object-oriented databases organize data by classes, attributes, methods, and:

面向对象的数据库通过类、属性、方法和以下方式组织数据：

a. objects

• 匹配

1. View that focuses on the actual format and location of the data.

视图，主要关注数据的实际格式和位置。

f. physical

2. Group of related characters.

相关字符组。

d. field

3. Type of processing in which data is collected over several hours, days, or even weeks and then processed all at once.

在数小时、数天甚至数周内收集数据，然后一次处理所有数据的处理类型。

b. batch

4. A data problem that often occurs when individual departments create and maintain their own data.

个别部门创建和维护自己的数据时经常发生的数据问题。

g. redundancy

5. Another name for a data dictionary.

数据字典的另一个名称。

i. schema

6. Type of database structure where fields or records are structured in nodes that are connected like the branches of an upside-down tree.

数据库结构的类型，其中字段或记录被组织在像倒置树的分支一样连接的节点中。

e. hierarchical

7. Type of database structure where the data elements are stored in different tables.

数据元素存储在不同表中的数据库结构类型。

h. relational

8. Two of the most significant advantages of multidimensional databases are conceptualization and processing _.

多维数据库的两个最显著的优点是概念化和处理_。

j. speed

9. Object-oriented databases organize data by classes, objects, methods, and _.

面向对象的数据库通过类、对象、方法和_来组织数据。

a. attributes

10. Type of database that uses communication networks to link data stored in different locations.

使用通信网络连接存储在不同位置的数据的数据库类型。

c. distributed

第 12 章 SYSTEMS ANALYSIS AND DESIGN

系统分析与设计

SYSTEMS ANALYSIS AND DESIGN 系统分析与设计

A system is a collection of activities and elements organized to accomplish a goal.

Systems analysis and design is a six-phase problem-solving procedure that makes up the systems life cycle. The phases are

系统是为实现一个目标而组织起来的活动和要素的集合。系统分析和设计是组成系统生命周期的六个阶段的问题解决过程。这些阶段为

- Preliminary investigation—identifying organization's problems or needs and summarizing in short report.

初步调查-发现组织存在的问题或需求，并总结成简短的报告。

- Systems analysis—studying present system in depth, specifying new requirements, and documenting findings.

系统分析-深入研究现有系统，指定新的需求，并记录发现。

- Systems design—designing new or alternative system to meet new requirements and creating a design report.

系统设计—设计新的或替代的系统以满足新的需求，并创建设计报告。

- Systems development—acquiring, developing, and testing needed hardware and software.

系统开发——获取、开发和测试所需的硬件和软件。

- Systems implementation—installing new system and training people.

系统实施—安装新系统和培训人员。

- Systems maintenance—periodically evaluating and updating system as needed.

系统维护—根据需要定期评估和更新系统。

Systems analysts are the computer professionals who typically follow the six-phase systems life cycle.

系统分析师是通常遵循六阶段系统生命周期的计算机专业人员。

PHASE 1: PRELIMINARY INVESTIGATION 第一阶段:初步调查

The preliminary investigation determines the need for a new information system. It is typically requested by an end user or a manager. Three tasks of this phase are defining the problem, suggesting alternative systems, and preparing a short report. 初步调查决定了对新信息系统的需求。它通常是由最终用户或管理人员请求的。这一阶段的三项任务是确定问题、提出替代制度和编写一份简短报告。

Defining the Problem 定义问题

The current information system is examined to determine who needs what information, when the information is needed, and why it is needed.

检查当前的信息系统是为了确定谁需要什么信息,什么时候需要信息,以及为什么需要信息。

If the existing information system is large, then a systems analyst conducts the survey. Otherwise, the end user conducts the survey.

如果现有的信息系统很大,那么由系统分析师进行调查。否则,由最终用户进行调查。

Suggesting Alternative Systems 建议选择系统

Some possible alternative systems are suggested. Based on interviews and observations made in defining the problem, alternative information systems are

identified.

提出了一些可能的替代系统。根据在确定问题时所做的访谈和观察，确定了可供选择的信息系统。

Preparing a Short Report 准备简短报告

To document and communicate the findings of Phase 1, preliminary investigation, a short report is prepared and presented to management.

为记录和沟通第一阶段(初步调查)的结果，准备一份简短的报告，并提交管理层。

PHASE 2: SYSTEMS ANALYSIS 阶段 2: 系统分析

In systems analysis, data is collected about the present system. The focus is on determining the requirements for a new system. Three tasks of this phase are gathering data, analyzing the data, and documenting the analysis.

在系统分析中，收集有关当前系统的数据。重点是确定新系统的需求。这个阶段的三个任务是收集数据、分析数据和记录分析。

Gathering Data 数据采集

Data is gathered by observation, interviews, questionnaires, and looking at documents. One helpful document is the organization chart, which shows a company's functions and levels of management.

通过观察、访谈、问卷调查和查阅文件来收集数据。组织结构图是一个有用的文件，它显示了公司的职能和管理水平。

Analyzing the Data 分析数据

There are several tools for the analysis of data, including top-down analysis, grid charts, and system flowcharts.

有几种用于数据分析的工具，包括自顶向下分析、网格图和系统流程图。

Documenting Systems Analysis 记录系统分析

To document and communicate the findings of Phase 2, a systems analysis report is prepared for higher management.

为记录和沟通第 2 阶段的结果，我们准备了一份系统分析报告，以供高级管理人员使用。

PHASE 3: SYSTEMS DESIGN 阶段 3: 系统设计

In the systems design phase, a new or alternative information system is designed.

This phase consists of three tasks:

在系统设计阶段，设计一个新的或可选的信息系统。这个阶段包括三个任务：

Designing Alternative Systems 设计选择系统

Alternative information systems are designed. Each alternative is evaluated for
设计了替代信息系统。对每个选项进行评估

- Economic feasibility—cost versus benefits; time for the system to pay for itself.
经济可行性-成本 vs 收益;是时候让系统自己买单了。

- Technical feasibility—hardware and software reliability; available training.
技术可行性——软硬件可靠性;提供培训。

- Operational feasibility—will the system work within the organization?
操作可行性——系统在组织内是否有效?

Selecting the Best System 选择最佳系统

Four questions should be considered when selecting the best system:

在选择最佳系统时应考虑四个问题：

- Will the system fit into an overall information system?
该系统是否适合整个信息系统?

- Will the system be flexible enough to be modified as needed in the future?
系统是否足够灵活，可以在将来根据需要进行修改?

- Will it be secure against unauthorized use?
它是否能防止未经授权的使用?

- Will the system's benefits exceed its costs?
这个系统的收益会超过成本吗?

Writing the Systems Design Report 撰写系统设计报告

To document and communicate the findings of Phase 3, a systems design report is prepared for higher management.

为记录和沟通第三阶段的研究结果，本处拟备了一份系统设计报告，以供高层管理人员使用。

PHASE 4: SYSTEMS DEVELOPMENT 第四阶段:系统开发

In the systems development phase, software and hardware are acquired and tested.
在系统开发阶段，获取和测试软件和硬件。

Acquiring Software 获取软件

Two ways to acquire software are purchasing off-the-shelf packaged software and designing custom programs.

获得软件的两种方法是购买现成的打包软件和设计定制程序。

Acquiring Hardware 获取硬件

Acquiring hardware involves consideration for future company growth, existing networks, communication capabilities, and training.

获取硬件涉及到对未来公司发展、现有网络、通信能力和培训的考虑。

Testing the New System 测试新系统

Using sample data, the new system is tested. This step can take several months for a complex system.

利用样本数据对新系统进行了测试。对于一个复杂的系统,这个步骤可能需要几个月的时间。

PHASE 5: SYSTEMS IMPLEMENTATION 阶段 5:系统实现

Systems implementation, also known as conversion, is the process of changing to the new system and training people.

制度实施又称转换,是对新制度进行变更和人员培训的过程。

Types of Conversion 句式转换

Four ways to convert are direct, parallel, pilot, and phased approaches.

四种转换方法是直接、并行、先导和分阶段的方法。

- Direct approach—abandoning the old system and starting up the new system; can be very risky and not recommended.

直接方式——摒弃旧体制,启动新体制;可能是非常危险的,不建议。

- Parallel approach—running the old and new side by side until the new system proves its worth; very low risk; however, very expensive; not generally recommended.

并行方法——新老系统并行运行,直到新系统证明其价值;非常低的风险;然而,非常昂贵;一般不推荐。

- Pilot approach—converting only one part of the organization to the new system until the new system proves its worth; less expensive but riskier than parallel conversion; recommended for situations with many people performing similar

operations.

试点方法——只将组织的一个部分转换为新系统，直到新系统证明其价值；比并行转换更便宜，但风险更大；适用于许多人执行类似操作的情况。

- Phased approach—gradually implementing the new system to the entire organization; less risky but more expensive than parallel conversion; recommended for situations with many people performing different operations.

分阶段方法——在整个组织内逐步实施新系统；比并联转换风险小但成本高；适用于许多人执行不同操作的情况。

Training 训练

Training is important, but often overlooked. Some people may train early as the equipment is being delivered so that they can adjust more easily. Sometimes a professional trainer is used; other times the systems analyst acts as the trainer. 训练很重要，但经常被忽视。有些人可能会在设备交付时提前训练，以便更容易地调整。有时会聘请专业的培训师；其他时候，系统分析师充当培训师。

PHASE 6: SYSTEMS MAINTENANCE 阶段 6: 系统维护

Systems maintenance consists of a systems audit followed by periodic evaluation. 系统维护包括系统审计和定期评估。

Systems Audit 系统检查

Once the system is operational, the systems analyst performs a systems audit by comparing the new system to its original design specifications.

一旦系统开始运行，系统分析师将通过将新系统与其原始设计规范进行比较来执行系统审计。

Periodic Evaluation 期刊评价

The new system is periodically evaluated to ensure that it is operating efficiently 新系统定期进行评估，以确保其有效运作

PROTOTYPING AND RAD 原型和快速

Due to time pressures, it is not always feasible to follow every phase of the systems life cycle. Two alternatives that require less time are prototyping and RAD.

由于时间压力，跟踪系统生命周期的每个阶段并不总是可行的。需要较少时间的两个替代方案是原型设计和快速开发。

Prototyping 样机研究

Prototyping means to build a model or prototype that can be modified before the actual system is installed. Typically, the development time for prototyping is shorter; however, it can be more difficult to manage the project and to control costs.

原型是指在实际系统安装之前建立一个可以修改的模型或原型。通常情况下，原型开发时间会更短；然而，管理项目和控制成本可能会更加困难。

Rapid Applications Development 快速应用开发

Rapid applications development (RAD) uses powerful development software, small specialized teams, and highly trained personnel. Typically, the development costs more. However, the time is much less and the quality is often better.

快速应用程序开发(RAD)使用强大的开发软件、小型的专门团队和训练有素的人员。通常，开发成本更高。然而，时间更少，质量往往更好。

CAREERS in IT 其中的职业

Systems analysts plan and design new systems or reorganize a company's computer resources to better utilize them. They follow the systems life cycle through all its steps. Either an advanced associate's degree or a bachelor's degree in computer science or information systems and technical experience are required. Salary range is \$53,000 to \$77,000.

系统分析师计划和设计新的系统或重新组织公司的计算机资源以更好地利用它们。它们遵循系统生命周期的所有步骤。需要高级副学士学位或计算机科学或信息系统和技术经验学士学位。薪资范围为 53,000 美元至 77,000 美元。

雨课堂

课后习题

- 选择

1. An information system is a collection of hardware, software, people, procedures, the Internet, and:

信息系统是硬件、软件、人员、程序、互联网和：

a. data

2. What is the first phase in the systems life cycle?

系统生命周期的第一个阶段是什么?

b. preliminary investigation

3. Which phase involves installing the new system and training people?

哪个阶段包括安装新系统和培训人员?

d. systems implementation

4. This phase is concerned about determining system requirements not with design.

这个阶段关注的是确定系统需求, 而不是设计。

b. systems analysis

5. Which systems analysis tool shows the relationship between input and output documents?

哪个系统分析工具显示输入和输出文档之间的关系?

c. grid chart

6. These tools relieve the systems analysts of many repetitive tasks, develop clear documentation, and, for larger projects, coordinate team member activities.

这些工具减轻了系统分析人员的许多重复任务, 开发了清晰的文档, 并且, 对于更大的项目, 协调了团队成员的活动。

b. CASE

7. Which phase is concerned with economic, technical, and operational feasibility?

哪个阶段与经济、技术和操作的可行性有关?

c. systems design

8. What type of feasibility evaluates whether the people within the organization will embrace or resist a new system?

什么样的可行性评估组织内的人是会接受还是抵制一个新系统?

c. operational

9. Which approach to conversion begins by trying out a new system in only one part of an organization?

哪种转换方法是从只在组织的一个部分尝试新系统开始的?

b. pilot

10. An alternative to the systems life cycle approach using powerful development

software, small specialized teams, and highly trained personnel.

系统生命周期方法的另一种选择，使用强大的开发软件，小型的专门团队，和受过高度训练的人员。

d. RAD

• 匹配

1. Systems analysis and design is a six-phase problem solving procedure for examining and improving an information _.

系统分析与设计是一个检验和改进信息系统的六个阶段的问题解决过程。

h. system

2. Systems life cycle phase that studies the present system in depth.

系统生命周期阶段，即深入研究本系统的阶段。

a. analysis

3. Systems analysis involves suggesting alternative _.

系统分析包括建议替代_。

g. solutions

4. The last and ongoing phase of the systems life cycle is systems _.

系统生命周期的最后一个和正在进行的阶段是系统_。

e. maintenance

5. The document that shows the levels of management and formal lines of authority is an _.

显示管理级别和正式权力线的文件是_。

j. organization char

6. This phase begins with designing alternative systems.

这个阶段从设计替代系统开始。

i. systems design

7. Another name for systems implementation.

系统实现的另一个名称。

b. conversion

8. The phase in which the old system is replaced and training begins.

替换旧系统和开始训练的阶段。

d. implementation

9. The four approaches to conversion are parallel, pilot, phased, and _.

转换的四种方法是平行的、试点的、阶段性的和_。

c. direct

10. The approach in which the new system is implemented gradually over a period of time.

在一段时间内逐步实施新制度的方法。

f. phased